

TRANSPONDER LANDING SYSTEM (TLS) MAINTENANCE MANUAL



August 18, 2021

DOCUMENT #020-00074 REV G

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Advanced Navigation and Positioning Corporation has attempted to describe in this manual all areas of possible danger to personnel in connection with the use of this equipment. This document refers to United States government and professional technical standards: Consult government standards prevailing in the region of installation to assure compliance with local requirements.

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ENGINEERING CHANGE ORDER RECORD

TLS MAINTENANCE MANUAL

Document #020-00074 Rev G

Revision	Issue Date	Revision Description	ECN #
A	08/10/12	TLS Manual Division & Updates	1916
B	2014Dec09	4.3 Updates	1935
C	2018Apr11	Update customer support phone number, add steps to re-align ASA	02046
D	2019Oct01	Added parallel port switch, combiner, multiple aircraft multiple GTU, GXT4 UPS, new rack diagrams, sensor 920-00400, added Genset maintenance, removed references to RSM and XP	ECA-00148
E	2020Jul8	Updated wording w.r.t. Base 2 Refresh	ECA-00177
F	2020Oct20	Update troubleshooting sections	ECA-00197
G	2021Apr29	Develop troubleshooting tables, update for auto-archive, update for inter-TLS communication	ECA-00276

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SECTION 1: INTRODUCTION

1.1 Document Overview

This document details the procedures and settings required for maintaining and optimizing Advanced Navigation and Positioning Corporation's (ANPC) Transponder Landing System (TLS).

1.2 ANPC TLS Manuals

The ANPC TLS is accompanied by a group of four primary manuals:

- Document # 020-00071, TLS Calibration Manual
- Document # 020-00072, TLS Operators Manual
- Document # 020-00073, TLS Installation Manual
- Document # 020-00074, TLS Maintenance Manual

The ANPC TTLS includes two additional manuals:

- Document 020-00076 Container System Deployment
- Document 020-00077 TTLS Packing Instructions

SECTION 2: STANDARDS AND TOLERANCES

2.1 Transponder Landing System/Instrument Landing System Parameters

The table below presents a list of parameters similar to those found in Instrument Landing System (ILS) Standards and Tolerances to assist the user in understanding system monitor functions. A complete list of BIT and Track IM elements is presented in Table 2-1.

Table 2-1 Definition of Terms

Standard	Standard is the optimum value assigned to the parameter.
Initial Tolerance Limit	Initial tolerance limit (ITL) is the maximum deviation from the standard value or range that is permissible at the time of initial commissioning of the system or after replacement of an LRU.
Operating Tolerance Limit	The operating tolerance limit is the maximum deviation from the system's standard value or range. Within this range, the equipment can continue to operate without adjustment or corrective maintenance. However, Corrective action by TLS maintenance personnel may be required.
Verification Method	Parameters identified in the tables below are tested by the following methods: Built-In Test (BIT) Track Integrity Monitor (IM) Periodic Flight Inspection (FI) Factory Acceptance Test (FAT) Site Acceptance Test (SAT) Completion of a periodic maintenance action (PM)
Key Performance Parameters	Key performance parameters are selected critical indicators that the system, sub-system, or equipment is performing its intended function. These key parameters are indicated with a symbol (→) in the tables below.

2.2 Interrogator Standards and Tolerances

Table 2-2 Interrogator Standards and Tolerance

Parameter	Standard	Tolerance / Limit		Verification Method
		Initial	Operating	
A. RF peak pulse power (P1 and P3):	As established by flight inspection	Same as standard	>50 percent of standard (power) >70.7 percent of standard (voltage)	FAT, BIT, PM
B. RF peak pulse power (P2):	As established by flight Inspection	Same as standard	>50 percent of standard (power) >70.7 percent of standard (voltage)	FAT, BIT, PM
C. P3 amplitude referenced to P1:	Equal amplitudes	-1 to +3 dB	Same as initial	FAT, SAT, PM
D. Transmitter return power (interrogation and suppression):	No return power	10 dB below forward-power	Same as initial	FAT, SAT, BIT
E. RF Pulse Frequency:	1030 MHz	Standard $\pm$ 100 kHz	Same as initial	FAT
F. Pulse timing:				
(1) Spacing P1 to P3 (mode 3/A)	8.0 μ s	Standard $\pm$ 0.2 μ s	Same as initial	FAT, SAT, PM, BIT
(2) Spacing P1 to P2	2.0 μ s	Standard $\pm$ 0.15 μ s	Same as initial	FAT, SAT, PM, BIT
(3) P1, P2, and P3 Pulse Width	0.8 μ s	Standard $\pm$ 0.1 μ s	Same as initial	FAT, SAT, PM, BIT
(4) Pulse Rise Time (all pulses)	50 ns to 100 ns	Same as standard	Same as standard	FAT, SAT, PM
(5) Pulse Rise Time (all pulses)	50 ns to 200 ns	Same as standard	Same as standard	FAT, SAT, PM

2.3 Localizer Standards and Tolerances

Table 2-3 Localizer Standards and Tolerances

Parameter	Standard	Tolerance / Limit		Verification Method
		Initial	Operating	
RF carrier power	As established by flight inspection	Same as standard	>50 percent of standard (power) >70.7 percent of standard (voltage)	FAT, BIT, IM, FI
B. Modulation levels:				
(1) 90 and 150 Hz SDM	40 percent	39 to 41 percent	36 to 44 percent	FAT, SAT, BIT, IM, FI
(2) Modulation equality	0 DDM	≤ 0.003 DDM	≤ 0.007 DDM	FAT, SAT, BIT, IM, FI
(3) 1020 HZ	10 percent	5 to 15 percent	same as initial	FAT, SAT, BIT, IM, FI
(4) A, B, C, 1 Tone	30 percent	25 to 35 percent	same as initial	SAT, BIT
C. Frequencies:				
(1) Carrier	Authorized	Standard ± .002 percent	Same as initial	FAT, SAT, FI
(2) 90 Hz tone	90 Hz	89.0 to 91.0 Hz	Same as initial	FAT, SAT
(3) 150 Hz tone	150 Hz	148.0 to 152.0 Hz	Same as initial	FAT, SAT
(4) 1020 Hz tone	1020 Hz	970 to 1070 Hz	Same as initial	FAT, SAT
D. Antenna system:				
VSWR	1:1	<2:1	Same as initial	BIT, IM
E. Monitor alarm points:				IM
(1) Tracking accuracy, i.e., shift from derived normal centerline	≤12 μA ≤8 μA	Same as standard Same as standard	Same as standard Same as standard	
- Alignment				
- Structure				
(2) Monitor shutdown delay	< 10 seconds	Same as standard	Same as standard	IM

2.4 Glide Slope Standards and Tolerances

Table 2-4 Glide Slope Standards and Tolerances

Parameter		Standard	Tolerance / Limit		Test Methods
			Initial	Operating	
→	RF carrier power	As established by flight inspection	Same as standard	>50 percent of standard (power) >70.7 percent of standard (Voltage)	FAT, BIT, IM, FI
→	Modulation levels:				
	(1) 90 and 150 Hz SDM	80 percent	78 to 82 percent	75 to 85 percent	FAT, BIT, IM, FI
	(2) Modulation equality	0 DDM	≤ 0.004 DDM	≤ 0.008 DDM	FAT, BIT, IM, FI
→	Frequencies:				
	(1) Carrier	Authorized	Standard ± .002 percent	Same as initial	FAT, SAT, FI
	(2) 90 Hz tone	90 Hz	89.0 to 91.0 Hz	Same as initial	FAT
	(3) 150 Hz tone	150 Hz	148.0 to 152.0 Hz	Same as initial	FAT
→	Antenna system:				
	VSWR	1:1	<2:1	Same as initial	BIT, IM
→	Monitor alarm points:				
	Tracking accuracy, i.e., deviation from the commissioned angle (CA) Alignment	≤CA+10 percent ≥CA-7.5 percent	Same as standard Same as standard	Same as standard Same as standard	IM IM IM
	Structure	≤30 µA	Same as standard	Same as standard	
	Structure	≤30 µA	Same as standard	Same as standard	
	(2) Monitor shutdown delay	< 6 seconds	Same as standard	Same as standard	IM

2.5 AOA Sensor / Cal/BIT Standards and Tolerances

Table 2-5 AOA Sensors and Cal/BIT Standards and Tolerances

Sensor / Cal/BIT Parameter	Standard	Tolerance / Limit		Test Methods
		Initial	Operating	
→ A) Cal/BIT test pulse amplitude:	As established at installation.	Same as standard	Standard ± 4 dB (power) Standard ± 100 mV (BIT meas voltage)	SAT, BIT, IM
B) Cal/BIT test pulse amplitude jitter:	0 dB	Standard ± 3 dB (power) Standard ± 75 mV (BIT Voltage)	Same as initial	SAT, BIT
C) Cal/BIT test pulse frequency:	1090 MHz	Standard ± 2 MHz	Same as initial	FAT, BIT, IM
→ D) Cal/BIT test pulse phase:	As established at installation.	Same as standard	Standard ± 8 counts for ASA Sensor. Standard ± 16 counts for ESA Sensor.	FAT, SAT BIT, IM,
E) Cal/BIT test pulse phase jitter:	0 counts	Standard ± 10 counts	Same as initial	SAT, BIT
F) Cal/BIT test pulse delta phase:	0 counts difference.	Standard ± 8 counts	Same as initial	FAT, SAT, BIT, IM
→ G) Cal/BIT test pulse Time of Arrival:				
→ (1) NEAR PULSE:	As established at installation.	Same as standard.	Standard ± 10 ns.	BIT, IM
(2) Far Pulse:	As established at installation.	Same as standard.	Standard ± 40 ns.	BIT, IM
(3) Jitter:	0 ns	Standard ± 8 ns	Same as initial	BIT
→ H) ESA Tilt Sensors:				
(1) TILT READINGS:	0 degrees	Standard ± 0.04 degrees.	Standard ± 0.11 degrees	BIT, IM
(2) Tilt Self Test:	≤ -0.9 degrees	Same as standard.	Same as standard.	BIT
(3) Tilt Sensor Temperature:	Ambient	Standard ± 5 °C.	Same as initial.	BIT
I) Sensor Temperature:	Ambient	Standard ± 5 °C	Same as initial.	BIT

Sensor / Cal/BIT Parameter	Standard	Tolerance / Limit		Test Methods
		Initial	Operating	
→ J) Sensor Receiver Power Sensitivity:	-70 dBm signal (700 mV)	Standard $\pm$ 4 dB. ($\pm$ 100 mV)	Same as standard.	BIT, IM
K) Sensor Selectivity:	0dB over $\pm$ 5 MHz	Standard $\pm$ 3 dB. ($\pm$ 9 cts)	Same as standard.	BIT
L) Sensor Frequency Meas:				
(1) Center Freq Accuracy	1090 MHz	Standard $\pm$ 2 MHz	Same as initial	BIT, IM
(2) Error over band	0 MHz	Standard $\pm$ 1 MHz	Same as initial	BIT

2.6 UPS Power Standards and Tolerances

Table 2-6 UPS Power Standards and Tolerances

Parameter	Standard	Tolerance / Limit		Test Methods
		Initial	Operating	
1. UPS-R Output Voltage:	120 VAC	108 to 132 VAC	Same as initial	FAT, SAT, PM
2. UPS-R Battery Operation:	> 60 minutes	Same as standard	Same as standard	FAT, SAT, PM
3. UPS-S Output Voltage:	120 VAC	108 to 132 VAC	Same as initial	FAT, SAT, PM
4. UPS-S Battery Operation:	> 60 minutes	Same as standard	Same as standard	FAT, SAT, PM

2.7 RCU Functions and Indications

Table 2-7 RCU Functions and Indications

Parameter	Standard	Tolerance / Limit		Test Methods
		Initial	Operating	
A. Remote Status Indication:	Appropriate Mode and State Indications	Same as standard	Same as standard	FAT, SAT, PM
B. Remote Entry:				
(1) CODE ENTRY	Ability to enter transponder ID code.	Same as standard	Same as standard	FAT, SAT, PM
(2) ACQUISITION RESET	Ability to reset acquisition	Same as standard	Same as standard	FAT, SAT, PM
C. ACQUISITION TIMEOUT	Transition out of acquire	Standard $\pm$ 5 seconds	Same as initial	FAT, SAT, PM

2.8 Tools and test equipment

The following tools and test equipment are required:

- Portable ILS receiver (PIR), antenna(s), chargers, and carrying case
- Sokkia theodolite with data stream capable of interfacing with the TLS FTM function, carry case, tripod, prism, and prism pole.

- Portable oscilloscope Tektronix TDS-340A or equivalent
- SMA 50 ohm termination
- Diode detector – SMA
- SMA attenuators 20, 10, 6, 3, 1 dB
- Coax cables, splitters, and adaptors in the interrogator test kit

2.9 TLS Critical and Sensitive Areas

The critical and sensitive areas for the TLS azimuth and elevation sensor subsystems must be protected from unrestricted aircraft and vehicular traffic. This will ensure guidance signal integrity during approach procedures and target quality during surveillance operations. During such operations, no activity by the vehicles noted below is allowed in the critical area.

Guidelines for determining accepted activity in the sensitive area are as follows:

- Small aircraft are allowed in the sensitive area.
 - ✓ Small aircraft with a length of 45' or less and a height of 15' or less
- Cars and small trucks are allowed in the sensitive area.
- Large trucks and general aviation-type aircraft may pass through the sensitive area but should not remain parked in the sensitive area.
 - ✓ Large trucks and general aviation (GA type) aircraft with a length of 60' or less and a height of 20' or less

2.9.1 Critical/Sensitive Areas

In addition, the critical and sensitive areas are drawn with geometric simplicity in mind and are larger than the actual area that needs to be controlled. Therefore, it may be acceptable to have roads or taxiways intersect the critical or sensitive areas without requiring the roads or taxiways to be controlled. This is typically true with respect to the corners of the critical or sensitive area corners.

2.9.2 Tower Personnel: Ceiling Visibility

Tower personnel will issue control instructions to ground aircraft and vehicles to avoid interfering with TLS signals. The control specifics are as follows:

- a. Weather conditions are less than a ceiling 800 feet and/or visibility less than 2 miles, vehicles and aircraft are not authorized over the area when an arriving aircraft is between the TLS final approach fix and the airport.
- b. Weather conditions at or above a ceiling 800 feet and/or runway visibility 2 miles, the tower should promptly issue an advisory if the area is not protected when an arriving aircraft advises that a “coupled” approach will be conducted.

2.9.3 Identifying Critical Areas

During the system installation, areas deemed to be critical and sensitive will be identified. The airport operator must mark all taxiways with standard hold short markings outside of the critical area, and signs must be posted on all roads outside of the critical area boundary to ensure that neither vehicular nor aircraft traffic interfere with TLS signals during an approach. Taxiways and roads in the sensitive area must be correctly marked for the allowed operations in the sensitive area.

2.9.4 Approach Operations: Critical/Sensitive Areas

ESA critical and sensitive areas for approach operations are depicted in Figure 2-1 and Figure 2-2. ASA critical and sensitive areas for approach operations are depicted in Figure 2-3 and Figure 2-4. A plan view and a profile view of each area are shown for each sensor. The three-dimensional critical or sensitive area is the intersection of the two views. Therefore, an object must violate the plan view and the profile view to be considered in the critical or sensitive area. Table 2-8 and Table 2-9 and Equations 1 and 2 present further

details for defining the critical and sensitive area. Vegetation within the critical and sensitive areas must be maintained in accordance with Periodic Maintenance procedures.

2.9.5 Surveillance Operations: Critical/Sensitive Areas

The TLS critical areas for surveillance operations are identical to those identified for approach operations. This is to ensure there are no obstructions in the close proximity of the antenna arrays.

A circle of 1200 feet radius around the antenna arrays is designated as the sensitive area for surveillance. Vehicle and aircraft movement through this zone is unrestricted. During the system installation process, parts of this sensitive area may be identified as “no-parking” zones for large vehicles and airplanes as identified above.

The criterion for identifying a no-parking zone is that the stationary vehicle would block a direct path signal from the service volume to a TLS antenna. For example, a taxiway adjacent to the ESA may have a no-stopping restriction, while a nearby apron adjacent to a building would not require any restrictions.

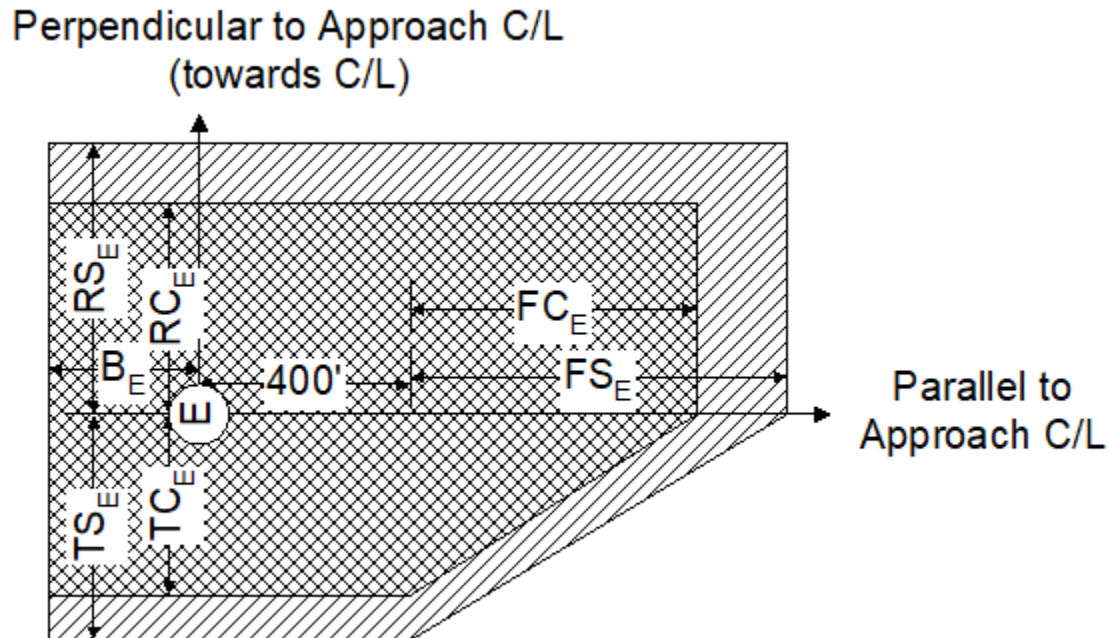


Figure 2-1 Plan View: ESA Critical and Sensitive Areas (not to scale)

The distance of the Critical Area (DC_G) and Sensitive Area (DS_G) is a function of the glide path angle (θ) and is given in Table 2-8 and Table 2-9. Note that these values are in addition to the 400' from the ESA to where the critical and sensitive areas taper inwards.

Table 2-8 Values for Critical Areas of ESA

Glide Path Angle (degrees)	FC _E (feet)	B _E (feet)	RC _E (feet)	TC _E (feet)
2.0° to less than 2.5°	1700	100	400	400
2.5° to less than 3.0°	650	50	150	50
3.0° to less than 3.5°	400	50	150	50
3.5° to 4.0°	150	50	150	50

Table 2-9 Values for Sensitive Areas of ESA

Glide Path Angle (degrees)	FS _E (feet)	B _E (feet)	RS _E (feet)	TS _E (feet)
2.0° to less than 2.5°	1300	100	400	400
2.5° to less than 3.0°	700	50	200	150
3.0° to less than 3.5°	500	50	200	150
3.5° to 4.0°	300	50	200	150

If desired, a more precise value for FSB_E can be calculated with Equation 1: $DS_G = 700 - 400((\theta) - 25)$ for $2.5^\circ \leq \theta \leq 3.6^\circ$.

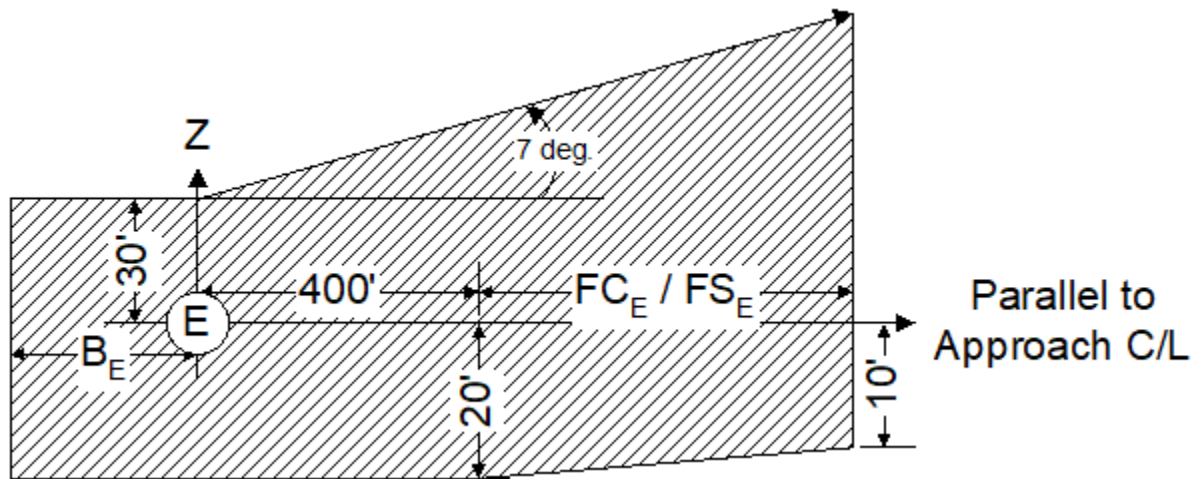


Figure 2-2 Profile View: ESA Critical & Sensitive Areas (not to Scale)

Perpendicular to Approach C/L
(towards C/L)

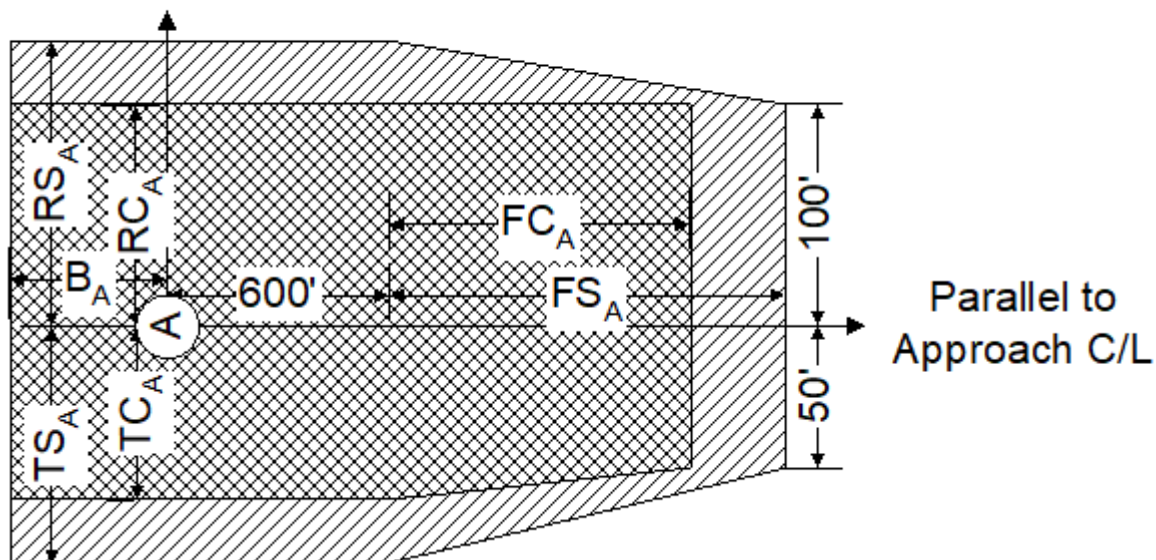


Figure 2-3 Plan View: ASA Critical & Sensitive Areas (not to scale)

The distance of the Critical Area (DC_L) and Sensitive Area (DS_L) are functions of the glide path angle (θ) and are given in Table 2-10 and Table 2-11. Note that these values are in addition to the 600' from the ASA to where the critical and sensitive areas taper inwards.

Table 2-10 Values for Critical Areas of ASA

Glide Path Angle (degrees)	FC_A (feet)	B_A (feet)	RC_A (feet)	TC_A (feet)
2.0° to less than 2.5°	2400	300	400	400
2.5° to less than 3.0°	900	150	100	100
3.0° to less than 3.5°	700	150	100	100
3.5° to 4.0°	200	150	100	100

If desired, a more precise value for FS_A can be calculated with Equation 2:

$$DSB_{GB} = 1200 - 800(\theta - 25) \text{ for } 2.5^\circ \leq \theta \leq 3.6^\circ.$$

Table 2-11 Values for Sensitive Areas of ASA

Glide Path Angle (degrees)	FS_A (feet)	B_A (feet)	RS_A (feet)	TS_A (feet)
----------------------------	---------------	--------------	---------------	---------------

2.0° to less than 2.5°	2400	300	400	400
2.5° to less than 3.0°	1200	150	200	200
3.0° to less than 3.5°	800	150	200	200
3.5° to 4.0°	400	150	200	200

If desired, a more precise value for FS_A can be calculated with Equation 3: $DS_G = 1200 - 800(\theta) - 25$ for $2.5^\circ \leq \theta \leq 3.6^\circ$.

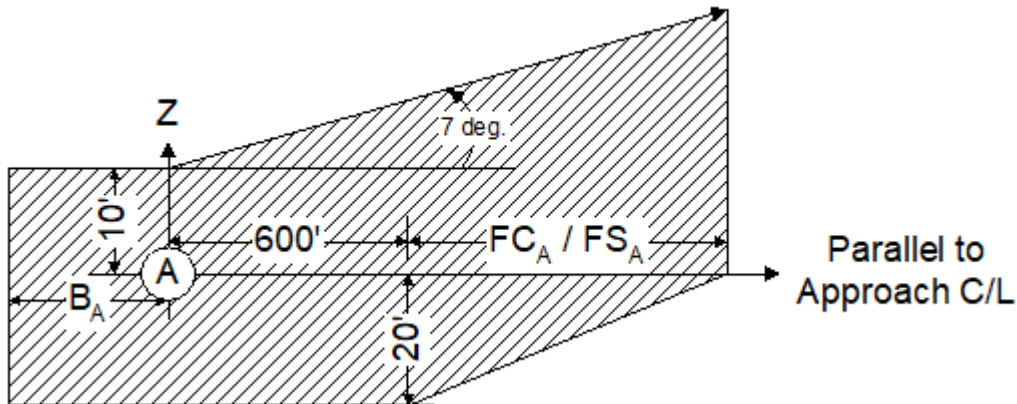


Figure 2-4 Profile View: ASA Critical & Sensitive Area (not to scale)

The critical area includes a perimeter defined by the layout of TLS system components and the entire area within this perimeter. The perimeter area includes a 15-foot radius around the Cal/Bit and a 40-foot radius around the Uplink and Interrogator assemblies.

SECTION 3: SYSTEM MAINTENANCE

3.1 Preventive Maintenance

The performance checks and maintenance tasks herein shall be performed immediately after initial installation and then at the required periodicity.

Table 3-1 lists the required maintenance procedure with cross-reference to the applicable tolerance and maintenance procedure over a 12-month calendar.

Only properly trained and certified technicians should make adjustments to the TLS.

Performance checks are designed to use Built-in-Test, external test equipment, and the Maintenance Interface Unit. When abnormal performance conditions are indicated during performance check procedures, refer to the corrective maintenance procedures.

3.2 Documentation Requirements

All periodic maintenance procedures require completion of the test, and any applicable data or observations shall be indicated in the Facility Maintenance Log Book. This documentation step is critical for the maintenance of a record of system operations and maintenance.

3.2.1 Separate Data Sheet

For some periodic maintenance procedures, a separate data sheet is provided to record test measurements. For these procedures, complete, sign, and file on-site the TLS Technical Performance Record for that test and indicate completion in the Facility Maintenance Log Book.

3.2.1.1 Personnel Safety

All personnel servicing LRUs must wear Electrostatic Discharge Safety (ESD) equipment: ESD wrist strap with an alligator clip.

Table 3-1 TLS Periodic Maintenance Schedule

Maintenance Item	Standards and Tolerances	Procedure #	Month											
			1	2	3	4	5	6	7	8	9	10	11	12
C-1 Archive of Data Files	Table 2-2, -3, -4, -5	3.3	X	X	X	X	X	X	X	X	X	X	X	X
C-2 RCU Functions Check and Monitor Shutdown Validation	Table 2-7	3.4												X
C-3 Maintenance of Critical Areas and Obstruction Lights	N/A	3.5	X	X	X	X	X	X	X	X	X	X	X	X
C-4 UPS Battery Check	Table 2-7	3.6						X						X
C-5 Air Filters	N/A	3.7			X			X			X			X
C-6 Check Guy Wire Tension	TLS	3.8						X						X
	TTLS	3.8	X	X	X	X	X	X	X	X	X	X	X	X
C-7 Rack Switch Control (RSC)	N/A	3.9			X			X			X			X
C-8 Antenna System Maintenance	N/A	0												X
C-9 Clean and Inspect Equipment	N/A	0												X
C-10 Interrogation Transmitters Check	Table 2-2	3.12												X
C-11 Guidance Transmitter Check	Table 2-3 and 2-4	3.13												X
C-12 Cable Repair and Waterproofing	N/A	3.14	As Required											
C-13 Generator Periodic Maintenance	N/A	1.25	See manufacturer's schedule, especially oil and filter changes											

* Procedure or test must be performed and successful completion documented in the Facility Maintenance Log Book.

3.3 C-1 Archive of Data Files

This procedure contains three elements: Review and recording of key performance data for the TLS, Archive of site data and transfer of data to ANPC for permanent archiving; and optionally, purging of data to maintain the operational hard disk space on Base station computers.

This procedure may be done more often if TLS approaches are conducted each day since the *.ram files from approaches are much larger than the *.bit files from BIT.

To remotely access the MIU or RCU's on a second TLS:

1. Open a Remote Desktop Connection from the maintenance room RCU (by opening the Windows start menu in the bottom left corner of the screen and typing 'Remote Desktop').
2. Enter "external.#LRU_NAME#" (where #LRU_NAME# is desired MIU or RCU connection) as the Computer, and press connect.
3. It will then prompt you for a user's credentials to connect to the LRU. This can only be done with a TLS technician's credentials. Enter the TLS technician credentials and press OK.
4. The Remote LRU's (MIU or RCU) screen should now appear.

3.3.1 Data Review and Recording

Do not change any nominal settings without coordinating with flight inspection. Monitor limits changes may be accomplished in coordination with a flight check to verify UHF/VHF alignment.

The following procedure describes the examination of the TLS data and recording of data for any parameters that are drifting toward an out of tolerance condition or parameters that are already out of tolerance. If LRUs are identified as out-of-tolerance, then refer to Corrective Maintenance.

Some parameters shift according to environmental changes throughout the year, and this may contribute to determining a better nominal to reflect a better average that will stay within monitor limits.

3.3.1.1 Tools and Special Instructions

None

3.3.1.2 Special Instructions

Record data in the "Technical Performance Record - Data Review and Recording C-1" in Appendix C.

3.3.1.3 Procedure

1. From the MI main window, press the Start button. Verify that the system transitions through Diagnostic and into Track mode as indicated in the MI "System Mode."
2. Depress the "System Status" button. Within the "View" options box, select "History." It may take a few seconds to generate the required log files.
3. Select each of the TLS unit icons that indicate an alert or alarm condition (yellow or red, respectively).
4. Review the displayed file for indications of system problems. For each TLS component, note and record alert and alarm conditions that appear excessive (non-transient, large number of alarms) and that may require a maintenance action on the TLS Technical Performance Record.
5. On the MI, stop the system software.

6. Open the MultiBIT tool and select "Open base1 interrog.bit" in the File menu.
7. Compare each parameter listed in the "Interrogator" section of the TLS Technical Performance Record and record any data that has become close to a monitor alert or alarm level. For the Data Range entry of the datasheet, record the minimum value measured and the maximum value measured (example: P1 Amplitude, 800 - 820 mV).
8. Open the file \\Base2\\c\\Run\\Bit**esa.bit** in MultiBIT.
9. Use the toolbar buttons to display the data for each of the parameters listed in the "ESA" section of the TLS Technical Performance Record and record any data that has become close to a monitor alert or alarm level. For the Data Range element of the datasheet, record the minimum value measured and the maximum value measured (example: High Phase, 232 - 240 cnts). Enable the "Cal/BIT Limits" button to display nominal values and alarm/alert limits.

Do Not enter individual transient values)
10. Open the file \\Base2\\c\\Run\\Bit**asa.bit** in MultiBIT.
11. Use the toolbar buttons to display the data for each of the parameters listed in the "ASA" section of the TLS Technical Performance Record and record any data that has become close to a monitor alert or alarm level. For the Data Range element of the datasheet, record the minimum value measured and the maximum value measured (example: High Phase, 232 - 240 cnts). Enable the "Cal/BIT Limits" button to display nominal values and alarm/alert limits.

Do Not enter individual transient values.
12. Open the file \\Base2\\c\\Run\\Bit**ata.bit** in MultiBIT.
13. Use the toolbar buttons to display the data for each of the parameters listed in the "ATA" section of the TLS Technical Performance Record and record any data that has become close to a monitor alert or alarm level. For the Data Range element of the datasheet, record the minimum value measured and the maximum value measured (example: Near TOA, 6123 - 6126 ns). Enable the "Cal/BIT Limits" button to display nominal values and alarm/alert limits.

Do Not enter individual transient values.)
14. Open the file \\Base2\\c\\Run\\Bit**uplink_1.bit** in MultiBIT.
15. Use the toolbar buttons to display the data for each of the parameters listed in the "Uplink" section of the TLS Technical Performance Record. Record any data that has become close to a monitor alert or alarm level.
16. For the Data Range element of the datasheet, record the minimum value measured and the maximum value measured (example: Loc Carrier, 2990 - 3010 mV). Enable the "Show Limits" button to display nominal values and alarm/alert limits.

Do Not enter individual transient values.

The SDM reading is displayed in decimal format (ex: 45% is presented as 0.45)
17. Switch to Base1 (QNX).
18. At the command line, type the following command: `chkfsys -u /dev/hd0*t77`. Depress the enter key.
19. If disk errors are indicated, record a summary of displayed faults. At the repair prompt, press the "Y" key to fix the fault. If no errors are indicated, record "None."

20. On the MI, press the Start button to start the system software. Verify that the system transitions through Diagnostic and into Track mode as indicated in the MI "System Mode."

3.3.2 Archiving Data

3.3.2.1 Tools and Materials Required

- External USB Flash Drive or another storage medium (optional)
- Drive mailer (optional)

3.3.2.2 Procedure

1. Press the Stop button from the Maintenance Interface (MI) main window if the system software is running. Wait for the MI to display "NO COMM WITH BASE SW," which indicates that system software on both bases has been shut down.
2. From the MI main window, press the Archive button. This will start the TLS data archive tool, which will retrieve all data files from the Bases and zip them into a file located on the MIU in the "C:\archives\< current date>[a-z]" directory (the [a-z] part is a character suffix that automatically sequences from 'a' to 'z' for each archive on a given day).
3. Press the "Run" button to initiate the archive, and when the archive status window indicates that it is done, press the "Close" button to exit.
4. Archiving can be configured to run periodically. On the MI, enter the System Setup, Settings window, and select the "Auto Archive" checkbox, and enter the interval in days.
5. Using Windows Explorer, copy the c:\archives\<current date> folder on the MIU to the USB Flash Drive.
6. Store the USB Flash Drive in the offsite data archiving facility.
7. Optional: mail the USB Flash Drive to the address below:

ANPC
489 N. 8th Street, Suite 203
Hood River, OR 97031
Attn: Customer Support

NOTE: Electronic transfer of data to an FTP site with prior coordination with ANPC is an acceptable alternative to mailed data. Additional procedures for creating a ZIP file of an archive folder must be followed when uploading data using FTP.

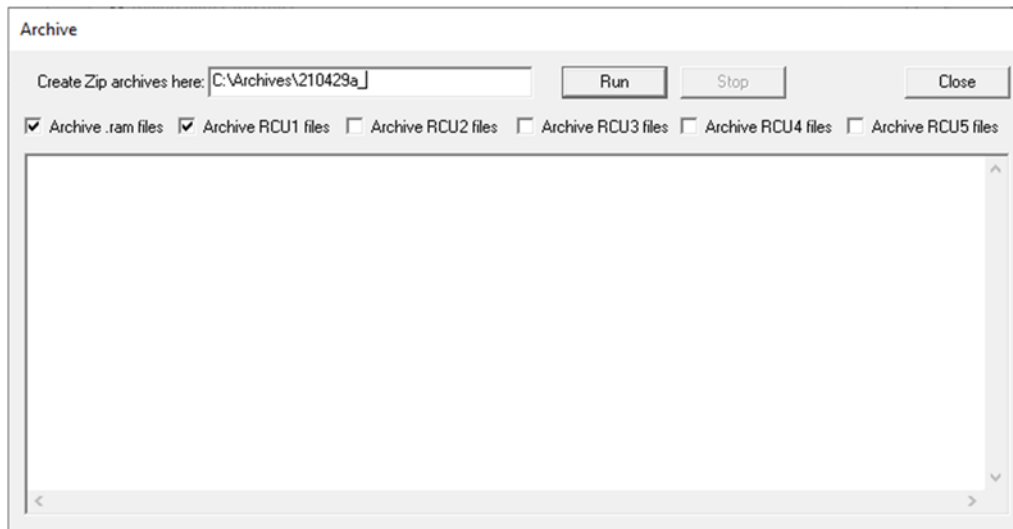


Figure 3-1 Archive Window

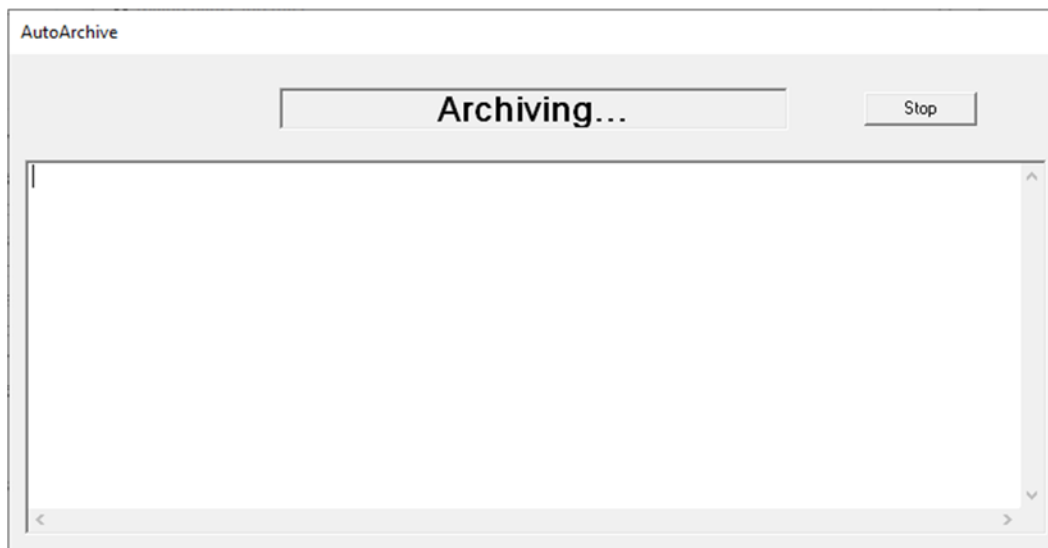


Figure 3-2 Auto Archive Window

3.3.3 Archive Data Documentation

1. Complete, sign, and file on-site the TLS Technical Performance Record for data review and recording in C-1.
2. Record the completion of C-1 in the Facility Maintenance Log Book.

3.4 C-2 RCU Functions Check and Monitor Shutdown Validation

This procedure verifies that RCU control functions are operational and the proper indication occurs at the RCU.

3.4.1 Tools and Materials Required

None

3.4.2 Special Instructions

Coordinate a short TLS service outage with appropriate airfield personnel who will notify ATC and other local and federal agencies as required.

Record Data in "Technical Performance Record - RCU Functions Check and Monitor Shutdown Validation C-2" in Appendix B.

3.4.3 Procedure

1. At the RCU interface, verify that the system is in Track Mode (System Mode indicates "TRACK"). Record this verification.
2. At the RCU, enter transponder code 7777 and press Acquire ILS. Verify that the system transitions to Acquire state (Track State indicates "ILS SEARCHING"). Record this verification.
3. Wait two minutes.
4. Depress the Reset command, acknowledge the reset action and verify that the system remains in Track mode. Record this verification.
5. At the TLS Base Station, disconnect the short BNC-BNC Patch Cable between the "Interrogation Sample In" and "Interrogation Sample Out" connectors on the Signal Test Points Panel.
6. On the MI, stop the system software.
7. On the MI, enter the System Setup, Settings window, change BIT Retries to 0. Depress "OK" to exit. Exit the setup section of the MI.
8. On the MI, start the system software. This will cause the system to run a diagnostic BIT routine. Verify that after the Diagnostic routine is completed, the MI System Mode indicator shows that the system is in Maintenance Mode. Record this verification.
9. At the RCU, verify an alarm indication on the RCU (System, Surveillance, and Guidance indicate "ALARM" in red). Record this verification.
10. At the TLS Base Station, re-connect the short BNC-BNC Patch Cable between the "Interrogation Sample In" and "Interrogation Sample Out" connectors on the Signal Test Points Panel.
11. On the MI, enter Test Mode, then exit Test mode. This will cause the system to run a diagnostic BIT routine. Verify that after the Diagnostic routine is completed, the MI System Mode indicator shows that the system is in Track Mode. Record this verification.
12. At the RCU, verify that the system is in Track Mode (System Mode indicates "TRACK"). Record this verification.
13. On the MI, stop the system software.
14. On the MI, enter the System Setup, Settings window, change BIT Retries back to the site retry setting. Depress "OK" to exit. Exit the setup section of the MI.
15. On the MI, start the system software.
16. Use the procedures of the appropriate airfield authority to notify them that the TLS is returned to service.

3.4.4 RCU Function and Monitor Shutdown Documentation

1. Complete, sign and file on-site the TLS Technical Performance Record for RCU functions check and monitor shutdown C-2.
2. Record the completion of C-2 in the Facility Maintenance Log Book.

3.5 C-3 Maintenance of Critical Areas and Obstruction Lights (Ob-lights)

This procedure ensures the TLS critical area is free of obstructions, vegetation, and snow, that critical area markings are in serviceable condition and ensures all obstruction lights are working.

If the approach glide path angle is less than 2.5°, then the area must be maintained to a maximum height of 2 inches instead of 1 foot.

3.5.1 Tools and Materials Required

- Climbing harness
- Replacement lights

3.5.2 Special Instructions

WARNING

Use of a climbing safety system is required when replacing obstruction light bulbs on the ESA or Uplink towers. Use of a bucket lift truck to ensure operator safety is suggested. If using a climbing harness, always use the two clip method to ensure constant attachment to the tower.

3.5.3 Procedure

1. Check all critical and sensitive areas for vegetation, snow buildup, or other objects that may interfere with the signals [Figure 3-3]. Coordinate the removal of any foreign objects, vegetation, or snow buildup that is higher than one foot above ground level and/or is obstructing the line of sight between the Cal/BIT and sensor antennas. Refer to Section 1 (Standards) for descriptions of the critical and sensitive areas.
2. Coordinate the removal of foreign objects, vegetation, or snow buildup taller than one foot inside the perimeter area defined by the layout of TLS system components and/or is obstructing the line of sight between the Cal/Bit and sensor antennas. Refer to Figure 3-3 for a typical TLS layout with vegetation and obstacle clearance zones shown. Note that the layout diagram below may differ from your site; however, the objective is to define a perimeter area using the outermost components of the system layout.
3. Coordinate the removal of foreign objects, vegetation, or snow buildup taller than one foot in an area to include a 15-foot radius around the Cal/Bit, ATA, and ASA assemblies and a 40-foot radius around the ESA, Uplink, and Base Shelter assemblies.
 - If grass cutting or snow removal is required, notify the appropriate airfield authority to schedule grounds maintenance work. Coordinate a TLS service outage with appropriate airfield personnel who will notify ATC and other local and federal agencies as required during the maintenance work.
4. Check obstruction lights for burned-out bulbs. Replace the light unit as necessary.
5. Check critical area markings (signs and hold short markings). Ensure signs are still in place, and in good condition, and are visible. Ensure the hold-short markings for the critical areas are visible and not excessively worn.
6. Visually inspect all system external components, looking for degradations, equipment structure or antenna shifts, etc. Possible shifts include frost heaves of system foundations, change in antenna

structure elements, missing structure hardware (mounting bolts, etc.), and damage from collisions with airport vehicles.

3.5.4 Maintenance of Critical Area and Obstruction Lights Documentation

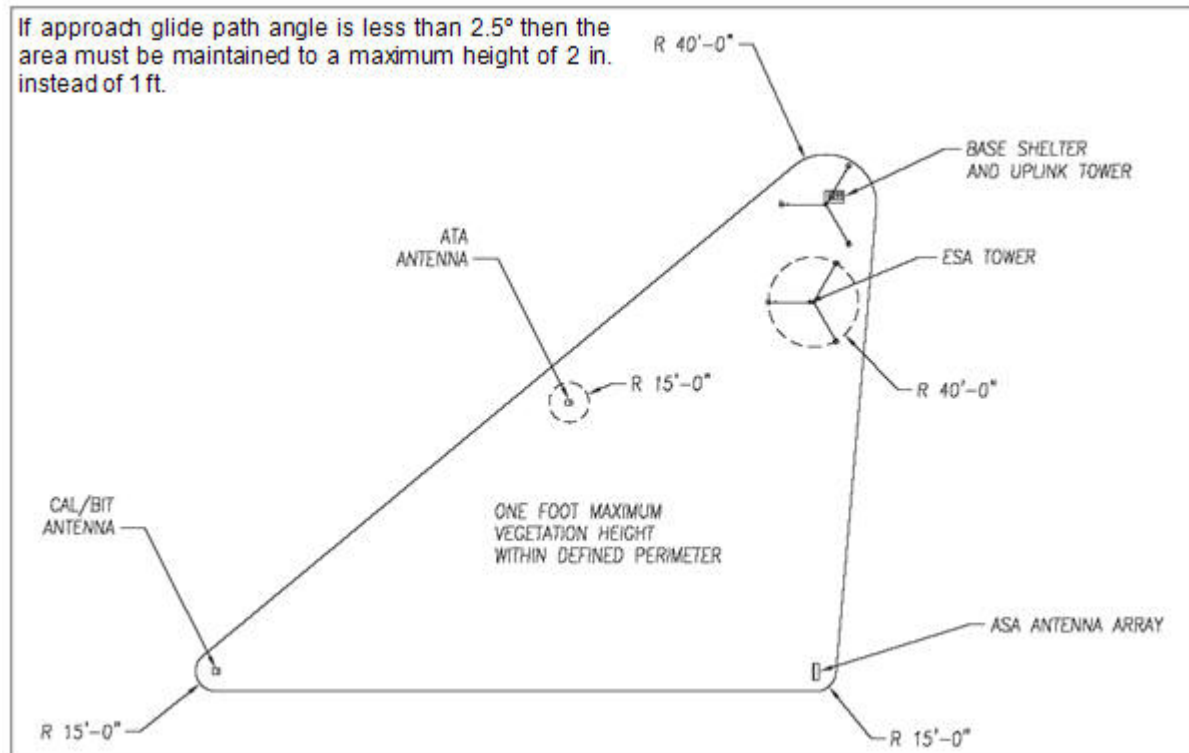


Figure 3-3 Vegetation and Obstacle Clearance Zones

3.5.1 Maintenance of Sensitive Area and Snow Clearance Objectives

This section is taken from FAA Advisory Circular 150/5200-30C, "Airport Winter Safety and Operations." The guidance on snow clearance for NAVAIDs is applicable to TLS and, as such, should be followed for maximum accuracy. While each antenna structure has a critical area, the entire system has a sensitive area in the approach direction, and care should be taken in keeping this field of view clear of obstructions, including snow build-up. The coned area in Figure 3-5 can be interpreted to be utilized for the ASA, ESA, and Uplink Tower each, and Figure 3-4 can be interpreted to be utilized around TLS sensitive areas.

The following guidance should not be interpreted to require specific equipment at any airport. It is presented to show efficient use of various equipment to optimize snow clearing operations. Some types of equipment may not be appropriate for some airports. Equipment and procedures used must be determined based on many factors, including but not limited to climate, number and types of operations, airport staffing, and the airport budget.

(1) A covering of snow and ice or drifts may affect visual and electronic NAVAIDs. Any snow or ice that affects the signal of electronic NAVAIDs should be removed. When clearing with rotary plows and displacement plows, special procedures need to take into account the location of all NAVAIDs, especially to protect the guidance signal of instrument landing systems (ILS). The SICP needs to address the following situations:

(i) Glideslope critical ground areas along the runway require that snow depths be limited in height to prevent signal loss or scattering. Figure 3-5 provides graphic representations of the glide slope ground snow clearance areas with prescribed snow depth limitations according to type of facility and aircraft approach

category. When snow depths exceed the specified depth limitations, minima are raised to the “localizer only” function until the conditions revert or are corrected. Two consecutive pilot reports of glide slope signal malfunctions generally result in raised minima (a NOTAM must be issued by the owner of the NAVAID). A few additional points should be considered. First, the 200-foot width dimension adjacent to the threshold might be wider for an antenna mast placed further out (see FAA Order 6750.49, Maintenance of Instrument Landing System (ILS) Facilities). Second, the snow clearance areas illustrated in the figures are minimal in size. Third, snow clearing activities should not allow snow banks, mounds, or ridges exceeding 2 feet to be placed along the edges of the prescribed snow clearance areas. Fourth, snow banks should not be placed off the approach ends of runways, especially for CAT II/III operations. Note: Snow banking operations need to take into account the guidance in Figure 3-5.

(ii) Visibility of signs (legibility) and lights should be maintained by certain prescribed clearing techniques or by performing post-clearing maintenance. Maintaining visibility can be better achieved by taking into account wind directions. For example, in crosswind conditions, cast in the downwind direction.

(iii) The snow depth height limitations noted in Figure 3-4 do not take into consideration airplane characteristics. That is, at some airports, airplane characteristics, such as engine clearances, may dictate lower snow banks than shown in Figure 3-4. The objective here is prevention by avoiding the introduction of hazardous snow banks, drifts, windrows, and ice ridges that could come into contact with any portion of the airplane wing or nacelle surface.

(2) If the airport’s operation involves the use of snow banks, their height profiles should be compatible with NAVAID ground requirements and offer sufficient clearance between airplane wings and engine nacelles to avoid structural damage to jet and propeller airplanes. Figure 3-4 shows desired maximum snow height profiles, which are based on airplane design groups (see AC 150/5300-13, Airport Design, for airplane design group categories.) Snow banking along terminal or cargo aprons likewise should comply with Figure

3-4 to prevent operational problems caused by ingestion of ice into turbine engines or by propellers striking the snow banks.

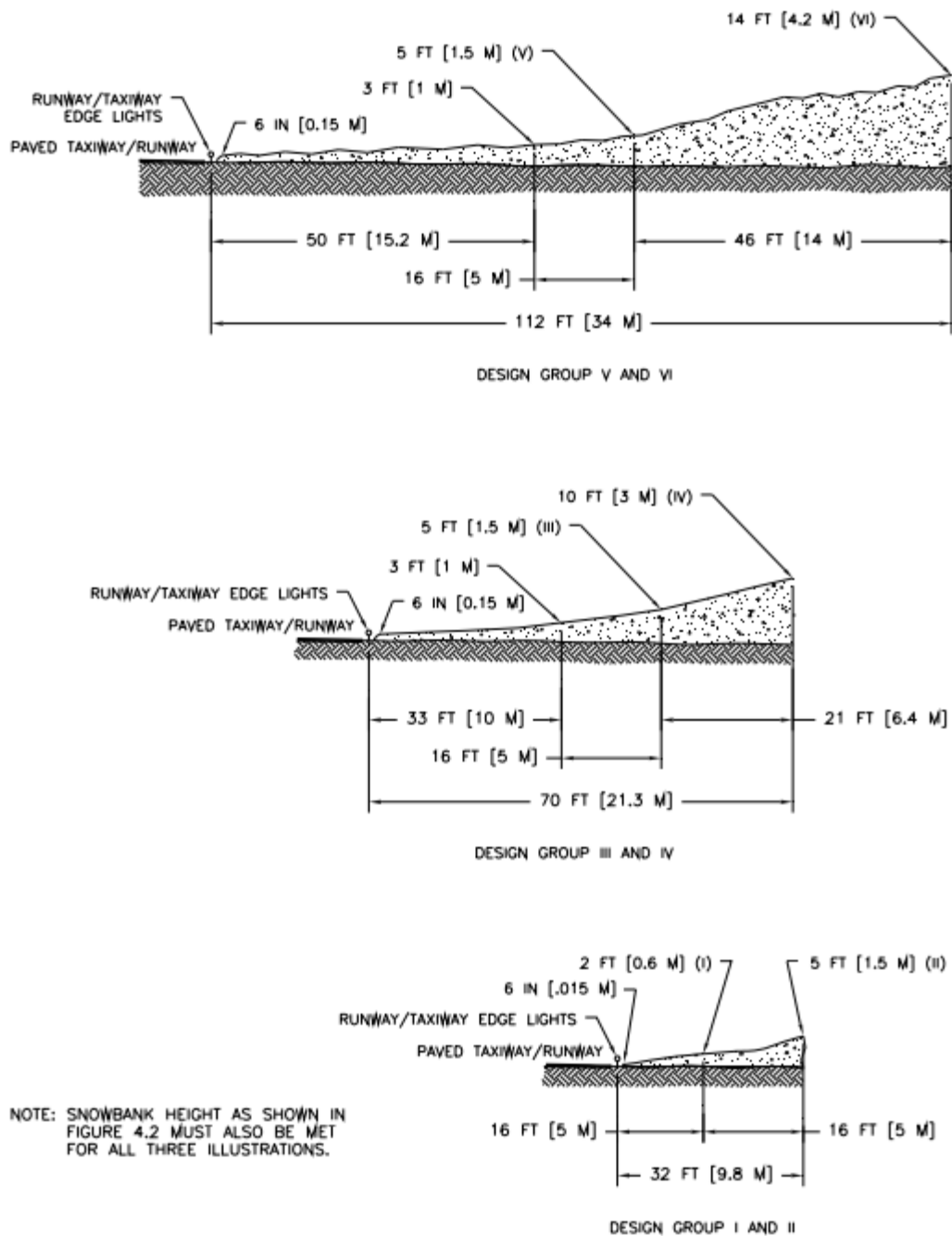
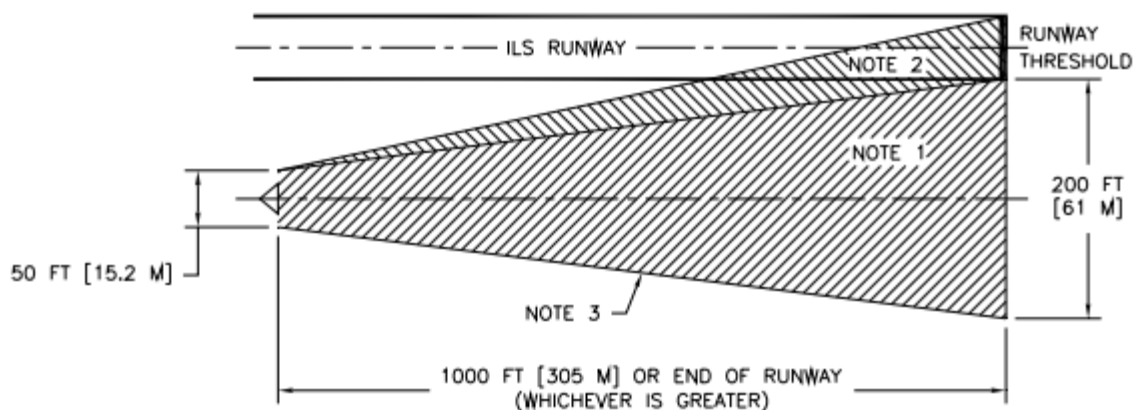


Figure 3-4: Snow Bank Profile Limits Along Edges of Sensitive Areas



NOTES:

1. CATEGORY I GLIDE SLOPE SNOW CLEARANCE AREA.
2. CATEGORY II AND III GLIDE SLOPE SNOW CLEARANCE AREA. THE AREA DEPICTED UNDER NOTE 1 SHALL ALSO BE CLEARED.
3. THE DEPTH OF SNOWBANKS ALONG THE EDGES OF THE CLEARED AREA SHALL BE LESS THEN 2 FEET.

ACTION TAKEN	SNOW DEPTH		
	SBR <6 IN [15 cm] NR. CEGS <18 IN [45 cm]	SBR 6 TO 8 IN [15 TO 20 cm] NR. CEGS 18 TO 24 IN [45 TO 60 cm]	SBR >8 IN [20 cm] NR. CEGS >24 IN [60 cm]
SNOW REMOVAL (SEE ABOVE FIGURE)	REMOVAL NOT REQUIRED RESTORE FULL SERVICE AND CATEGORY.	ILS CATEGORY I REMOVE SNOW 50 FT [15M] WIDE AT MAST WIDENING TO 200 FT [60M] WIDE AT 1000 FT [300M] OR END OF RUNWAY TOWARD MIDDLE MARKER. ILS CATEGORIES II AND III AS ABOVE PLUS WIDEN THE AREA TO INCLUDE A LINE FROM THE MAST TO THE FAR EDGE OF RUNWAY THRESHOLD.	
NO SNOW REMOVAL	RESTORE FULL SERVICE AND CATEGORY.	ALL CATEGORIES RESTORE TO CATEGORY I SERVICE. CATEGORY D AIRCRAFT MINIMA RAISED TO LOCALIZER ONLY. TYPICAL NOTAM TEXT: "DUE TO SNOW ON THE IXXX (APPROPRIATE IDENTIFIER) GLIDE SLOPE, MINIMA TEMPORARILY RAISED TO LOCALIZER ONLY FOR CATEGORY D AIRCRAFT" IF APPLICABLE. "CATEGORY II NA"* OR "CATEGORY II/III NA".	ALL CATEGORIES APPROACH RESTRICTED TO LOCALIZER ONLY MINIMA. TYPICAL NOTAM TEXT: "DUE TO SNOW ON THE IXXX (APPROPRIATE IDENTIFIER) GLIDE SLOPE, MINIMA TEMPORARILY RAISED TO LOCALIZER ONLY.

* NA (NOT AUTHORIZED)

Figure 3-5: ILS CAT I Snow Clearance Area Depth Limitations

3.6 C-4 UPS Battery Check

This procedure verifies that the UPS batteries remain healthy.

3.6.1 Accuracy Requirement

Accuracy in this check requires that the system must NOT have been operating on battery power during any time in the previous seven hours. There may be up to three UPS installed with the Reserve Rack and Rack Switch option installed.

3.6.2 Tools and Materials Required

- Digital Multi-Meter (DMM)

3.6.3 Special Instructions

Record all measured in the "TLS Technical Performance Record - UPS Output Voltage C-4" form in Appendix C.

3.6.4 UPS UPS-R and UPS-S Battery Check

Each UPS is equipped with automatic battery test features [Figure 3-6]. The automatic test occurs every 14 days if the utility power has not been interrupted. Should the battery fail this test, the red Fault indicator LED and the A and C diagnostic LEDs will illuminate, and an alarm will sound. Record indication of battery LED status on C-4.

3.6.5 Common Procedure for UPS-R and UPS-S

1. Using the DMM, measure the AC output voltage of the UPS at the UPS return to rack plug on the shelter wall. The measured voltage shall be within the range of 108 to 132 volts. Record the measured DMM voltage.
2. To initiate a battery test on models prior to GXT4, press the ON button for at least one second while operating from utility power with no alarm conditions present. With the GXT4, follow menu selections to initiate a battery test.

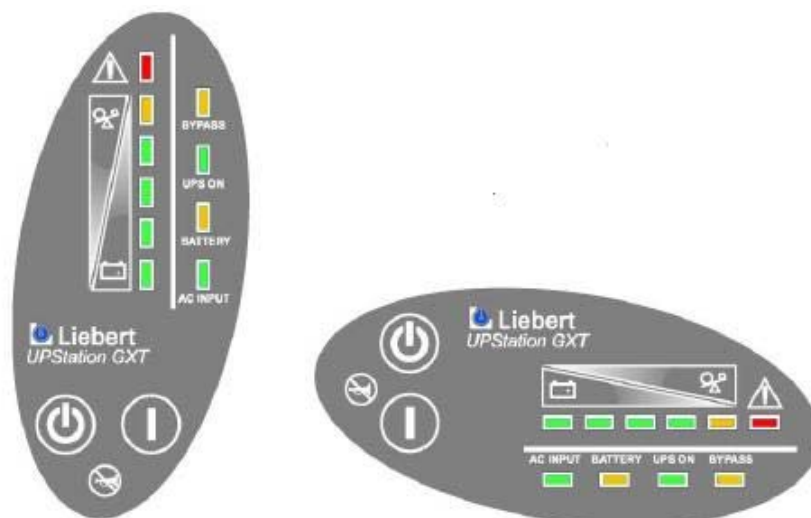


Figure 3-6 UPS GXT1 and 2 Front Panel: Indicator LEDs and on/off switch



Figure 3-7 UPS GXT3 Front Panel: Indicator LEDs and on/off switch



Figure 3-8 UPS GXT4 Front Panel: Indicator with LED menu and intuitive interface

3. For models before GXT4, if only three of the five Battery LEDs illuminate, allow the UPS to recharge the batteries for 24 hours. The GXT4 indicates battery health.
4. After 24 hours, retest the batteries. After the batteries have been retested, if only three of the five Battery LEDs illuminate, contact ANPC or an authorized local partner for service information.
5. Record verification that the batteries pass the automatic battery self-test.
6. Disconnect the UPS from the supply power so that the UPS switches to the battery for 20 minutes. Record this verification.
7. Using the DMM, measure the AC output voltage of the UPS at the UPS return to rack plug on the shelter wall. The measured voltage shall be within the range of 108 to 132 volts. Record the measured DMM voltage.

The following procedure is to be used with the GXT5 UPS model:

1. Using the DMM, measure the AC output voltage of the UPS at the UPS return to rack plug on the shelter wall. The measured voltage should be within the range of 108 to 132 volts. Record the measured DMM voltage.
2. To initiate a battery test, from the main menu, select the control icon and press *Enter*. Use the arrow buttons to move the cursor to the *Start/Stop Battery Manual Test*, then press *Enter* to select the control.



Figure 3-9 UPS GXT5 Front Panel and User Interface



Figure 3-10 UPS GXT5 Main Menu

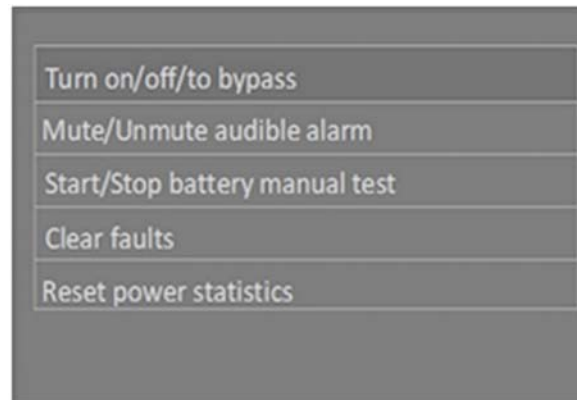


Figure 3-11 UPS GXT5 Control Menu

3. Record verification that the batteries pass the automatic battery self-test.
4. Using the DMM, measure the AC output voltage of the UPS at the UPS return to rack plug on the shelter wall. The measured voltage shall be within the range of 108 to 132 volts. Record the measured DMM voltage.

3.6.6 UPS Battery Check Documentation

1. Complete, sign, and file on-site the TLS Technical Performance Record for UPS Battery Check C-4.
2. Record the completion of C-4 in the Facility Maintenance Log Book.

3.7 C-5 Air Filters

This procedure ensures that all TLS air filters are clean and free of obstructions.

3.7.1 Tools and Materials Required

The number of air filters will double for installations with a Reserve Rack and/or Dual Environmental Control Units (ECU).

- Three foam air filters (such as ANPC part no. 340-00022).
- Two computer air filters (such as ANPC part no. 190-00321).
- One computer air filter (such as ANPC part no. 190-00320).
- One ECU air filter (Part no. dependent on ECU make and model).
- Two sensor cabinet A/C filters if equipped with A/C
- Bristle brush.
- Vacuum cleaner or compressed air source.
- #2 Phillips screwdriver.
- TLS Periodic Maintenance Record.

3.7.2 Procedure

1. Replace foam filters on the front panel of the Interrogation Transmitter and the GTU.
2. Remove the outer filter cover and foam filter. Discard the old filter.
3. Install a new foam filter and replace the outer filter cover.

Replace filters on each Base.

1. Depending on the model, either open the front panel or open the two captive screws on the front faceplate.
2. Loosen the captive screws on the air filter cover (Polywell filter only). Remove the filter.
3. Install new air filter, discard the old filter.
4. Replace the filter cover and front panel.

Replace Filter on Environmental Control Unit (ECU).

1. Remove filter cover by loosening the latch on both sides (may vary depending on the make and model of ECU).
2. Remove the filter and install a new air filter. Discard the old filter.
3. Replace filter cover.

3.7.3 Air Filters Documentation

Record the completion in Table B-5 of the Facility Maintenance Log Book.

3.8 C-6 Check/Adjust Guy Wire Tension

This procedure ensures that the ESA and Uplink tower guy wires are within the proper tension range. This procedure is the same for both the fixed TLS and TTLS (transportable), only differing by frequency as indicated by the Preventative Maintenance schedule. This procedure should also be used to bring the ESA tower back into tilt limits, if necessary, after a tilt alarm.

This procedure should be repeated every 2 months for the first six (6) months after SAT.

3.8.1 Tools and Materials Required

- Cable Tensiometer or equivalent
- Two 10" adjustable wrenches

3.8.2 Special Instructions

1. Notify the appropriate airfield authority that the TLS is going out of service for maintenance procedures.
2. Shut down system software.

3.8.3 Procedure

1. Following manufacturer's instructions, check the tension on each ESA and Uplink tower guy wire:
 - Top guy wires tension should read 660 lbs. ± 60 lbs.
 - Bottom guy wires tension should read 400 lbs. ± 40 lbs.
2. Check that the guy wires have a minimum of one (1) inch of threads exposed inside the coupler, as depicted in the figure below. This is to ensure that the turnbuckles do not strip out the remaining threads on the rods in high winds. If the turnbuckles do not have the correct amount of threads exposed, adjust the turnbuckles till all turnbuckles are within tolerance. Perform the following procedure steps to get the guy wires back into tolerance with regards to tension.
3. If guy wire(s) are not within tolerance, use adjustable wrenches to adjust turnbuckles until guy wire(s) are at the proper tension. Adjusting one guy wire may affect tensions on the other guy wires, so keep rechecking all guy wires to ensure balanced tensions are obtained.
4. Manually command a few BIT tests from the MI test mode screen, then using the BIT tool, check ESA tower tilt to ensure ESA is still within limits, optimally within ± 0.05 degrees of the nominal setting but no more than ± 0.1 degrees of nominal. If not within these limits, adjust appropriate guy wires to get both tensions and tilt values correct.

3.8.4 Guy Wire Tension Documentation

Record the completion of C-6 in the Facility Maintenance Log Book.

3.9 C-7 Rack Switch Controller

This section covers the periodic inspection required for the RSC. This procedure ensures that the stand-by rack (if equipped) is in proper working order. Follow the schedule as laid out in the Preventative Maintenance table

3.9.1 Tools and Materials Required

- Remote Control Unit
- RSC Periodic Inspection Form

3.9.2 Special Instructions

Notify the airfield authority that the TLS is going out of service for maintenance procedures.

3.9.3 Procedure

Control of the Rack Switch is accessed through the RCU.

- From the RCU software, select the "Control" drop-down, located in the upper-left-hand corner.
- Select "Switch Rack"
- Monitor the system as the currently active rack shuts down and switches over to the second rack.
- Ensure that the system enters "Track" and that there are no alerts or alarms.

3.10 C-8 Antenna Maintenance

C-8 Antenna System Maintenance

This procedure ensures all TLS antennas and cabling are free from corrosion and physical damage.

3.10.1 Tools and Materials Required

- Wire Brush
- Open-end Wrench
- Permanent Marker
- Lock-Out/Tag-Out Warning Tag

3.10.2 Special Instructions

1. Coordinate a TLS service outage with appropriate airfield personnel who will notify ATC and other local and federal agencies as required.
2. Related Maintenance: CABLE REPAIR AND WATERPROOFING PROCEDURE (C-12).

3.10.3 Procedure

CAUTION

Tag the main circuit breaker with a lock-out/tag-out warning tag.

1. Power down all equipment.
2. Physically inspect all system antennas for damage and deterioration.
3. If damage or deterioration is found on any antenna, arrange with ANPC field support for repair or a replacement.
4. Check all antennae mounting hardware for tightness and corrosion. Tighten loose bolts. Remove corrosion with the wire brush and spot paint structures only if needed, DO NOT paint antennas or sensor enclosures.

WARNING

Use of a climbing safety system is required when replacing obstruction light bulbs on the ESA or Uplink towers. ANPC recommends the use of a bucket lift truck to ensure operator safety. If using a climbing harness, always use a two-clip method to ensure constant attachment to the tower.

5. Physically inspect the entire length of all antenna cables.
6. Check for kinks, twists, and cuts. Re-run cable to remove kinks and twists. Minor cuts that do not cut into the cable conductor may be repaired with the cable repair kit. Any cable with cuts into the cable conductor must be replaced.
7. Check connectors for corrosion and waterproof seal. If no corrosion is noted but moisture is present, remove old waterproofing and replace it. If corrosion is present, remove it with a wire brush and waterproof it using the cable repair kit.
 - * Use Cable Repair and Waterproofing Procedure to repair and/or waterproof cables and connectors [section 3.10.3.1].
8. Inspect antennas to ensure that none have shifted position.

9. If equipped with antenna deicing, power on the antenna heater subsystem and verify that all antennas are warm to the touch. Inspect the antenna for any breaks in the surface, which expose the heating element and repair as necessary. The normal impedance between pins A and B is 45 ohms for all dBSystems 5100 D/7 antennas. The normal impedance between pins A and B for the localizer and glideslope antennas is 145 ohms.
10. Restore power to all TLS components, start system software and ensure normal operation.
11. Notify the appropriate airfield authority (using their procedures) that the TLS is returned to service.

3.10.3.1 Antenna Maintenance: Paint Touch-up Procedure

Use necessary protective equipment –safety glasses, latex gloves, etc.

- a. Remove any loose, soft, or contaminated materials from the area that will be repaired using a wire brush or sandpaper.
- b. Apply degreaser to free the surface of debris, dust, dirt, grease, and oil rinse and repeat.
- c. Slowly Mix in a 60:40 ratio ABRI Iron Bond 111 Powder (60) into ABRI Iron Bond Emulsion(40) in a clean container.
- d. Brush on a thin layer of the mixture and allow to dry (approx. 4 hours) before applying a second coat. Continue to agitate the solution during application.
- e. Apply the finish coat (Orange or White) with a brush.

3.10.4 Antenna System Maintenance Documentation

Record the completion of Table B-8 in the Facility Maintenance Log Book.

3.11 C-9 Clean and Inspect Equipment

This procedure ensures the TLS hardware remains clean and physically secure.

3.11.1 Tools and Materials Required

- Rags
- Bristle brush
- Vacuum cleaner
- #2 Phillips screwdriver
- Wire brush
- Lubricating Oil
- Lock-Out/Tag-Out Warning Tag
- Touch up paint
- Air filter supply

3.11.2 Special Instructions

Coordinate a TLS service outage with appropriate airfield personnel who will notify ATC and other local and federal agencies as required.

3.11.3 Procedure

1. Power down all equipment.
2. Inspect all sites for rodent, snake, bird, or insect infestation. Arrange for proper authorities to remove such infestations if found.

At the ESA, ASA, and Cal/Bit sites:

3. Check sensor boxes for structural damage. Check the sensor cover seal for deterioration and leaks; if the integrity of the sensor enclosure has been compromised, coordinate the replacement of the sensor.
4. Clean and lubricate the sensor cover latches. Close and secure the cover.
5. Inspect obstruction lights for a cracked lens, damaged or degraded seals, moisture, and corrosion. Check the power cable for deterioration. Replace burned-out bulbs.
6. Inspect array structures for rust, corrosion, paint chips, or scratches. Remove corrosion with the wire brush, prime, and paint as necessary.
7. Inspect the lightning arrestor path for deterioration, corrosion, and tight connections; clean and repair as necessary.

At the ASA, ATA, and Cal/BIT sites:

- a. Check all frangible couplings for breakage/deterioration. (Attention to detail required here- broken couplings may not be apparent).

At the ESA site, inspect the guy wires.

- b. At both ends of the guy wires, check for corrosion and the condition of the safety wrap. Repair as necessary.
8. Ensure that the ground anchors are solid.

At the base station:

9. Clean and inspect the exterior of the base equipment, keyboard, and monitor.
10. Dust the outside of the equipment and vacuum the bottom of the equipment rack.
11. Clean and lubricate the shelter door hinge.
12. Inspect equipment ground for corrosion, deterioration, and tight connections.
13. Inspect the cable access port seals. Ensure rodents, snakes, and insects cannot enter the shelter through the ports.
14. Ensure an adequate supply of all required air filters is available.

At the RCU:

15. Clean and inspect the Base, keyboard, monitor, and uninterruptible power supply.

Test completion:

16. Restore power to all TLS components and ensure normal operation.
17. Notify the appropriate airfield authority (following their procedures) that the TLS is returned to service.

3.11.4 Clean and Inspect Equipment Documentation

Record the completion of C-9 in the Facility Maintenance Log Book.

3.12 C-10 Interrogation Transmitters Check

This procedure ensures that the Interrogation Transmitter pulse characteristics are within tolerance. Record all measured values on the C-10 Interrogation Transmitters Check sheet located in Appendix C.

3.12.1 Tools and Materials Required

- SMA 50 Ω termination
- Oscilloscope (Tektronix TDS-340A or equivalent)
- SMA attenuators (20 dB, 10 dB, 6 dB)
- Crystal detector equivalent
- Coax cables, splitters, and adapters as needed
- 30 dB coupler, N connectors

3.12.2 Special Instructions

Verify that all test equipment has a valid/current calibration sticker prior to use. All measured values are to be recorded in the section Interrogation Transmitters Appendix C.

If a reserve rack is installed, this procedure will need to be performed on the reserve Interrogation Transmitter(s) as well.

3.12.3 Signal Test Points

1. Connect the “T0” output on the rack “SIGNAL TEST POINTS” panel to the oscilloscope CH 1 through a 50 Ω terminator using a 6’ BNC – M to BNC – M test cable. Use the following oscilloscope settings:

Table 3-2 Signal Test Points

Horizontal Scale	50 nSec/Div
Vertical Scale	1 V/Div
Coupling (Vertical Menu)	DC
Mode (Trigger Menu)	Normal
Trigger Source (Trigger Menu)	CH 1
Trigger Level (Trigger Menu)	1.4 V

2. On the MIU, start the system software if it is not running. With the system in either Maintenance or Track mode, enter Test mode by selecting Test Mode from the Tools menu on the MI.

Unless otherwise specified, the following applies to all pulse measurements:

- Pulse width is measured from the 50% level on the leading edge to the 50% level on the trailing edge;
 - Rise time is measured from the 10% level to the 90% level on the rising edge;
 - Fall time is measured from the 90% level to the 10% level on the falling edge;
 - Pulse amplitude is measured with reference to ground;
 - Pulse spacing or delay is measured from the 50% level of the first pulse’s leading edge to the 50% level of the second pulse’s leading edge.
3. Start BIT by clicking on the “Command BIT” button on the “Test Mode” window. Measure and record the data on the data table as measured from the oscilloscope CH 1.

4. Move the test cable from T0 output on the rack “SIGNAL TEST POINTS” panel to the “CAL-BIT CONTROL” output on the rack “SIGNAL TEST POINTS” panel; the other end will still be connected to the oscilloscope CH 1 through a 50Ω terminator.
5. Start BIT by clicking on the “Command BIT” button on the “Test Mode” window. Measure and record the data on the data table as measured from the oscilloscope CH 1.
6. Move the test cable from CAL-BIT CONTROL output on the rack “SIGNAL TEST POINTS” panel to the “INTERROGATION CONTROL” output on the rack “SIGNAL TEST POINTS” panel; the other end will still be connected to the oscilloscope CH 1 through a 50Ω terminator. Change the oscilloscope Horizontal Scale to 100 nSec/Div.
7. Start BIT by clicking on the “Command BIT” button on the “Test Mode” window. Measure and record the data on the data table as measured from the oscilloscope CH 1.
8. Move the test cable from INTERROGATION CONTROL output on the rack “SIGNAL TEST POINTS” panel to the “T0” output on the rack “SIGNAL TEST POINTS” panel; the other end will still be connected to the oscilloscope CH 1. Using the second BNC – M / BNC – M test cable, connect the “CAL-BIT CONTROL” to the oscilloscope CH 2 through 50Ω terminators, per Figure 3-12. Use the following settings:

Table 3-3 Signal Test Points: Oscilloscope Settings Cal/BIT CNTRL

	CH 1	CH 2
Horizontal Scale	1 μSec/Div	1 μSec/Div
Vertical Scale	1 V/Div	1 V/Div
Coupling (Vertical Menu)	DC	DC
Mode (Trigger Menu)	Normal	
Trigger Source (Trigger Menu)	CH1	
Trigger Level (Trigger Menu)	1.4V	

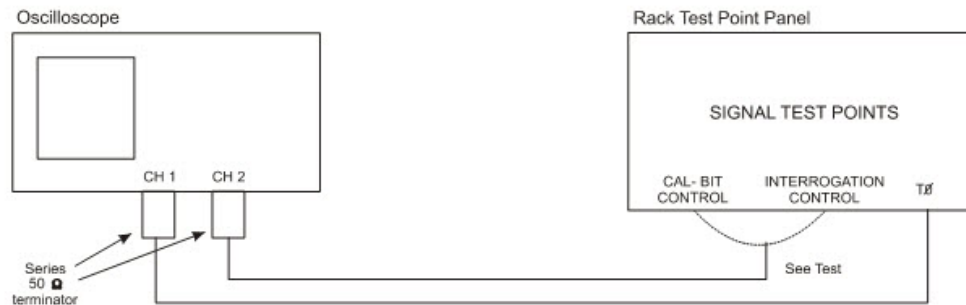


Figure 3-12 Oscilloscope Setup

9. Command BIT, and measure then record delay from “T0” (CH 1) to the “CAL-BIT CONTROL” (CH 2).
10. Change the oscilloscope CH 2 input from the “CAL-BIT CONTROL” signal to the “INTERROGATION CONTROL” signal on the “SIGNAL TEST POINTS” panel.
11. Command BIT, measure and record the P1 (CH 2 first pulse) to P3 (CH 2 second pulse) pulse spacing.
12. Change the oscilloscope horizontal scale to 10 nSec.
13. Command BIT, measure and record the delay from “T0” (CH 1) to P1 (CH 2 first pulse).

14. Change the following oscilloscope settings.

Table 3-4 Signal Test Points: Oscilloscope Settings Change

Horizontal Scale	50 mSec/Div
Acquisition Mode (Acquire Menu)	peak detect
Trigger	Channel 1

15. Command BIT and verify that during the first part of BIT, that T0 period is approximately 20 to 25 mSec and that it changes to approximately 50 to 55 mSec during the latter portion of BIT. Record this verification.

3.12.4 Interrogation Transmitter Output Verification

If possible, when using the detector, use an adapter instead of a cable to keep the detector output to the oscilloscope as short as possible to minimize its effect on the measured signal. Use the transmitter test setup for the interrogation and suppression test [Figure 3-13], taking care to include at least two 1 dB attenuators on the end connected directly to the interrogator

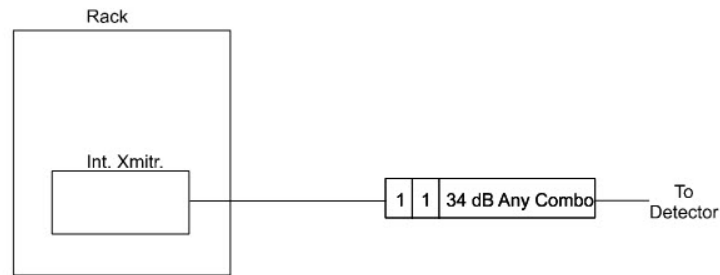


Figure 3-13 Transmitter Test Setup

1. With T0 still connected to the oscilloscope CH 1, connect the RF detector to the interrogation transmitter's attenuation string. Then connect the output of the detector to the oscilloscope CH 2 through a 50Ω terminator. Change to the following oscilloscope settings:

Table 3-5 Signal Test Points: Interrogation Transmitter Output Verification

	CH 1	CH 2
Horizontal Scale	100 nSec/Div	100 nSec/Div
Vertical Scale	1 V/Div	50 mV/Div
Acquisition Mode (Acquire Menu)	Sample	

2. On the MIU, command BIT by clicking on the “Command BIT” button on the “Test Mode” window. Measure and record the data on the data table as measured from the oscilloscope CH 2 (Set the oscilloscope channel 2 to have an 8 μSec delay to display the P3 pulse).
3. Change the oscilloscope horizontal scale to 1 μSec/Div. Command BIT by clicking on the “Command BIT” button on the “Test Mode” window. Measure and record the pulse spacing from P1 (CH 2 first pulse) to P3 (CH 2 second pulse).
4. Change the oscilloscopes' Horizontal scale to 50 nSec/Div.
5. Command BIT two times by clicking twice on the “Command BIT” button on the “Test Mode” window. Measure and record the pulse spacing from T0 (CH 1 pulse) to P1 (CH 2 first pulse).

3.12.4.1 Suppression Transmitter Pulse Characteristics

1. With T0 still connected to the oscilloscope CH 1, move the RF detector test cable to the suppression (P2) pulse attenuation string. Change the oscilloscope horizontal scale to 100 nSec/Div and set channel 2 to have a 2.2 μ Sec delay to display the P2 pulse.
2. Command BIT by clicking on the “Command BIT” button on the “Test Mode” window. Measure and record the data on the data table as measured from the oscilloscope CH 2.
3. Change the oscilloscope horizontal scale to 500 nSec/Div. Command BIT by clicking on the “Command BIT” button on the “TLS Test Mode” window. Measure and record the pulse spacing from T0 (CH 1 pulse) to P2 (CH 2 pulse).
4. Using the data recorded in step 12 of 3.12.4 (for T0 to P1 delay) and step 3 of 3.12.3 (for T0 to P2 delay), perform the calculation on the data sheet to obtain the P1 to P2 delay.
5. Exit the test mode by clicking on the “Close” button in the “Test Mode” window.
6. Stop the system software from the MIU by pressing the Stop button on the MI.

3.12.5 Interrogation Transmitters Check Documentation

1. Complete, sign, and file on-site the TLS Technical Performance Record for Interrogation Transmitters C-10.
2. Record the completion of C-10 in the Facility Maintenance Log Book.

3.13 C-11 Guidance Transmitter Unit (GTU) Check

The following procedure ensures that the GTU is configured properly.

3.13.1 Uplink Monitor

Note: See Figure 3-14 GTU Test Interface for the remainder of this section.

Note: The sign of DDM values indicated on the PIR is opposite the values indicated in the GTU Test tool. The GTU transmit calibration uses the sign indicated on the PIR for values measured and entered in the GtuTxCal tool

Note: In the GTU Test tool, the “Use transmit calibration” checkbox must be checked when verifying the transmit calibration and performing a monitor calibration.

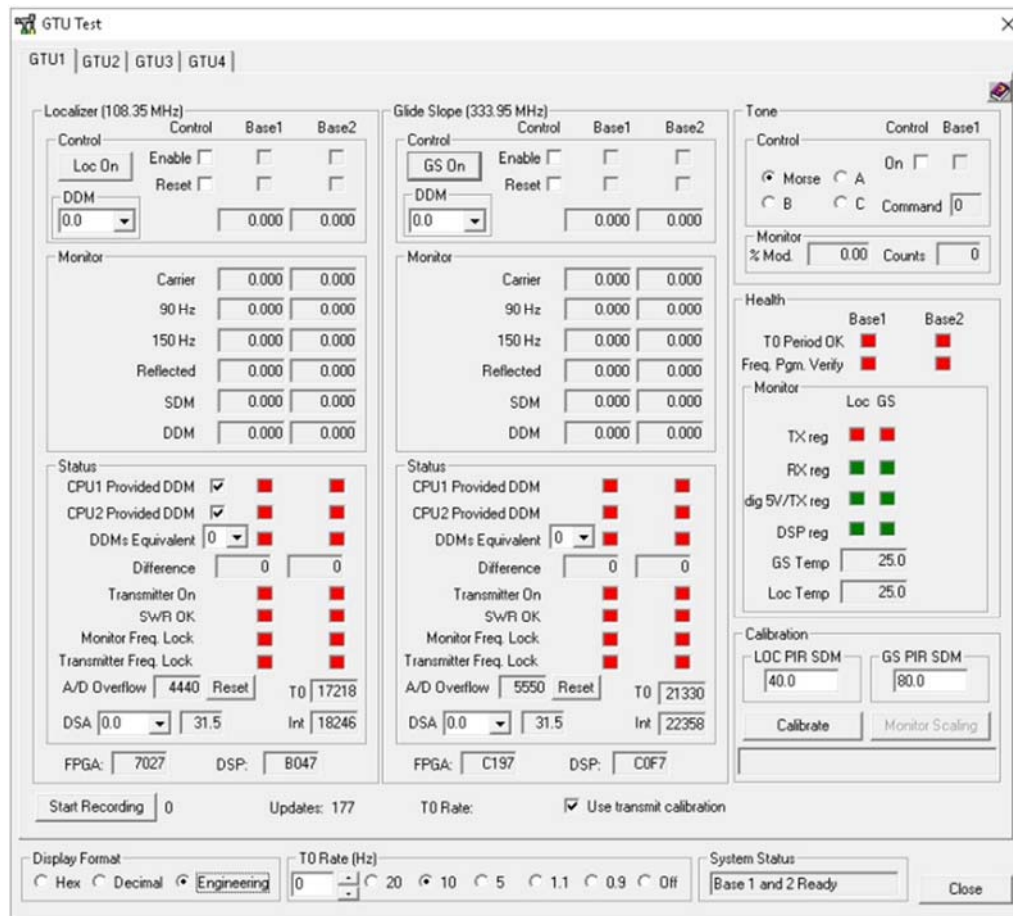


Figure 3-14 GTU Test Interface

1. Assemble the PIR and position it in front of the uplink antenna array at a distance of 30 feet from the base of the uplink tower.
2. Enable both the LOC and GS transmitters via the GTU Test Interface.
3. Ensure the carrier power to be between 2100 and 2300 mV for both localizer and glideslope.

If they are out of tolerance, complete the following:

- a. To accomplish this tuning, read the carrier power level for the localizer and, if it is below 2200mV, then reduce the DSA attenuation value. If the carrier is above 2200mV, then increase the DSA attenuation value.
 - b. Repeat this process for glide slope.
 - c. Record the DSA and carrier amplitude on the test datasheet for both localizer and glideslope
4. Allow 15 minutes transmitter ON time prior to continuing evaluation/test. If the GTU has already achieved the normal operating temperature and has been on for 15 minutes, continue to the next step now.
 5. Power on the PIR and ensure the PIR frequency select knobs are selected to the frequency of the localizer transmitter. Record both the Monitor Carrier levels from the GTU Test window and the PIR SDM readings for the localizer and glideslope

6. In the Calibration section of the GTU Test Interface, input the LOC PIR SDM and GS PIR SDM as measured above. Then depress the “Calibrate” button.
7. When the calibration is complete, the System Status window will display “Calibration Complete, Run Monitor Scaling.”
8. Depress the “Monitor Scaling” button in the Calibration section of the GTU Test Interface.
9. In the Monitor Scaling window, click on the calculate buttons for both the localizer and Glideslope sides, and verify that the calculated values transfer down to the “Calculated Scaling” windows. Click on the “Send to Bases” button. Verify that the Base1 and Base2 read-backs are the same as the control fields. (The Offset numbers will be rounded to the closest integer).
10. In the Monitor Scaling window, depress the “Close” button. Record successful GTU calibration.
11. For the localizer signal, change DDM control settings per the list below and record monitor SDM, monitor DDM, PIR SDM, and PIR DDM values. SDM values for the localizer shall be $40 \pm 1\%$, and DDM values shall be setting $\pm .002$. Record validation in Appendix C. PIR readings on DDM has reversed polarity.
 - i. **+20, +18, +15, +12, +09, +06, +03, 0, -03, -06, -09, -12, -15, -18, -20**
12. For the glide slope signal, change DDM control settings per the list below and record monitor SDM, Monitor DDM, PIR SDM, and PIR DDM values. SDM values for glide slope shall be $80.0 \pm 2\%$, and DDM values shall be setting $\pm .004$. Record validation in Appendix C.
 - i. **+24, +22, +2, +18, +15, +12, +09, +06, +03, 0, -03, -06, -09, -12, -15, -18, -2, -22, -24**
13. Record GTU monitor values as listed in Appendix C.
14. On the GTU test tool, click on “Close” to exit the test tool.

3.13.2 Guidance Transmitter Check Documentation

1. Complete, sign, and file on-site the TLS Technical Performance Record for GTU C-11.
2. Record the completion of C-11 in the Facility Maintenance Log Book.

3.14 C-12 Cable Repair and Waterproofing

This procedure provides special guidance for the repair and maintenance of all TLS cabling when required.

3.14.1 Tools and Materials Required

1. Scotch 3M 130C splicing tape
2. Scotch 3M 33+ tape
3. Coldshrink™ connector sealer and provided vinyl-backed mastic tape

3.14.2 Special Instructions

Coordinate a TLS service outage with appropriate airfield personnel who will notify ATC and other local and federal agencies as required.

3.14.3 Procedure

1. Cuts to cable insulation:

Replace the cable if the inner cable shielding is cut.

- a. Wrap the cable two inches below and above the cut with 3M 130C splicing tape.

Note: Always wrap the cable in a clockwise fashion to ensure that the connection is not loosened by the weatherproofing wraps.

- b. Completely cover the 3M splicing tape with a layer of 3M 33+ electrical tape.

2. Removing Coldshrink™ Connector Sealing:

- a. Using a sharp blade, score the 3M Coldshrink™ from the connector end to the cable end to a depth of about $\frac{3}{4}$ of the way through the Coldshrink™ thickness. Be very careful not to cut through and into the cable.
- b. At the connector end of the scored Coldshrink™, make a $\frac{1}{2}$ inch cut through the entire thickness of the Coldshrink™. Again, be careful not to cut through the Coldshrink™ and into the cable.
- c. Grasp the cut end of the Coldshrink™ and pull - the Coldshrink™ should now tear along the score line.
- d. Inspect the cable and connector for any damage.

3. Waterproof a connector:

- a. Cut a 4-inch piece of 3M Coldshrink™ (cut an 8" factory piece in half). Put the Coldshrink™ over the connector end of the cable, with the factory cut end towards the cable connector.
- b. Connect the coaxial connector. Ensure proper fit and tightness.
- c. If the diameter of the coaxial cable is less than $\frac{1}{2}$ inch, then a tape buildup is required to facilitate the use of the Coldshrink™: use either the vinyl-backed mastic tape provided with the Coldshrink™ or 3M 130C splicing tape to wrap the existing heat shrink (rubber connector boot) enough so that the cable diameter is a minimum thickness of $\frac{1}{2}$ of an inch. Always wrap the cable in a clockwise fashion to ensure that the connection is not loosened by these weatherproofing wraps.
- d. Push the 3M Coldshrink™ up and over the connection and onto the mounting flange. Maintain the position of the Coldshrink™, and pull the plastic core tail slowly, allowing the Coldshrink™ to collapse over the connector and taped area while forming a flush seal against the mating surface.
- e. Inspect the Coldshrink™ connection and verify that the seal at the connector end and at the taped area are uniform and create a good seal.

4. Notify the appropriate airfield authority, following their procedures, that the TLS is returned to service.

3.14.4 Cable Repair and Weatherproofing Documentation

Record the completion of C-12 in the Facility Maintenance Log Book, being sure to identify cables repaired and weatherproofed.

3.15 C-13 Generator Periodic Service

Follow the manufacturers' recommendations indicated below. Refer to the Generator manual to perform the maintenance. Record on the datasheet provided the successful completion of generator maintenance.

SERVICE INTERVALS

Observe the following for service and maintenance.

The lubricating oil change intervals listed in the table below are for Classes CF, CE and CD lubricating oils of API classification with a low-sulfur fuel in use. If the CF-4 or CG-4 lubricating oil is used with a high-sulfur fuel, change the lubricating oil at shorter intervals than recommended in the table below depending on the operating condition.

No.	Check point	Interval											Ref. page	
		First 50 hours	Every 50 hours	Every 100 hours	Every 200 hours	Every 400 hours	Every 600 hours	Every 800 hours	Every 1500 hours	Every 3000 hours	Every 1 year	Every 2 years		
1	Check of fuel pipes and clamp bands		○										26	@
2	Change of engine oil	⊙		○									28	⊙
3	Cleaning of air cleaner element			○									29	*1 @
4	Check of battery electrolyte level			⊙							○		33	
5	Check of fan belt tension			○									36	
6	Check of radiator hoses and clamp bands				○								32	
7	Check of intake air hose				○								—	@
8	Replacement of oil filter cartridge	⊙			○								28	⊙
9	Replacement of fuel filter cartridge or element					○							—	@
10	Cleaning of fuel filter			○									27	
11	Cleaning of water jacket (radiator interior)						○						32	
12	Replacement of fan belt							○					36	
13	Check of valve clearance							○					—	
14	Replacement of air cleaner element										○		29	*2 @
15	Check of damage in electric wiring and loose connections										○		35	
16	Check of fuel injection nozzle injection pressure								○				—	*3 @
17	Check of injection pump									○			—	*3 @
18	Replacement of fuel pipes, fuel gauge and clamp bands											○	26	*3 @
19	Replacement of radiator hoses and clamp bands											○	32	
20	Replacement of battery											○	33	
21	Change of radiator coolant (L.L.C.)											○	31	
22	Replacement of intake air hose											○	—	*4 @

IMPORTANT :

- The jobs indicated by ⊙ must be done after the first 50 hours of operation.

*1 Air cleaner should be cleaned more often in dusty conditions than in normal conditions.

*2 After 6 times of cleaning.

*3 Consult your local KUBOTA Dealer for this service.

*4 Replace earlier if necessary.

- The items listed above (@ marked) are registered as emission related critical parts by KUBOTA in the U.S. EPA non-road emission regulation. As the engine owner, you are responsible for the performance of the required maintenance on the engine according to the above instruction. Please see the Warranty Statement in detail.

- When the battery is used for less than 100 hours in a year, check its electrolyte yearly. (for refillable battery's only)

Figure 3-15 Generator Periodic maintenance schedule

3.16 Corrective Maintenance

This section contains information to troubleshoot and perform the on-site repair of the TLS system. This information is designed to facilitate the detection, location, and repair of faults within the TLS system to the Line Replaceable Unit (LRU). An LRU is a functional module that is replaceable on-site.

* Return all defective LRUs to ANPC for repair. Refer to section 3.21 for instructions on this procedure.

All maintenance activities pertaining to the TLS must conform to U.S. regulatory guidelines.

3.17 Tools and Test Equipment

The TLS Install /Maintenance Equipment Kit provides the tools and equipment that are needed for on-site corrective maintenance of the TLS system.

3.18 Fault Detection Tools

3.18.1 System Status Display and Base Log

The Maintenance Interface (MI) provides LRU status via the System Status selection from the main menu. The following points will allow efficient troubleshooting and identification of LRU faults:

- The System Status screen, with its color-coded LRU buttons and LRU-specific log file messages, provides the starting point for troubleshooting the TLS.
- When an alarm in BIT occurs, this screen will turn the associated LRU red.
- Clicking on a red LRU will bring up a window displaying the message ID of the alarm. By looking up this message ID number in the Error Code Table, troubleshooting that fault can be found.
- BIT and Track alarm message IDs will have an associated line in the Error Code Table.

3.18.1.1 Alert Conditions

If a parameter moves into an alert condition, this indicates that the parameter should be monitored closely to notice any trends that continue. If the condition continues, this may be indicative of a change to the system that may warrant more investigating. At that point, preventative action is required to prevent the system from shifting to an Alarm state.

3.18.1.2 Troubleshooting Requirement

For message IDs that alert only, there will be a "N/A" listed under the Error Code Table column and do not require any troubleshooting.

3.18.1.3 Base Log Files

The log files, which contain information about the various LRUs, can be viewed from the MI System Status window by pressing the "Base Log" buttons; there are separate buttons for Base1 and Base2. Alerts and alarms are listed in date/time sequence, oldest to newest. See Table 3-6 for an example of the base log. By reviewing the base log, the maintenance technician can look for trends of alerts and alarms that can lead to an outage and as a troubleshooting aid to help in LRU isolation when an outage has occurred. All data included in the file should be reviewed. Additionally, the TLS Operator Log should be reviewed for pilot complaints.

Table 3-6 Sample Error Log

System error logs are saved in the Run\Log folders of the base station computers.

<i>Date</i>	<i>Time</i>	<i>Type</i>	<i>Msg</i>	<i>LRU ID</i>	<i>SW ID</i>	<i>Message text ID</i>
11/11/1997	11:10:15		INFO	0000	00	UL
11/11/1997	11:10:15		INFO	0000	00	UL
11/11/1997	11:10:15		INFO	0000	00	UL
11/11/1997	11:10:15		ALERT	3051	07	UL
11/11/1997	11:10:15		ALERT	3020	07	UL
11/11/1997	11:10:16		ALERT	3011	07	UL
11/11/1997	11:10:16		ALARM	3012	07	UL
11/11/1997	11:10:16		ALERT	3011	07	UL
11/11/1997	11:10:16		ALARM	3012	07	UL
11/11/1997	11:10:16		ALERT	3020	07	UL
11/11/1997	11:10:27		ALERT	3040	07	UL
11/11/1997	11:10:28		ALERT	3043	07	UL
11/11/1997	11:10:30		ALERT	3044	07	UL
11/11/1997	11:10:31		ALERT	3041	07	UL
11/11/1997	11:10:37		ALERT	3042	07	UL

START OF NEW LOG MESSAGES

UL--bit tone control bit mismatch
UL--bit tone 0.188184 mv power out of range
UL--bit localizer sdm 0.800000 out of range
UL--bit localizer carrier power 0.188184 mv out of range
UL--bit glideslope sdm 1.600000 out of range
UL--bit glideslope carrier power 0.188184 mv out of range
UL--bit GTU 5V 0.188184 out of range
UL--bit GTU 5V 0.188184 out of range
UL--bit GTU Localizer 24V 0.188184 out of range
UL--bit GTU Glideslope 24V 0.188184 out of range
UL--bit GTU 15V 0.188184 out of range
UL--bit GTU -15V 0.188184 out of range

3.18.2 Test Mode

Test Mode can be used as a troubleshooting tool when TLS monitor shutdown prevents fault isolation, e.g., when a faulty data parameter instead of hardware failure causes the shutdown. Test Mode can only be entered from the MIU computer. Test Mode cannot be entered from the RCU, and the RCU is disabled while the TLS is in Test Mode. The TLS monitoring executive function is disabled when in Test Mode. Guidance can be out of tolerance when test mode is used. The system does not respond to Alert and Alarms during Test mode, but all Alerts and Alarms are logged during Test mode.

CAUTION

Never use Test Mode as a monitor inhibit for normal operation.
An aircraft should only be used for Test track with prior planning and coordination and only in VMC conditions.

3.19 Troubleshooting Support Data

This section contains the primary troubleshooting table and LRU reference figures. Use the troubleshooting table with the procedures referenced to understand and resolve problems that arise.

Table 3-7 – Troubleshooting based on software error code from the file base.log

Error Code	Software Identifier	Description of Error	Possible Reason	Possible Solutions and corrective action reference
100	LM_ID_BAD_DATA_FILES	A settings file (approach, cal, limits, settings, or tilt data) has a checksum error.	When the generation of the .dat files happened, the process was interrupted, or the base was not connected to the MIU.	Ensure the configuration process is followed all the way to completion and the saving of the files. Run Network Diagnostics and repair any dropped connections. Reinstall .dat files from the archive See 3.20.4 Base Computer Hardware/Software Fault Isolation
101	LM_ID_DIFF_DATA_FILES	Checksum differs between CPU1 and CPU2 for one of the data files (approach, cal, survey, settings, monitor_limits, or tilt).	When the generation of the .dat files happened, the process was interrupted, or the base was not connected to the MIU.	Ensure the configuration process is followed all the way to completion and the saving of the files Run Network Diagnostics and repair any dropped connections. Reinstall .dat files from the archive See 3.20.4 Base Computer Hardware/Software Fault Isolation
102	LM_ID_BAD_HOSTS_FILE	A hosts file was not found on the computer.	A host file was deleted, or a folder containing the host file changed	Ensure the path to the host file is correct Reinstall hosts file. See SW installation instruction
103	LM_ID_BAD_SERVICES_FILE	A services file was not found on the computer.	The file was deleted, or the folder containing the services file changed	Ensure the path to the services file is correct Reinstall services file. See SW installation instruction
110	LM_ID_FAILED_INIT	Indicates a process initialization error.	The network connection to the sensors may not be operating	Power on the ESA and ASA electronics Ping the ESA and ASA See 3.20.8 Measurement Sensor Fault Isolation
111	LM_ID_REGISTER_CONTROL_PROCESS	There was a "Process Registration" error—a possible problem with an operating system.	This is a developmental message used by ANPC for troubleshooting	See 3.20.4.1 General Troubleshooting for Base1 See 3.20.4.2 General Troubleshooting for Base2

Error Code	Software Identifier	Description of Error	Possible Reason	Possible Solutions and corrective action reference
				Contact ANPC for assistance with this message
112	LM_ID_CREATE_TIMER	System timer fault. Possible problem with the operating system.	This is a developmental message used by ANPC for troubleshooting	See 3.20.4.1 General Troubleshooting for Base1 See 3.20.4.2 General Troubleshooting for Base2 Contact ANPC for assistance with this message
120	LM_ID_CREATE_DIR	Directory creation error.	The permissions to a folder may have been changed to read-only	Change the folder permissions to allow writing
130	LM_ID_UNKNOWN_MESSAGE	A message was sent to a process that was unknown.	Unknown cause, this is a developmental message used by ANPC for troubleshooting	This message does not result in an alert or alarm, and no action is necessary to resolve this message
140	LM_ID_VERSION_MISMATCH	The version of the software did not match.	When the software boots, software and firmware revisions are checked on all devices and must match or continued operation is not possible	Install the same version of software on all devices See SW installation instruction
1001	LM_ID_HIGH_PHASE (Post SW 6.0) LM_ID_HIGH_PHASE_A (Pre SW 6.0)	The CAL/BIT signal received at the AOA sensor has phase measurement error exceeding tolerance for the indicated resolution	Change to ground surface conditions between the antennas, grass height, snow accumulation Change to antenna alignment Change to AOA sensor electronics	Cut the grass, remove the excess snow See 3.5 C-3 Maintenance of Critical Areas Restore the antenna alignment if this occurs for the ASA, See 3.20.10 ASA Antenna Re-alignment If this occurs for the ESA, See 3.20.11 ESA Antenna Re-alignment Replace the sensor electronics See 3.21 LRU Replacement
1002	LM_ID_MED_PHASE (Post SW 6.0)	The CAL/BIT signal received at the AOA sensor has phase measurement error exceeding	Change to ground surface conditions between the	Cut the grass, remove the excess snow See 3.5.4 Maintenance of Critical Area

Error Code	Software Identifier	Description of Error	Possible Reason	Possible Solutions and corrective action reference
	LM_ID_MED_PHASE_A (Pre SW 6.0)	tolerance for the indicated resolution	antennas, grass height, snow accumulation Change to antenna alignment Change to AOA sensor electronics	Restore the antenna alignment See 3.20.11 ESA Antenna Re-alignment Replace the sensor electronics See 3.21 LRU Replacement
1003	LM_ID_MED_PHASE_C_deleted LM_ID_MED_PHASE_C (Pre SW 6.0)	The CAL/BIT signal received at the AOA sensor has phase measurement error exceeding tolerance for the indicated resolution	Change to ground surface conditions between the antennas, grass height, snow accumulation Change to antenna alignment Change to AOA sensor electronics	Cut the grass, remove the excess snow See 3.5.4 Maintenance of Critical Area Restore the antenna alignment See 3.20.11 ESA Antenna Re-alignment Replace the sensor electronics See 3.21 LRU Replacement
1004	LM_ID_LOW_PHASE (Post SW 6.0) LM_ID_LOW_PHASE_C (Pre SW 6.0)	The CAL/BIT signal received at the AOA sensor has phase measurement error exceeding tolerance for the indicated resolution	Change to ground surface conditions between the antennas, grass height, snow accumulation Change to antenna alignment Change to AOA sensor electronics	Cut the grass, remove the excess snow See 3.5 C-3 Maintenance of Critical Area Restore the antenna alignment if this occurs for the ASA, See 3.20.10 ASA Antenna Re-alignment If this occurs for the ESA, See 3.20.11 ESA Antenna Re-alignment Replace the sensor electronics See 3.21 LRU Replacement
1010	LM_ID_CHANA_FREQ	The CAL/BIT signal received at an AOA sensor has a frequency measurement that exceeds the tolerance	The CAL/BIT signal frequency exceeds tolerance. An AOA sensor frequency measurement exceeds the tolerance	If both AOA sensors have frequency alarm, replace the CAL/BIT If one AOA sensor has a frequency alarm, replace the AOA sensor See 3.21 LRU Replacement
1020	LM_ID_AOA_LOW_PWR	The CAL/BIT signal received at an AOA sensor has an	Change to ground surface conditions between the	Cut the grass, remove the excess snow See 3.5.4 C-3 Maintenance of Critical

Error Code	Software Identifier	Description of Error	Possible Reason	Possible Solutions and corrective action reference
		amplitude error on the indicated channel	antennas, grass height, snow accumulation Degraded cable attachment point at antenna or sensor Damaged antenna, water inside the antenna	Area Repair or replace the cable See 3.14 C-12 Cable repair and waterproofing Repair the antenna, drain the water, repair the seal to the obstruction light mounting bracket See 3.10 C-8 Antenna Maintenance
1021	LM_ID_AOA_MED_PWR	The CAL/BIT signal received at an AOA sensor has an amplitude error on the indicated channel	Change to ground surface conditions between the antennas, grass height, snow accumulation Degraded cable attachment point at antenna or sensor Damaged antenna, water inside the antenna	Cut the grass, remove the excess snow Repair or replace the cable See 3.14 C-12 Cable repair and waterproofing Repair the antenna, drain the water, repair the seal to the obstruction light mounting bracket See 3.10 C-8 Antenna Maintenance
1022	LM_ID_AOA_HIGH_PWR	The CAL/BIT signal received at an AOA sensor has an amplitude error on the indicated channel	Change to ground surface conditions between the antennas, grass height, snow accumulation Degraded cable attachment point at antenna or sensor Damaged antenna, water inside the antenna	Cut the grass or remove the excess snow See 3.5.4 C-3 Maintenance of Critical Area Repair or replace the cable See 3.14 C-12 Cable repair and waterproofing Repair the antenna, drain the water, repair the seal to the obstruction light mounting bracket. See 3.10 C-8 Antenna Maintenance
1023	LM_ID_AOA_REF_PWR	The CAL/BIT signal received at an AOA sensor has an amplitude error on the indicated channel	Change to ground surface conditions between the antennas, grass height, snow accumulation	Cut the grass or remove the excess snow See 3.5.4 C-3 Maintenance of Critical Area

Error Code	Software Identifier	Description of Error	Possible Reason	Possible Solutions and corrective action reference
			Degraded cable attachment point at antenna or sensor Damaged antenna, water inside the antenna	Repair or replace the cable See 3.14 C-12 Cable repair and waterproofing Repair the antenna, drain the water, repair the seal to the obstruction light mounting bracket See 3.10 C-8 Antenna Maintenance
1030	LM_ID_AOA_LOW_PWR_JTR	The CAL/BIT signal received at an AOA sensor has an amplitude jitter that exceeds tolerance on the indicated channel	There could be a reflective object in the critical area	Remove the reflector from the critical area See 3.5.4 C-3 Maintenance of Critical Area
1031	LM_ID_AOA_MED_PWR_JTR	The CAL/BIT signal received at an AOA sensor has an amplitude jitter that exceeds tolerance on the indicated channel	There could be a reflective object in the critical area	Remove the reflector from the critical area See 3.5.4 C-3 Maintenance of Critical Area
1032	LM_ID_AOA_HIGH_PWR_JTR	The CAL/BIT signal received at an AOA sensor has an amplitude jitter that exceeds tolerance on the indicated channel	There could be a reflective object in the critical area	Remove the reflector from the critical area See 3.5.4 C-3 Maintenance of Critical Area
1033	LM_ID_AOA_REF_PWR_JTR	The CAL/BIT signal received at an AOA sensor has an amplitude jitter that exceeds tolerance on the indicated channel	There could be a reflective object in the critical area	Remove the reflector from the critical area See 3.5.4 C-3 Maintenance of Critical Area
1040	LM_ID_AOA_NOISE	The CAL/BIT signal received at an AOA sensor has an amplitude jitter that exceeds tolerance on the indicated channel	There could be a reflective object in the critical area	Remove the reflector from the critical area See 3.5.4 C-3 Maintenance of Critical Area
1050	LM_ID_AOA_LOW_PH_JTR	The CAL/BIT signal received at an AOA sensor has phase	There could be a reflective object in the critical area	Remove the object from the critical area Cut the vegetation that may be

Error Code	Software Identifier	Description of Error	Possible Reason	Possible Solutions and corrective action reference
		measurement jitter that exceeds tolerance on the indicated channel	There could be vegetation within or on the edges of the critical area AOA sensor electronics LRU is failing	reflecting the signal See 3.5.4 C-3 Maintenance of Critical Area Replace the AOA sensor electronics LRU See 3.21 LRU Replacement
1051	LM_ID_AOA_MED_PH_JTR	The CAL/BIT signal received at an AOA sensor has phase measurement jitter that exceeds tolerance on the indicated channel	There could be a reflective object in the critical area There could be vegetation within or on the edges of the critical area AOA sensor electronics LRU is failing	Remove the object from the critical area Cut the vegetation that may be reflecting the signal See 3.5.4 C-3 Maintenance of Critical Area Replace the AOA sensor electronics LRU See 3.21 LRU Replacement
1052	LM_ID_AOA_HIGH_PH_JTR	The CAL/BIT signal received at an AOA sensor has phase measurement jitter that exceeds tolerance on the indicated channel	There could be a reflective object in the critical area There could be vegetation within or on the edges of the critical area AOA sensor electronics LRU is failing	Remove the object from the critical area Cut the vegetation that may be reflecting the signal See 3.5.4 Maintenance of Critical Area Replace the AOA sensor electronics LRU See 3.21 LRU Replacement
1060	LM_ID_TOA_NEAR	The CAL/BIT signal received at an AOA sensor has a Near TOA value that exceeds the tolerance	There may be a reflector in the critical area The AOA sensor electronics LRU may have failed	Remove the object from the critical area Cut the vegetation that may be reflecting the signal See 3.5.4 Maintenance of Critical Area Replace the AOA sensor electronics LRU See 3.21 LRU Replacement
1061	LM_ID_TOA_FAR	The CAL/BIT signal received at an AOA sensor has a Far	There may be a reflector in the critical area	Remove the object from the critical area Cut the vegetation that may be reflecting the signal

Error Code	Software Identifier	Description of Error	Possible Reason	Possible Solutions and corrective action reference
		TOA value that exceeds the tolerance	The AOA sensor electronics LRU may have failed	See 3.5.4 Maintenance of Critical Area Replace the AOA sensor electronics LRU See 3.21 LRU Replacement
1062	LM_ID_TOA_JTR	The CAL/BIT signal received at an AOA sensor has a TOA jitter value that exceeds the tolerance	The F1 trigger pulse needs to be adjusted to find the lowest TOA jitter There may be a reflector in the critical area The AOA sensor electronics LRU may have failed	Adjust the F1 pulse to reduce the jitter 020-00073 4.19.1 TOA F1 Tuning Remove the object from the critical area Cut the vegetation that may be reflecting the signal See 3.5.4 Maintenance of Critical Area Replace the AOA sensor electronics LRU See 3.21 LRU Replacement
1101	LM_ID_DEL_PH_HIGH	The CAL/BIT signal received at the AOA sensor has phase measurement difference between channels that exceeds tolerance for the indicated resolution	The sensor electronics has failed	 Replace the sensor electronics See 3.21 LRU Replacement
1102	LM_ID_DEL_PH_MED	The CAL/BIT signal received at the AOA sensor has phase measurement difference between channels that exceeds tolerance for the indicated resolution	The sensor electronics has failed	 Replace the sensor electronics See 3.21 LRU Replacement
1103	LM_ID_DEL_PH_LOW	The CAL/BIT signal received at the AOA sensor has phase measurement difference between channels that exceeds tolerance for the indicated resolution	The sensor electronics has failed	 Replace the sensor electronics See 3.21 LRU Replacement
1301	LM_ID_AOA_TEMP	An AOA sensor temperature has exceeded the tolerance.	The AOA sensor cooling is off	Turn on the AOA sensor cooling heat exchanger or A/C as appropriate

Error Code	Software Identifier	Description of Error	Possible Reason	Possible Solutions and corrective action reference
			The AOA sensor A/C requires service The AOA sensor cooling has failed Solar load is too great and or combined ambient temperature exceeds the design limit of +70C	Service the A/C Repair or replace the AOA sensor heat exchanger or A/C as appropriate Turn off the sensor to avoid damage to the electronics.
1302	LM_ID_AOA_12V	An AOA sensor 12V power supply has failed.	The power supply has reached the end of life	No action is required for this alert condition unless a key performance parameter TOA or Phase fails; if a key parameter fails, replace the AOA sensor electronics LRU See 3.21 LRU Replacement
1303	LM_ID_AOA_18V	An AOA sensor 18V power supply has failed.	The power supply has reached the end of life	No action is required for this alert condition unless a key performance parameter TOA or Phase fails; if a key parameter fails, replace the AOA sensor electronics LRU See 3.21 LRU Replacement
1304	LM_ID_AOA_5V	An AOA sensors 5 V power supply has failed.	The power supply has reached the end of life	No action is required for this alert condition unless a key performance parameter TOA or Phase fails; if a key parameter fails, replace the AOA sensor electronics LRU See 3.21 LRU Replacement
1305	LM_ID_AOA_MINUS12V	An AOA sensors' 12V power supply has failed.	The power supply has reached the end of life	No action is required for this alert condition unless a key performance parameter TOA or Phase fails; if a key parameter fails, replace the AOA sensor electronics LRU See 3.21 LRU Replacement
1306	LM_ID_AOA_MINUS5V	An AOA sensors' 5 V power supply has failed.	The power supply has reached the end of life	No action is required for this alert condition unless a key performance

Error Code	Software Identifier	Description of Error	Possible Reason	Possible Solutions and corrective action reference
				parameter TOA or Phase fails; if a key parameter fails, replace the AOA sensor electronics LRU See 3.21 LRU Replacement
1307	LM_ID_AOA_15VRF	An AOA sensor 15 V power supply has failed.	The power supply has reached the end of life	No action is required for this alert condition unless a key performance parameter TOA or Phase fails; if a key parameter fails, replace the AOA sensor electronics LRU See 3.21 LRU Replacement
1308	LM_ID_AOA_5VRF	An AOA sensor 5 V power supply has failed.	The power supply has reached the end of life	No action is required for this alert condition unless a key performance parameter TOA or Phase fails; if a key parameter fails, replace the AOA sensor electronics LRU See 3.21 LRU Replacement
1309	LM_ID_2PT5V	An AOA sensor 5 V power supply has failed.	The power supply has reached the end of life	No action is required for this alert condition unless a key performance parameter TOA or Phase fails; if a key parameter fails, replace the AOA sensor electronics LRU See 3.21 LRU Replacement
1310	LM_ID_2V	An AOA sensor 2 V power supply has failed.	The power supply has reached the end of life	No action is required for this alert condition unless a key performance parameter TOA or Phase fails; if a key parameter fails, replace the AOA sensor electronics LRU See 3.21 LRU Replacement
1350	LM_ID_TILT_PARA	The tilt sensor detected a parallel shift in the ESA tower position.	The ESA tower tilt has changed due to anchor settling or changes to guy wire tension	Return the tower back to original install condition by tensioning the guy wires See 3.8 C-6 Check and adjust guy wire tension Verify tilt sensors are operating normally

Error Code	Software Identifier	Description of Error	Possible Reason	Possible Solutions and corrective action reference
				Check tilt sensor cables for damage See 020-00073 4.17 Tilt sensor configuration
1351	LM_ID_TILT_PERPEN	The tilt sensor detected a perpendicular shift in the ESA tower position.	The ESA tower tilt has changed due to anchor settling or changes to guy wire tension	Return the tower back to original install condition by tensioning the guy wires See 3.8 C-6 Check and adjust guy wire tension Verify tilt sensors are operating normally Verify tilt sensor cables for damage See 020-00073 4.17 Tilt sensor configuration
1352	LM_ID_TILT_SELF_PARA	A tilt sensor self-test detected an error in the parallel sensor.	The tilt sensor cable may be damaged The tilt sensor may be failing,	Check cables for damage; see C-12 Replace the tilt sensor with a spare
1353	LM_ID_TILT_SELF_PERPEN	A tilt sensor self-test detected an error in the perpendicular sensor.	The tilt sensor cable may be damaged The tilt sensor may be failing,	Check cables for damage See 3.14 C-12 Cable repair and waterproofing Replace the tilt sensor with a spare See 3.20.12 Tilt Sensor Fault Isolation
1354	LM_ID_TILT_TEMP	A tilt sensor temperature is beyond limits	The ambient temperature may be beyond the tilt sensors operating limits. The tilt sensor may be failing The tilt sensor cable may be damaged	Verify that the system temperature is within operational limits Check cables for damage This is not a critical failure, and the system can continue to operate
2001	LM_ID_TRK_AZ_UPDATE	Not enough azimuth measurement updates on the track.	The aircraft transponder is off	Ensure that operations space is unimpeded by obstacles that interfere with line of sight

Error Code	Software Identifier	Description of Error	Possible Reason	Possible Solutions and corrective action reference
			Lost line-of-sight to aircraft The aircraft is beyond the operational RF range / the transponder may have a low gain or low power output Synchronous garble has interfered with transponder plot extraction	Enable side lobe suppression P2 See 020-00073 4.14 System software startup Figure 4-4 systems settings
2002	LM_ID_TRK_EL_UPDATE	Not enough elevation measurement updates on the track.	The aircraft transponder is off Lost line-of-sight to aircraft The aircraft is beyond the operational RF range / the transponder may have a low gain or low power output Synchronous garble has interfered with transponder plot extraction	Ask the pilot to turn on the transponder Ensure that the approach plate includes accurate notes about the signal coverage Verify that the interrogator output is sufficient Schedule a maintenance appointment for the aircraft Enable side lobe suppression P2 See 020-00073 4.14 System software startup Figure 4-4 systems settings
2003	LM_ID_TRK_RANGE_UPDATE	Not enough range measurement updates on the track.	The aircraft transponder is off Lost line-of-sight to aircraft	Ask the pilot to turn on the transponder Ensure that the approach plate includes accurate notes about the signal coverage Verify that the interrogator output is sufficient

Error Code	Software Identifier	Description of Error	Possible Reason	Possible Solutions and corrective action reference
			The aircraft is beyond the operational RF range / the transponder may have low gain or low power output Synchronous garble has interfered with transponder plot extraction	See 3.12.C-10 Interrogation transmitter check Schedule a maintenance appointment for the aircraft Enable side lobe suppression P2 See 020-00073 4.14 System software startup Figure 4-4 systems settings
2010	LM_ID_TRK_HYP_COMPARE	Track Hyperbolic measurement compare failed.	The calibration process has not been performed	Complete calibration process; See 020-00071
2011	LM_ID_TRK_LOW_LOC_COMP	Error in the comparison to the Localizer low accuracy measurement.	The calibration process has not been performed	Complete calibration process; See 020-00071
2012	LM_ID_TRK_LOW_GS_COMP	Error in the comparison to the glide slope low accuracy measurement.	The calibration process has not been performed	Complete calibration process; See 020-00071
2013	LM_ID_TRK_MEDIUM_HIGH	Error in the comparison between medium and high accuracy measurements.	The calibration process has not been performed	Complete calibration process; See 020-00071
2020	LM_ID_TRK_AZ_ACC	Azimuth acceleration error (track jumped in azimuth).	The calibration process has not been performed	Complete calibration process; See 020-00071
2021	LM_ID_TRK_EL_ACC	Elevation acceleration error (track jumped in elevation).	The calibration process has not been performed	Complete calibration process; See 020-00071
2022	LM_ID_TRK_RANGE_ACC	Range acceleration error (track jumped in range).	The calibration process has not been performed	Complete calibration process; See 020-00071
2023	LM_ID_TRK_AZ_STRUCT	Structure/jitter in the azimuth track exceeds tolerances.	The calibration process has not been performed	Complete calibration process; See 020-00071
2024	LM_ID_TRK_EL_STRUCT	Structure/jitter in the elevation track exceeds tolerances.	The calibration process has not been performed	Complete calibration process; See 020-00071

Error Code	Software Identifier	Description of Error	Possible Reason	Possible Solutions and corrective action reference
2030	LM_ID_TRK_AZ_VOL	Azimuth volume alarm (check if the aircraft flew out of azimuth volume).	The aircraft departed the service volume	This is an informational message, and no corrective action is needed. The aircraft has departed the service volume
2031	LM_ID_TRK_EL_VOL	Elevation volume alarm (check if aircraft flew out of elevation volume).	The aircraft departed the service volume	This is an informational message, and no corrective action is needed The aircraft has departed the service volume
2042	LM_ID_TRK_UPDATE_TIME	Time out on the update rate	The track coasted for more than the allowed time	The aircraft may not have line-of-sight to the system, and no corrective action is possible to ground equipment; if this occurs during flight test, ask the pilot to fly towards the system
2045	LM_ID_TRK_CAPACITY	The system is at the capacity on the number of tracks that the system can track	More than 100 aircraft have been detected, and the system will track and display the closest 100 aircraft only.	This is an informational message, and no corrective action is needed.
2046	LM_ID_TRK_MHT_CONFIDENCE	Software track processing confidence level low on the track; The track was dropped.	The calibration process has not been performed or can be improved	Complete calibration process; See 020-00071
2047	LM_ID_TRK_MISSED_AGREE	mismatch on the differential TOA compare	The calibration process has not been performed	Complete calibration process; See 020-00071
2048	LM_ID_TRK_PP_DROP	Pulse Processing track was dropped, the algorithm could not associate any more replies to the track, and the track was discontinued	The aircraft is not within line of sight to the system The transponder reply may not be consistent or could be marginally out of tolerance if there was intermittent tracking of the aircraft	It is possible to examine RAM data using ram.exe and determine if the transponder is within tolerance See ANPC doc 020-00089 for assistance with this analysis
2049	LM_ID_TRK_COAST	Track coast limit was exceeded; there were no updates to the track for more than the allowed interval	The aircraft has departed the service volume The aircraft is no longer line-of-sight	There is no corrective action, the track coast limit is a common information message in the base.log

Error Code	Software Identifier	Description of Error	Possible Reason	Possible Solutions and corrective action reference
2050	LM_ID_RCU_COMPARE	The RCU integrity test of PAR track position failed	The RCU track position test exceeds tolerance due to a failed RCU	Replace the RCU See 3.20.13 RCU Troubleshooting Procedures
2051	LM_ID_RCU_TIMEOUT	Lost the network connection to the RCU	The network path to the RCU was interrupted RCU was shut off	Ensure the network path to the RCU is uninterrupted and the RCU is pingable Turn on the RCU. Realign wireless link antennae See RCU troubleshooting 3.20.13.1
3010	LM_ID_ULBIT_DDM	GTU BIT DDM compare failed.	The DDM difference between the transmitted value and the monitor value exceeded the tolerance; the message will indicate if this occurred for the localizer or glideslope. The GTU should be calibrated.	Complete the GTU calibration process using a PIR See 3.13 C-11 GTU verification
3011	LM_ID_ULBIT_SDM	GTU BIT SDM is out of range.	The SDM has drifted and exceeds tolerance; this is a hardware fault and cannot be corrected in the field.	See 3.20.5 Guidance Transmitter Unit (GTU) Fault Isolation Replace the GTU See LRU Replacement 3.21
3012	LM_ID_ULBIT_CARRIER	GTU BIT power is out of range.	The GTU BIT power feedback loop has recorded an increase or decrease in amplitude. An increase in amplitude can be caused by a change in the environment or the cable and does not indicate a severe problem since increased amplitude can only improve the service coverage. If the power has decreased, then examine the cable connections and ensure that	Clean and waterproof the cable connections Replace the damaged cable See 3.14 C-12 Cable repair Replace the GTU See 3.13 C-11 GTU verification See 3.20.5 GTU Fault Isolation

Error Code	Software Identifier	Description of Error	Possible Reason	Possible Solutions and corrective action reference
			they are waterproof and the cables are not damaged.	
3013	LM_ID_ULBIT_REFL_POWER	GTU BIT reflected power out of range	Cable connections may be loose or contain moisture There may be damage to the antenna	Verify cable connections are clean, dry, and secured with waterproofing tape or cold shrink See 3.20.9 Coaxial Cables Fault Isolation Verify that the antenna has no physical damage, repair or replace antenna if it is damaged See 3.20.5 Guidance Transmitter Unit (GTU) Fault Isolation
3020	LM_ID_ULBIT_TONE_DB	GTU BIT tone modulation out of range.	The tone modulation hardware has drifted; this cannot be repaired in the field.	See 3.20.5 Guidance Transmitter Unit (GTU) Fault Isolation Replace the GTU See 3.21 LRU Replacement
3021	LM_ID_ULBIT_T0_Period	GTU BIT did not detect a T0 period; synch pulse failure.	The T-Zero cable is disconnected or damaged	Connect or replace the T-Zero BNC cable to the GTU See 3.20.5 Guidance Transmitter Unit (GTU) Fault Isolation
3022	LM_ID_ULBIT_CPU_UPDATE	Self-test failure; GTU BIT did not detect a CPU update failure.	The test of the dual dissimilar CPU integrity monitor failed	Verify that all cable connections are correct See 3.20.5 Guidance Transmitter Unit (GTU) Fault Isolation
3023	LM_ID_ULBIT_DDM_COMPAR E	Self-test failure; GTU BIT did not detect a DDM compare failure.	The test of the dual dissimilar CPU integrity monitor failed	Verify that all cable connections are correct See 3.20.5 Guidance Transmitter Unit (GTU) Fault Isolation
3030	LM_ID_ULBIT_TEMP_INIT	GTU BIT internal temperature is out of range.	The GTU temperature has exceeded the monitor limit because the shelter temperature is not controlled	Repair the ECU, A/C, or heater to return the shelter temperature with normal operating ranges.

Error Code	Software Identifier	Description of Error	Possible Reason	Possible Solutions and corrective action reference
3031	LM_ID_ULBIT_TEMP_EXT	Temperature from the pressure sensor is beyond tolerances	The shelter environmental control unit (ECU) has failed. The pressure sensor has a faulty temperature sensor	Repair the heating or cooling units, ensure any other temperature mitigations are implemented If the environmental control units are within spec and the shelter temperature is nominal - If the pressure sensor is suspected of failure, replace the pressure sensor See 3.21 LRU Replacement
3040	LM_ID_ULBIT_5V	GTU 5 V power supply out of range.	The power supply has reached the end of life	If a key performance parameter such as SDM, DDM, or output power is in alarm, then replace the GTU
3041	LM_ID_ULBIT_15V	GTU 15 V power supply out of range.	The power supply has reached the end of life	If a key performance parameter such as SDM, DDM, or output power is in alarm, then replace the GTU
3042	LM_ID_ULBIT_MINUS15V	GTU -15 V power supply out of range.	The power supply has reached the end of life	If a key performance parameter such as SDM, DDM, or output power is in alarm, then replace the GTU See 3.21 LRU Replacement
3043	LM_ID_ULBIT_LOC_24V	GTU Localizer 24V power supply out of range.	The power supply has reached the end of life	If a key performance parameter such as SDM, DDM, or output power is in alarm, then replace the GTU See 3.21 LRU Replacement
3044	LM_ID_ULBIT_GS_24V	GTU glide slope 24V power supply out of range.	The power supply has reached the end of life	If a key performance parameter such as SDM, DDM, or output power is in alarm, then replace the GTU See 3.21 LRU Replacement
3045	LM_ID_ULBIT_SUPPLY_BITS	Hardware failure, LRU needs service	The GTU has failed	Replace the GTU See 3.21 LRU Replacement
3050	LM_ID_ULBIT_TONE_CODE	GTU BIT output does not match commanded value.	The GTU has failed	Replace the GTU See 3.21 LRU Replacement
3051	LM_ID_ULBIT_TONE_ONOFF	GTU BIT tone control BIT mismatch.	The self-test of the GTU tone has failed	Replace the GTU See 3.21 LRU Replacement
3052	LM_ID_UPLINK_T0	T-zero synchronization signal failed at the GTU.	T-zero cannot be detected on the GTU	Connect the t-zero cable between the ICC and the GTU

Error Code	Software Identifier	Description of Error	Possible Reason	Possible Solutions and corrective action reference
				See 3.20.4.3 Base1 Timing Signal Fault Isolation
3053	LM_ID_UPLINK_SWR	Too much signal reflected back from the antenna (a standing wave ratio error).	Cable path from GTU to antennae could be damaged The antenna could be damaged	Check cable path for damage/ loose connections replace cables/ re-tighten connectors See 3.20.9 Coaxial Cables Fault Isolation Repair or replace the antenna
3054	LM_ID_UPLINK_GTU_ID	ID mismatch, cannot distinguish the GTU ID	Communications fault with Base1 or Base2.	Verify rack cable connections See 3.20 Fault Isolation, LRU Inspection
3055	LM_ID_UPLINK_NO_DDM	One of the base computers did not provide a DDM value.	Communications fault with Base1 or Base2.	Verify rack cable connections See 3.20 Fault Isolation, LRU Inspection
3056	LM_ID_UPLINK_DDM_NOMAT	The DDM value between the two CPUs did not match.	Communications fault with Base1 or Base2.	Verify rack cable connections See 3.20 Fault Isolation, LRU Inspection
3057	LM_ID_UPLINK_FAILURE	This error is logged if the uplink software cannot determine why the glideslope is failing.	Communications fault with Base1 or Base2.	Verify rack cable connections See 3.20 Fault Isolation, LRU Inspection
3061	LM_ID_UPLINK_NT_IO_ERR	The Windows operating system could not establish a communication port to the GTU.	Communications fault with Base1 or Base2.	Verify rack cable connections See 3.20 Fault Isolation, LRU Inspection
3062	LM_ID_UPLINK_MORSECODE	A non-alphanumeric Morse code identifier was requested.	Incorrect Morse code configuration, one or more characters cannot be broadcast using Morse code	Change the Morse code to a series of characters that can be broadcast using Morse code
3070	LM_ID_ULMON_SDM	GTU Monitor SDM is out of range.	The SDM has drifted and exceeds tolerance; this is a hardware fault and cannot be corrected in the field.	Replace the GTU See 3.21 LRU Replacement

Error Code	Software Identifier	Description of Error	Possible Reason	Possible Solutions and corrective action reference
3071	LM_ID_ULMON_DDM	GTU Monitor DDM compare failed.	Communications fault with Base1 or Base2.	Verify rack cable connections See 3.20 Fault Isolation, LRU Inspection
3073	LM_ID_ULMON_DDM_VOLUME	GTU Monitor DDM volume edge test failed.	The GTU has not been calibrated, or the calibration could be improved	Calibrate the GTU using a PIR See 3/13 C-11 GTU verification
4031	LM_ID_SENSOR_DATA_CHECKSUM	Corrupted data. The computed checksum at the base does not match what was sent from the sensor.	A configuration file has been changed, and the checksum does not match the contents of the file	Retrieve the last good archive from the MIU If a configuration file was changed intentionally and has been verified to be a valid change, run the checksum tool on the file and use send-to-bases to install the file
4032	LM_ID_PHASE_SHIFT	A failure has occurred in the Sensor PRM Phase discriminator.	This is a BIT check of the PRM and is not field repairable.	Replace the AOA sensor electronics See 3.21 LRU Replacement
4033	LM_ID_PHASE_STUCK_BIT	An RF channel on the AOA is not returning an expected value.	This is a BIT check of the PRM and is not field repairable.	Replace the AOA sensor electronics See 3.21 LRU Replacement
4034	LM_ID_PHASE_FIFO	This monitors for a stuck bit in the HW phase FIFO on the REV-A sensor.	This is a BIT check of the PRM and is not field repairable.	Replace the AOA sensor electronics See 3.21 LRU Replacement
5010	LM_ID_INT_P_WIDTH	Interrogator pulse width out of range	The interrogator self-test of pulse width is out of tolerance; this is not adjustable in the field, the hardware is failing	Perform the interrogation control card test to isolate the cause to either the ICC or the interrogator, and then replace the failed hardware See 3.12 C-10 Interrogation transmitter check
5020	LM_ID_INT_P1P2_SPACE	Interrogator P1/P2 spacing out of range	The interrogator self-test of time between P1 and P2 is out of tolerance; this is not adjustable in the field, the hardware is failing	Perform the interrogation control card test to isolate the cause to either the ICC or the interrogator See 3.12 C-10 Interrogation transmitter check

Error Code	Software Identifier	Description of Error	Possible Reason	Possible Solutions and corrective action reference
5021	LM_ID_INT_P1P3_SPACE	Interrogator P1/P3 spacing out of range	The interrogator self-test of time between P1 and P3 is out of tolerance; this is not adjustable in the field, the hardware is failing	Perform the interrogation control card test to isolate the cause to either the ICC or the interrogator See 3.12 C-10 Interrogation transmitter check
5030	LM_ID_INT_P1_P3_AMP	Interrogator P1/P3 amplitude out of range.	The interrogator self-test of amplitude for P1 and P3 is out of tolerance; this is not adjustable in the field, the hardware is failing	Replace the interrogator If it is acceptable to operate at a reduced interrogation power, then the integrity monitor limit may be adjusted on the TI settings dialog to remain in operation until the interrogator can be swapped for a spare. See 3.21 LRU Replacement
5031	LM_ID_P2_AMP	Interrogator P2 amplitude out of range	The interrogator self-test of amplitude for P2 is out of tolerance; this is not adjustable in the field, the hardware is failing	Disable P2 side lobe suppression if it is operationally acceptable and the airspace is not heavily traffic Replace the interrogator See 3.21 LRU Replacement
5032	LM_ID_INT_STATUS	Interrogator status bits indicate a failure	The interrogation control card (ICC) in Base1 is not properly installed or has failed	Replace or reseal the ICC card in Base1 See 3.20.4.3 Base1 Timing Signal Fault Isolation Replace Base1 See 3.21 LRU Replacement
6010	LM_ID_PRESSURE	General failure error on the pressure sensor one of the following may be at fault (communications, invalid temperature, or pressure)	The pressure sensor communication cable is not connected. The pressure sensor power supply is not connected	Verify cables are connected Replace the pressure sensor with a spare See 3.21 LRU Replacement
7001	LM_ID_UPS_SHUTDOWN	Error logged when the system software is shut down due to low battery minutes remaining on the UPS.	The utility power is disconnected There is a utility power outage	Connect the utility power Leave the system off until utility power is restored

Error Code	Software Identifier	Description of Error	Possible Reason	Possible Solutions and corrective action reference
7010	LM_ID_RACK_SWITCH_STAT US	General failure error on the Rack switch.	The Rack Switch is powered off The Rack Switch cables are not connected The Rack switch has failed	Power on the Rack Switch Connect all cables to the Rack Switch Replace the rack switch See 3.22 Rackswitch troubleshooting
8000	LM_ID_LRU_LINK_DOWN	Reports an RCU link failure.	The RCU is not powered on The RCU link is not powered on The RCU cables are not connected	Power on the RCU Power on the link for the RCU Connect the RCU cables See RCU troubleshooting 3.20.13.1
8010	LM_ID_CPU_DISK_SPACE	CPU's hard drive is full.	Archives have filled up the drive RAM collection is turned on and capturing too many files. Remove non-TLS files	Remove old archives to other storage space or Delete unused archives See 3.3 C-1 Archive of data files Turn off RAM collection in the MI settings. Remove any non-TLS files See 3.3 C-1 Archive of data files
8011	LM_ID_BASE_TEMPERATURE	The base temperature out of tolerance	The shelter's ECU has failed. The base's temperature sensor has failed	Repair the heating or cooling units, ensure any other temperature mitigations are implemented If the environmental control units are within spec and the shelter temperature is nominal - replace the base unit
9001	LM_ID_NO_RECEIVER	No network receiver for the message.	Unknown cause, this is a developmental message used by ANPC for troubleshooting	This message does not result in an alert or alarm, and no action is necessary to resolve this message
9002	LM_ID_NETWORK_MSG_ERROR	Network message error.	Unknown cause, this is a developmental message used by ANPC for troubleshooting	This message does not result in an alert or alarm, and no action is necessary to resolve this message

Error Code	Software Identifier	Description of Error	Possible Reason	Possible Solutions and corrective action reference
9003	LM_ID_BASE_TIME	Base Station's time differs from a sensor.	The clocks may not be synchronized	Use the SW tool Sync_clocks to synchronize the clocks 020-00073 4.13 System CPU clock synchronization
9004	LM_ID_MULTIPLE_REPLIES	The sensor received multiple replies to an interrogation.	More than one transponder is set to the code that was entered at the RCU	Ask the pilot to set a unique transponder code
9005	LM_ID_INTERROG_TIMEOUT	An interrogation timeout occurred.	Network cables are not connected properly	Verify all TLS components are powered on and can ping See 3.20.4.3 Base1 Timing Signal Fault Isolation Verify correct network connections See 3.20 Fault Isolation, LRU Inspection
9006	LM_ID_INTERROG_TIMEOUT_NET	Network connection errors between LRUs.	Network cables are not connected properly	Verify all TLS components are powered on and can ping See 3.20 Fault Isolation, LRU Inspection Verify correct network cable connections See 3.20.4.3 Base1 Timing Signal Fault Isolation
9007	LM_ID_INTERROG_TIMEOUT_GAOA	Interrogation timeout at the ESA.	The fiber optic cable that transmits the T-zero pulse to the ESA is not connected	Verify connectivity of the T-zero fiber cable by tracing from the output of the ICC to the rack switch, to the interface panel, and finally at the sensor See 3.20.4.3 Base1 Timing Signal Fault Isolation
9008	LM_ID_INTERROG_TIMEOUT_LAOA	Interrogation timeout at the ASA.	The fiber optic cable that transmits the T-zero pulse to the ASA is not connected	Verify connectivity of the T-zero fiber cable by tracing from the output of the ICC to the rack switch, to the interface panel, and finally at the sensor See 3.20.4.3 Base1 Timing Signal Fault Isolation

Error Code	Software Identifier	Description of Error	Possible Reason	Possible Solutions and corrective action reference
9009	LM_ID_INTERROG_TIMEOUT_INTERROG	Interrogation timeout at the Interrogator.	The interrogator control card (ICC) is in the wrong slot The ICC cables are not connected to the interrogator The ICC has failed	Re-install the ICC in the correct slot Verify connection of the cables from the ICC to the interrogator See 3.20.4.3 Base1 Timing Signal Fault Isolation
9010	LM_ID_INTERROG_TIMEOUT_RETRY	Retrying the interrogation after a timeout (an attempt to recover from code 9005).	Network cables may not be connected properly	Verify all TLS components are powered on and can ping Verify correct network connections See 3.20.3 Ethernet Communications Fault Isolation
9011	LM_ID_UNEXPECTED_MSG	An unexpected message was received by the base software.	Unknown cause, this is a developmental message used by ANPC for troubleshooting	This message does not result in an alert or alarm, and no action is necessary to resolve this message

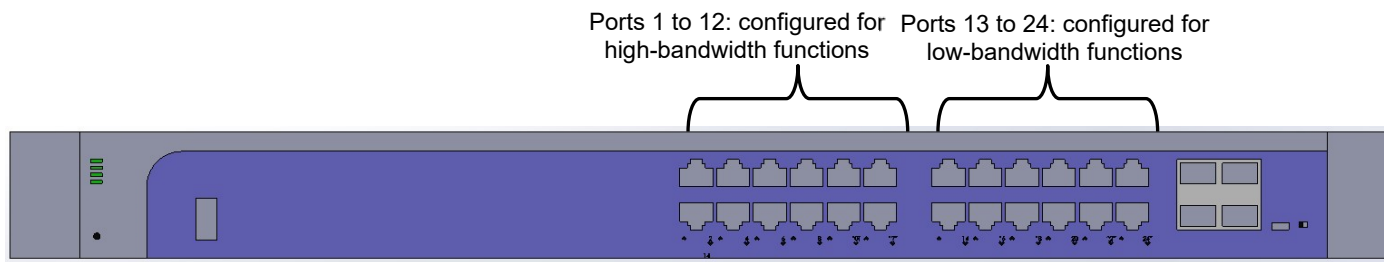


Figure 3-16 HP Model: Network Switch

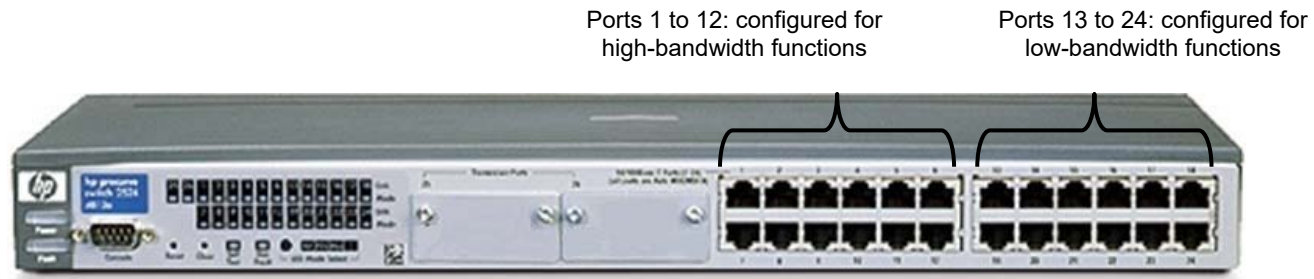


Figure 3-17 1U Netgear Ethernet Switch

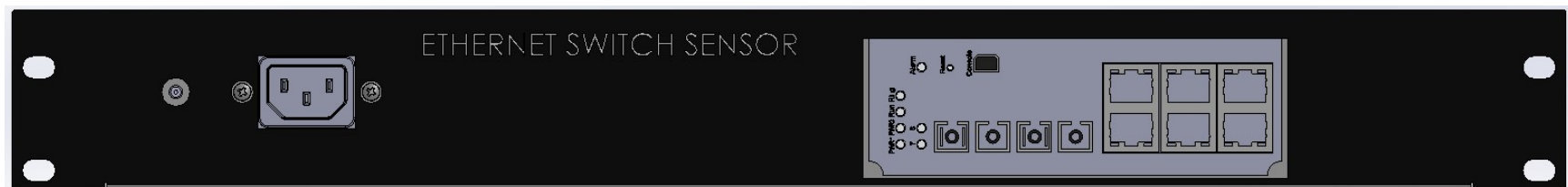


Figure 3-18 Sensor Ethernet Switch

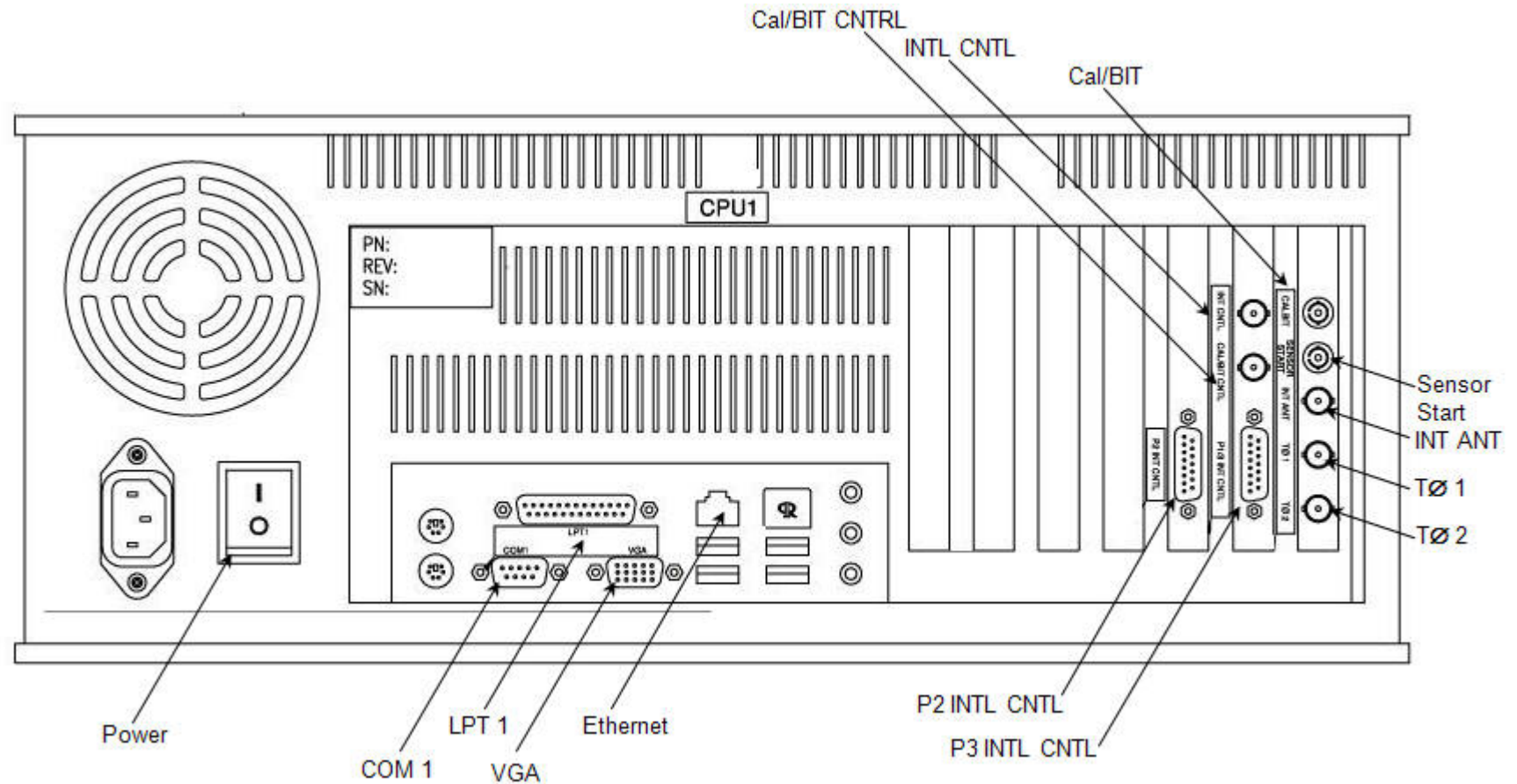


Figure 3-19 4U Base1 Backplane

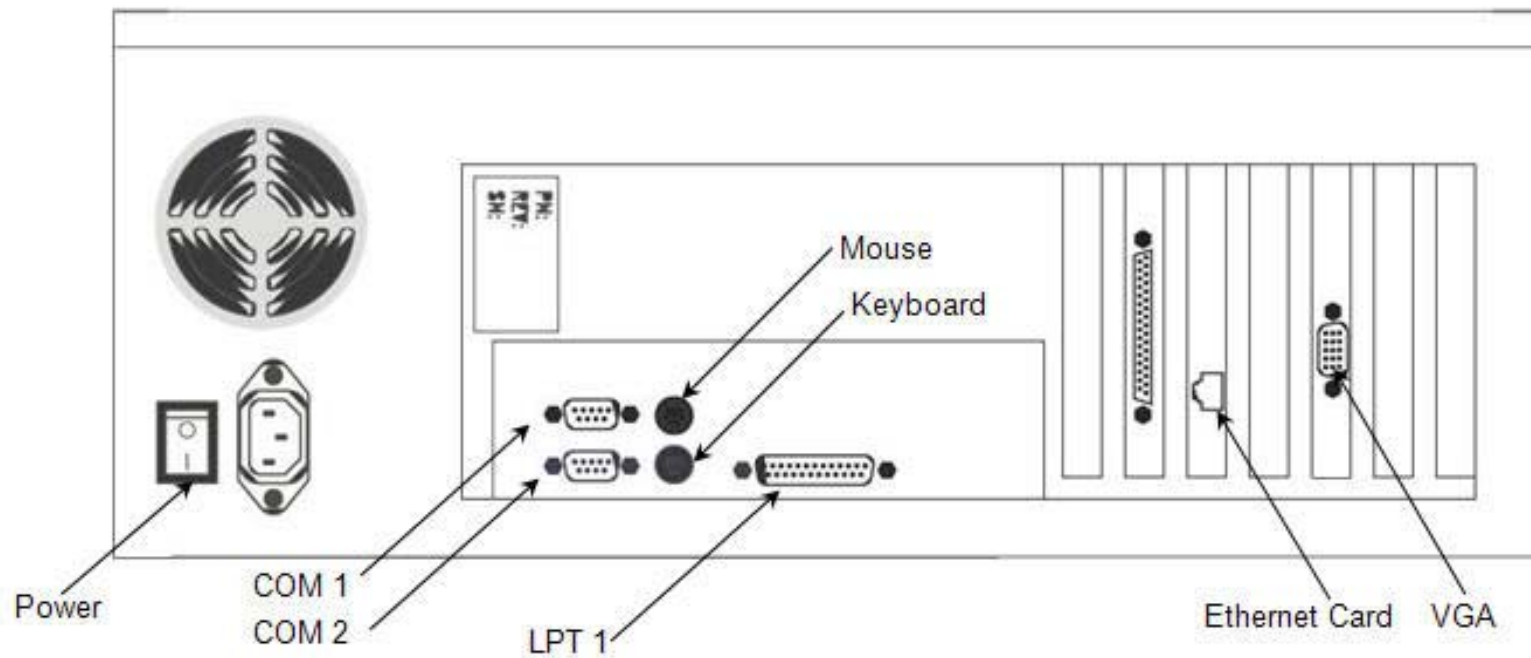


Figure 3-20 4U Base2 Backplane I/O

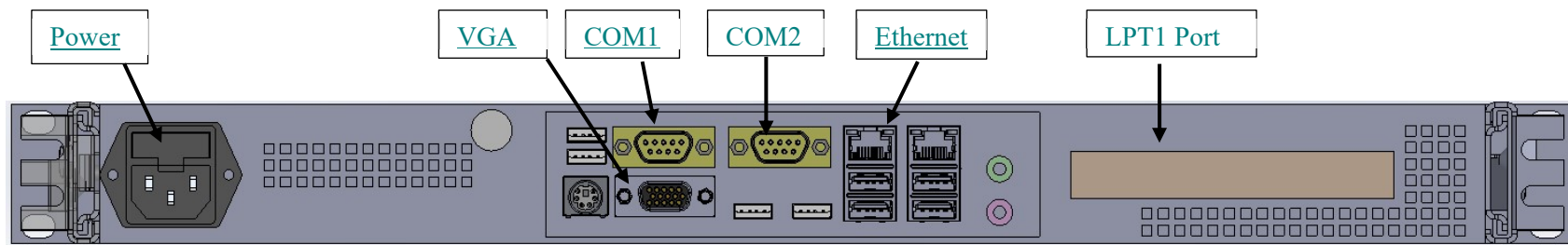


Figure 3-21: 1U Base2 Backplane I/O

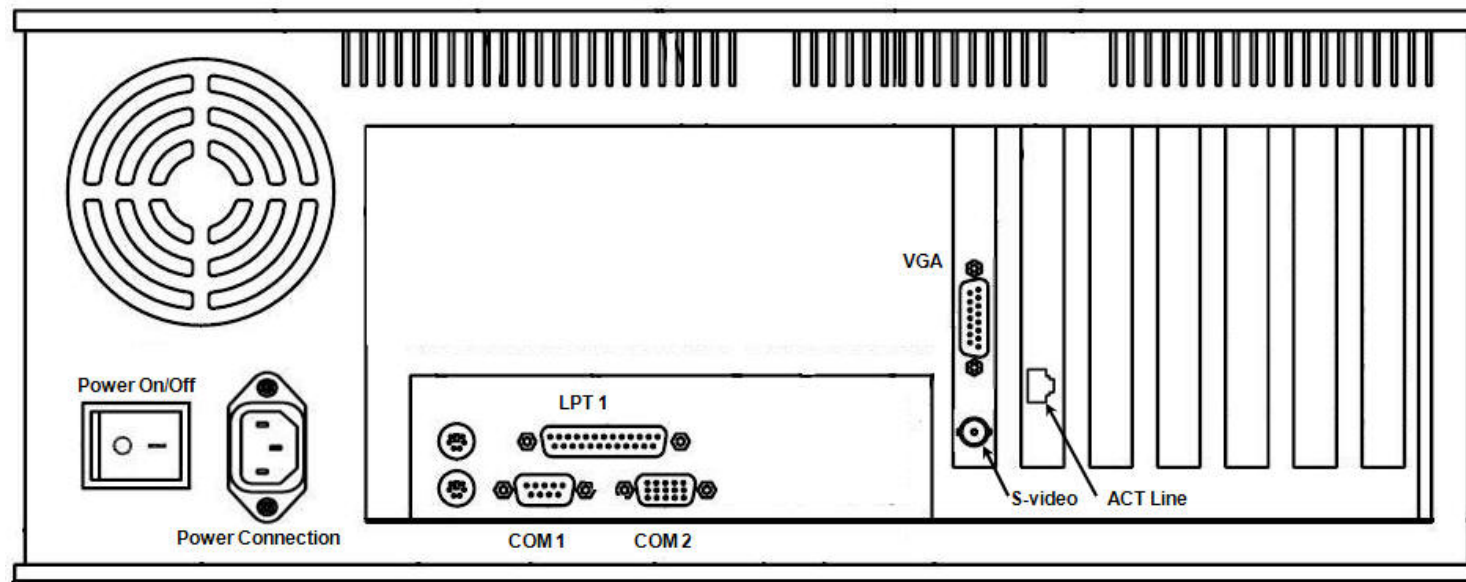


Figure 3-22 4U MIU Backplane

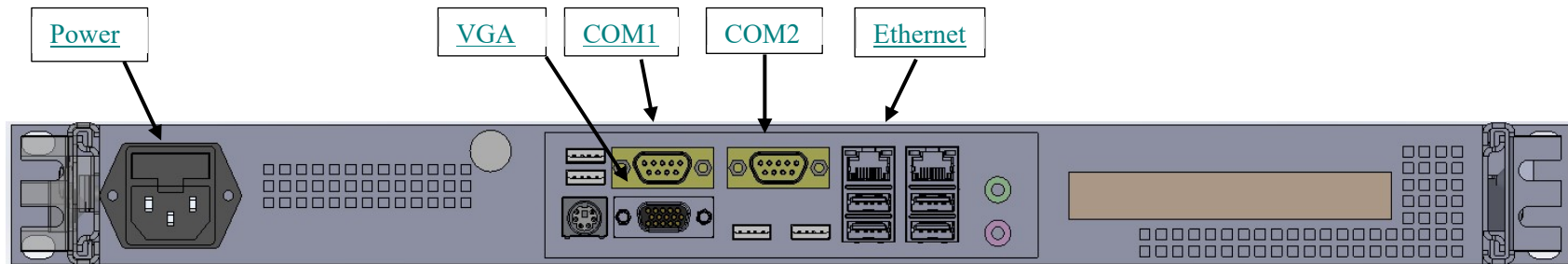


Figure 3-23: 1U MIU Backplane

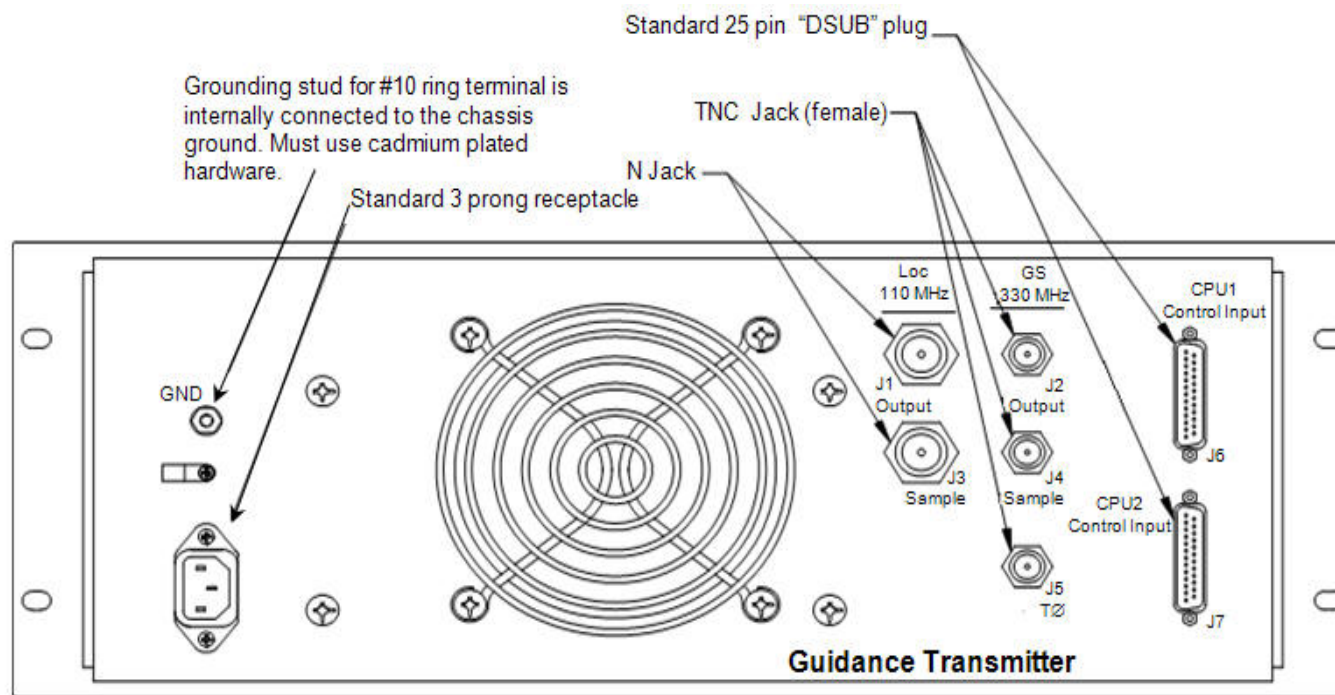


Figure 3-24 GTU 4U Backplane for 920-00101

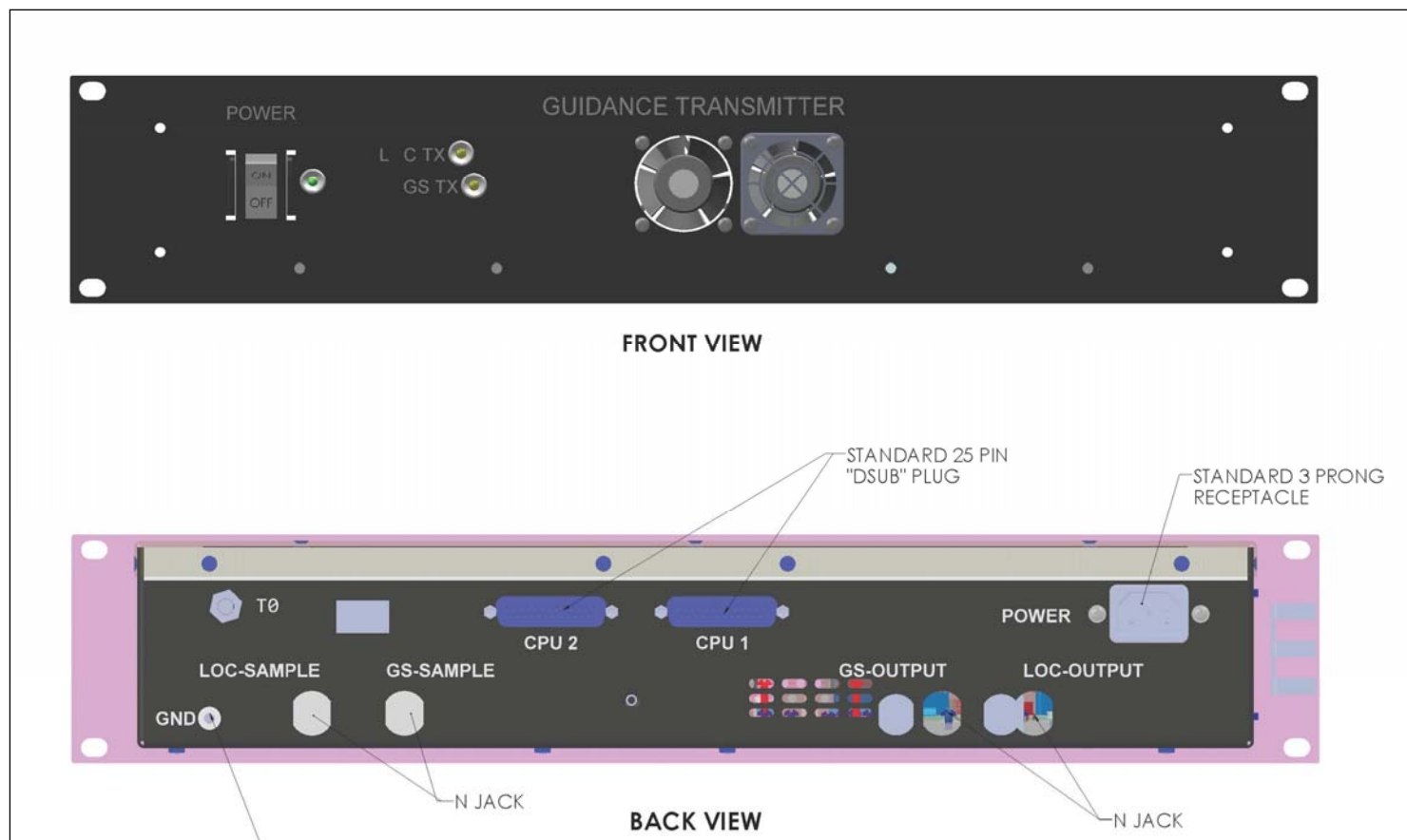


Figure 3-25 GTU 2U Backplane for 920-00402

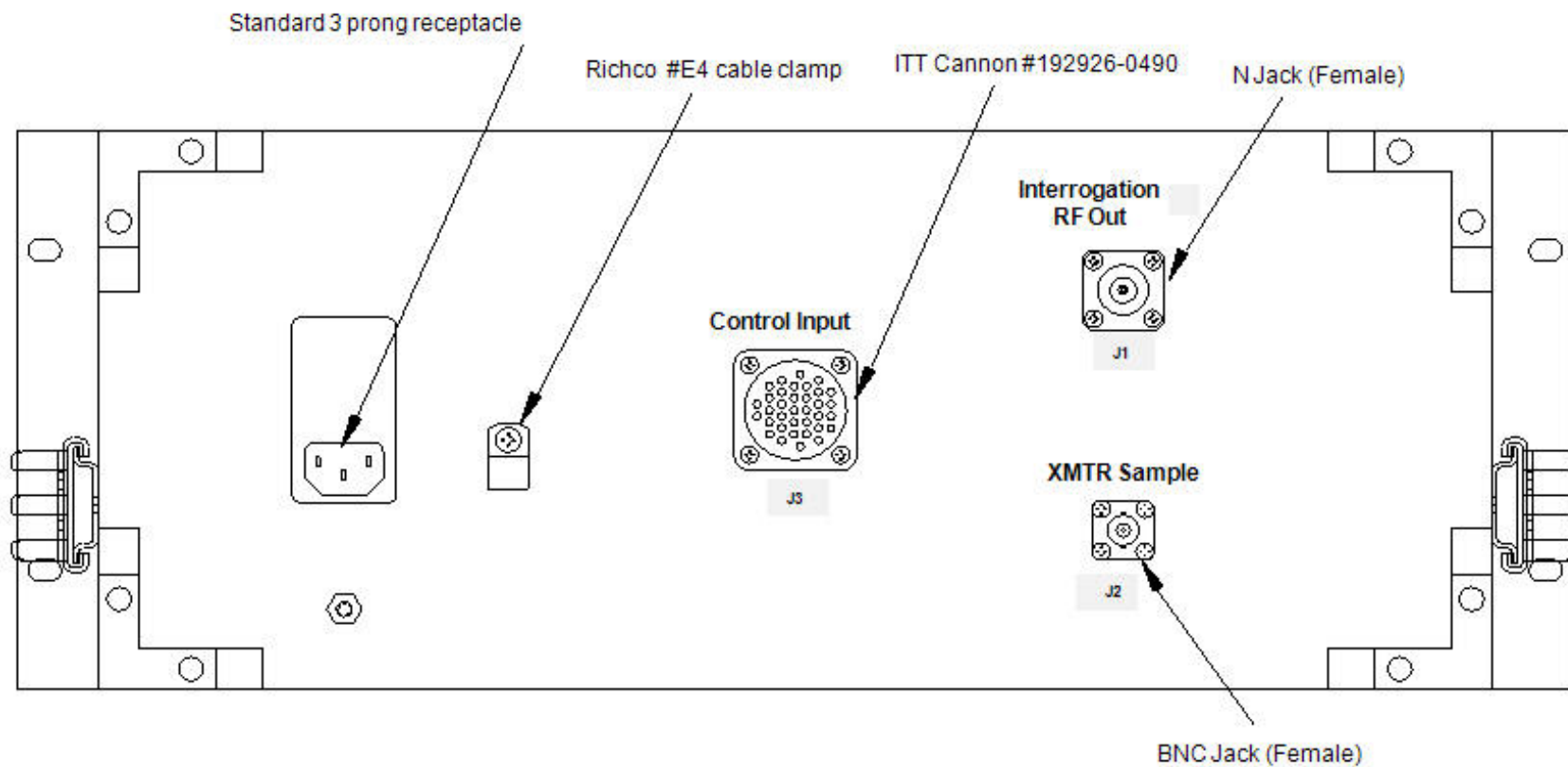


Figure 3-26 Backplane of the 4U Interrogation and Suppression Transmitters 870-00004

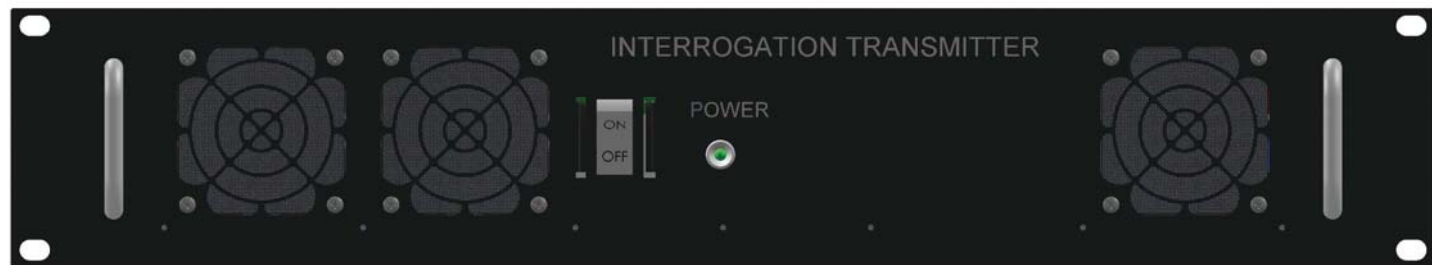
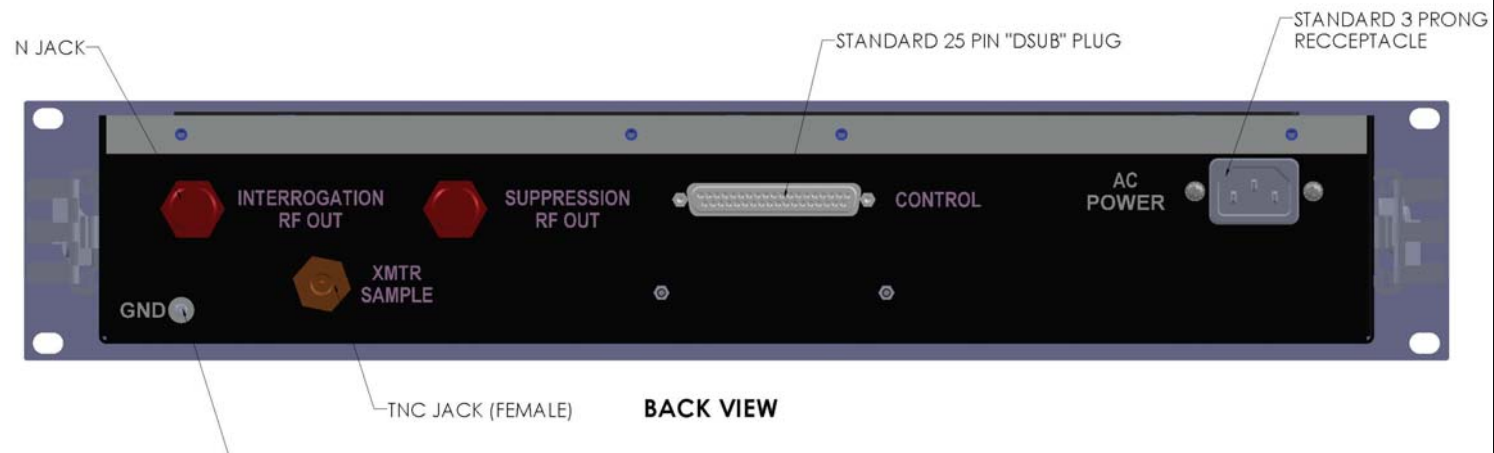
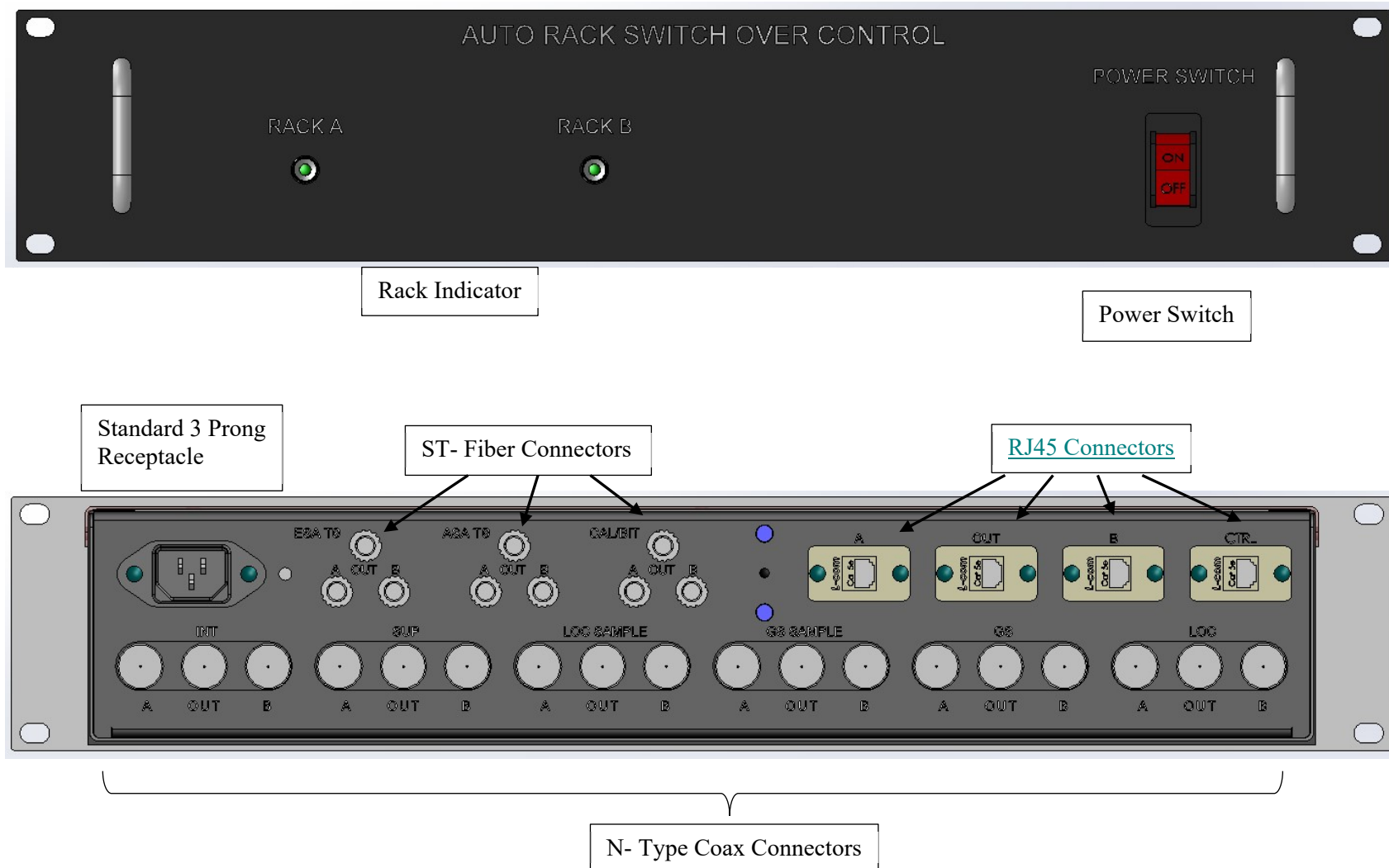
**FRONT VIEW****BACK VIEW**

Figure 3-27 Backplane of the 2U Interrogation and Suppression Transmitters 920-00586



VV

Figure 3-28 2U RackSwitch Front and Backplane

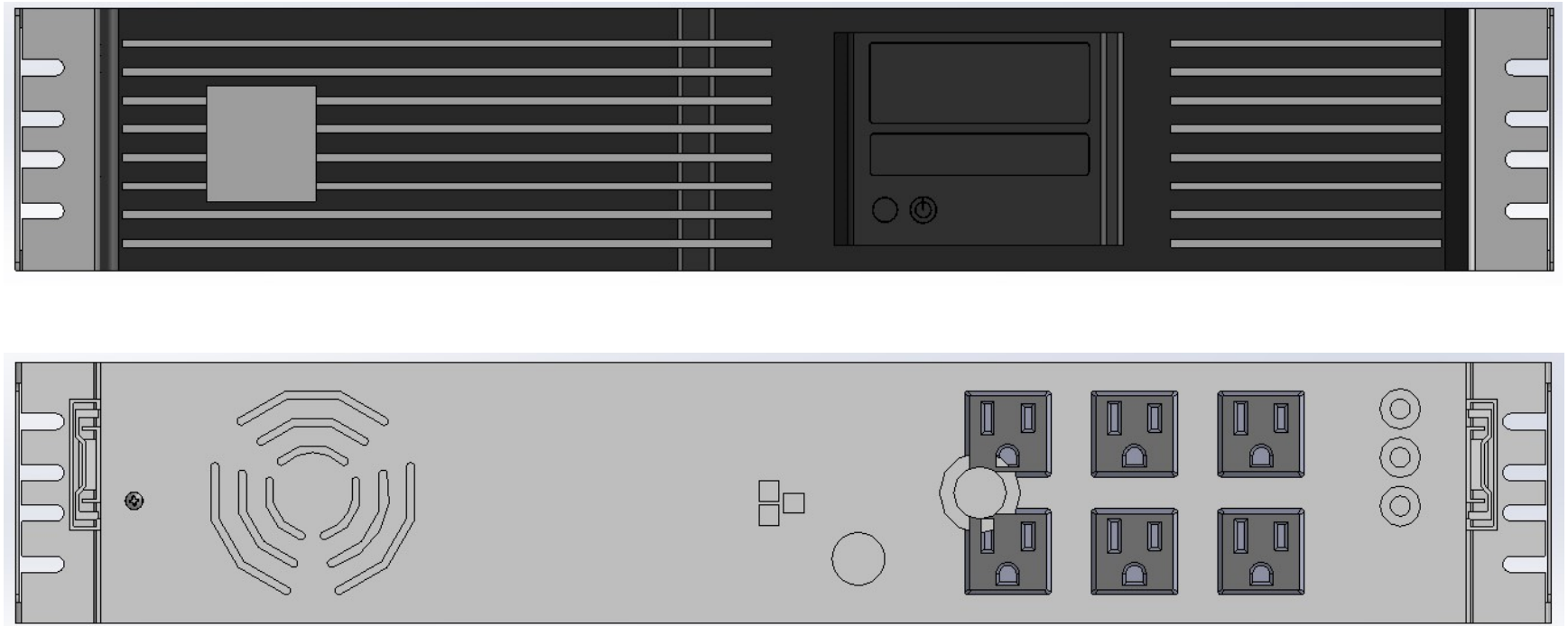


Figure 3-29 UPS Front and Backplane

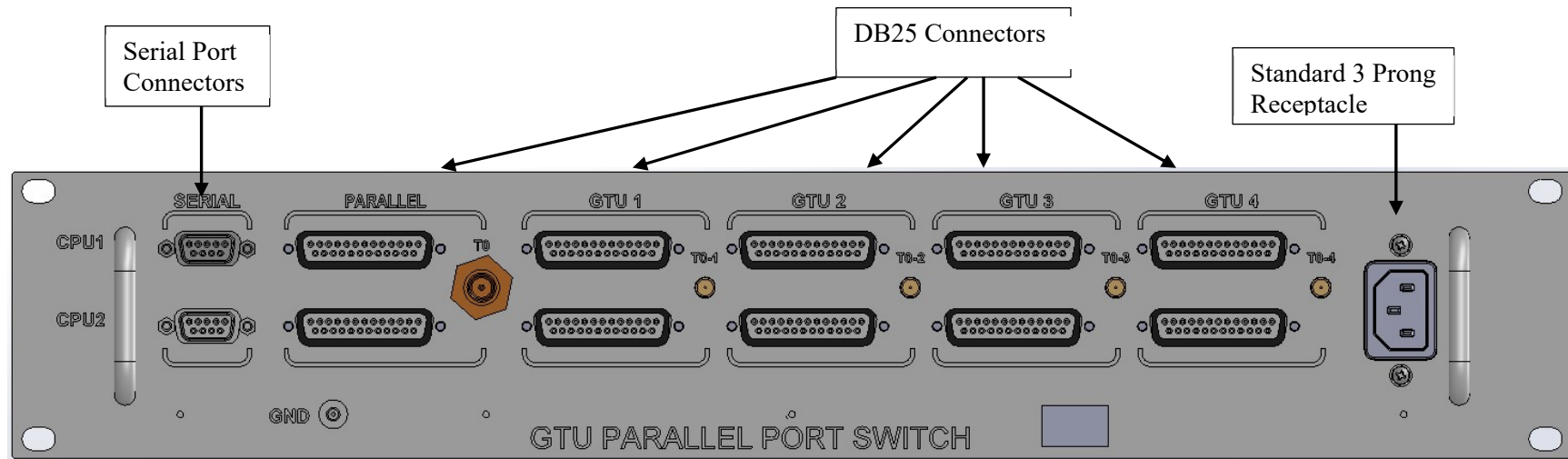


Figure 3-30 GTU Parallel Port Switch Backplane

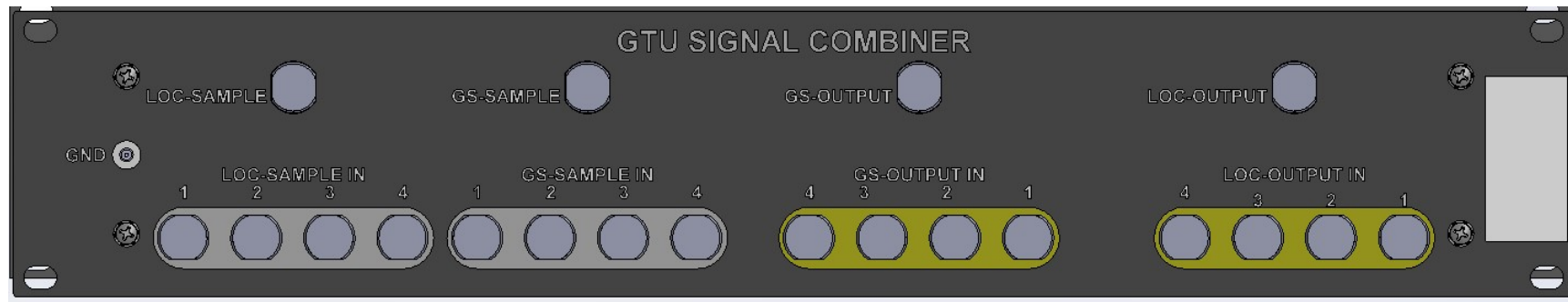


Figure 3-31 GTU Signal Combiner Backplane

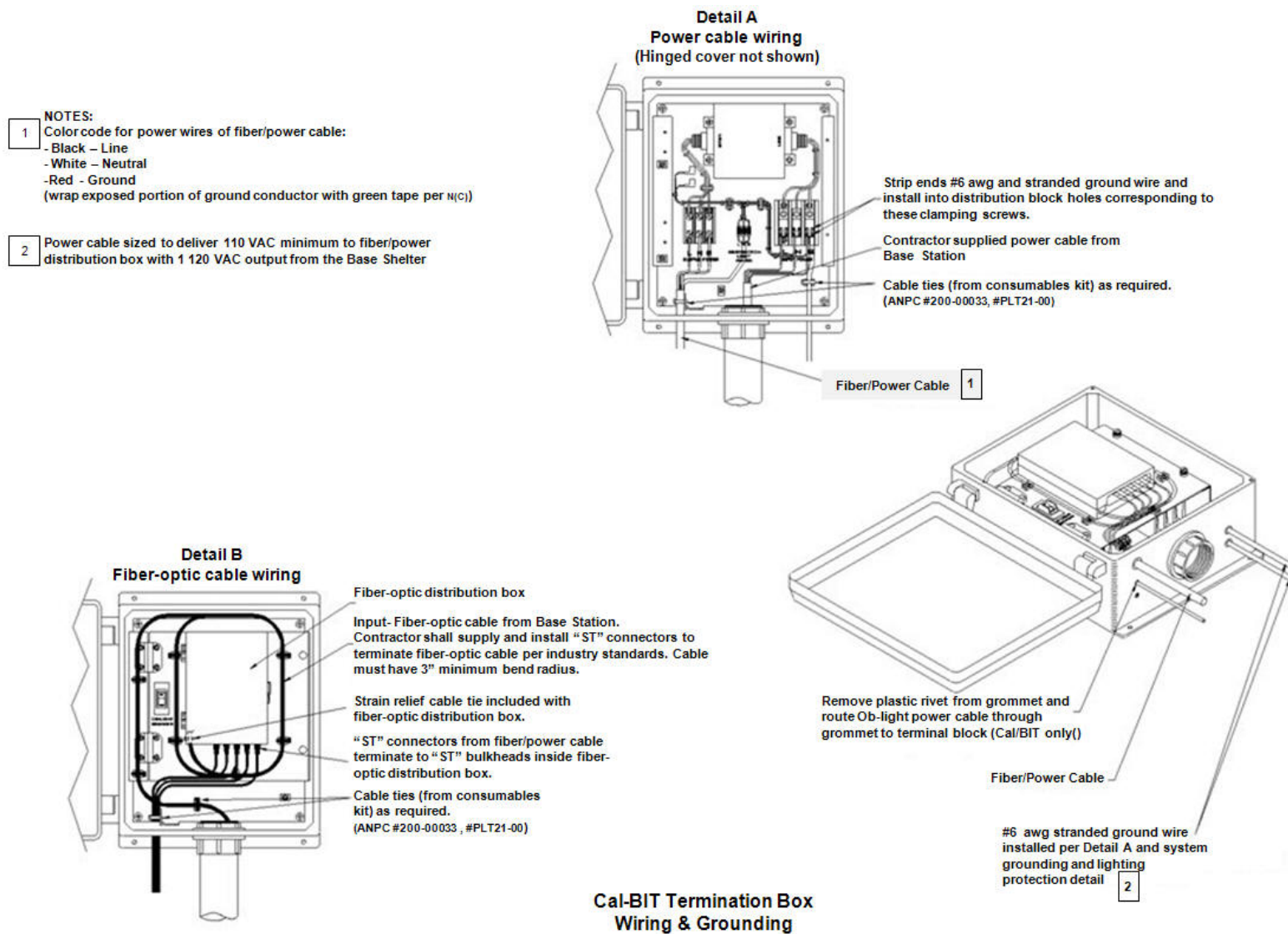


Figure 3-32 Termination Box Wiring Cal/Bit Transmitter
For Cal/BIT 920-00563, see drawing for junction box is 920-00684

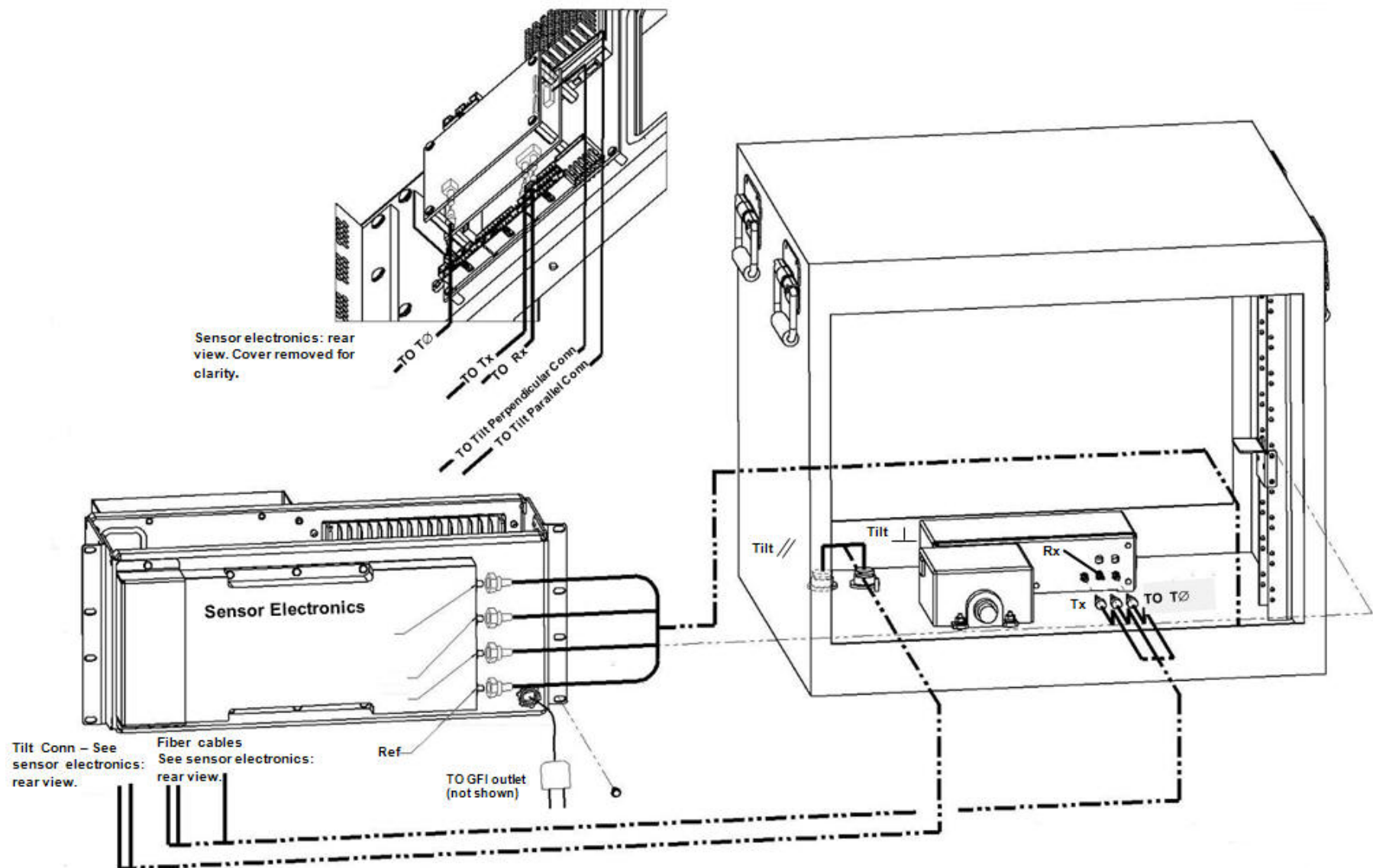


Figure 3-33 Sensor Housing I/O for 920-00256

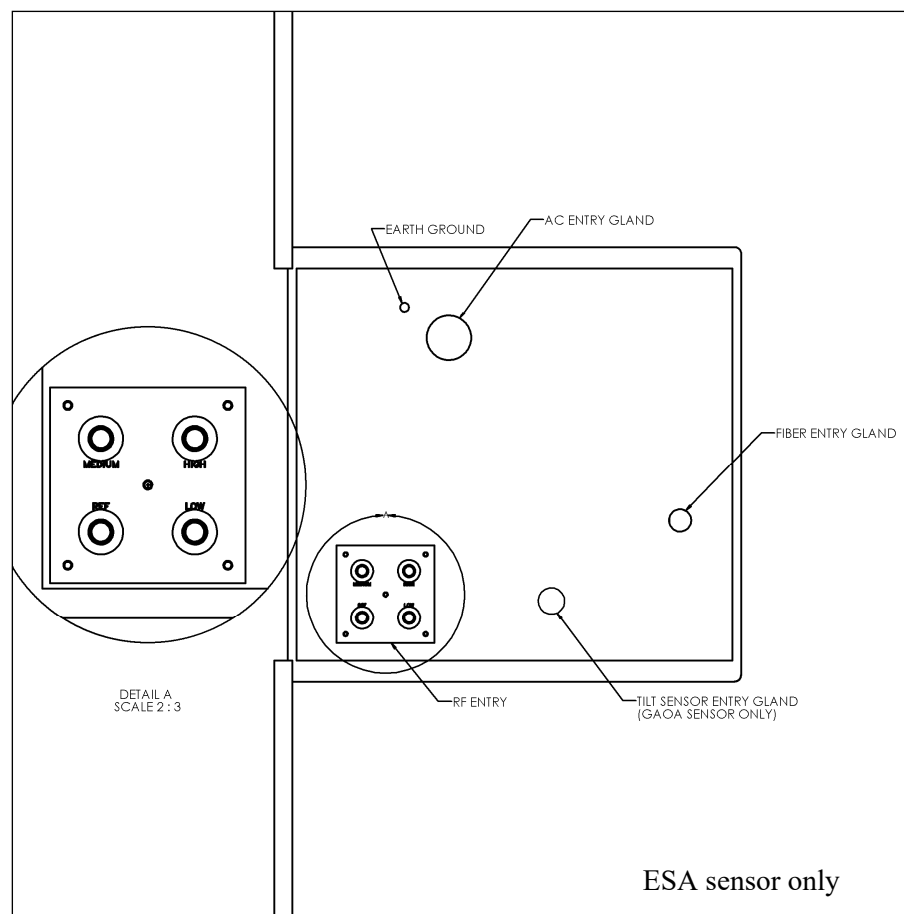


Figure 3-34 Sensor Housing I/O for Sensor 920-00256



Figure 3-35 Sensor Housing I/O for Sensor 920-00400

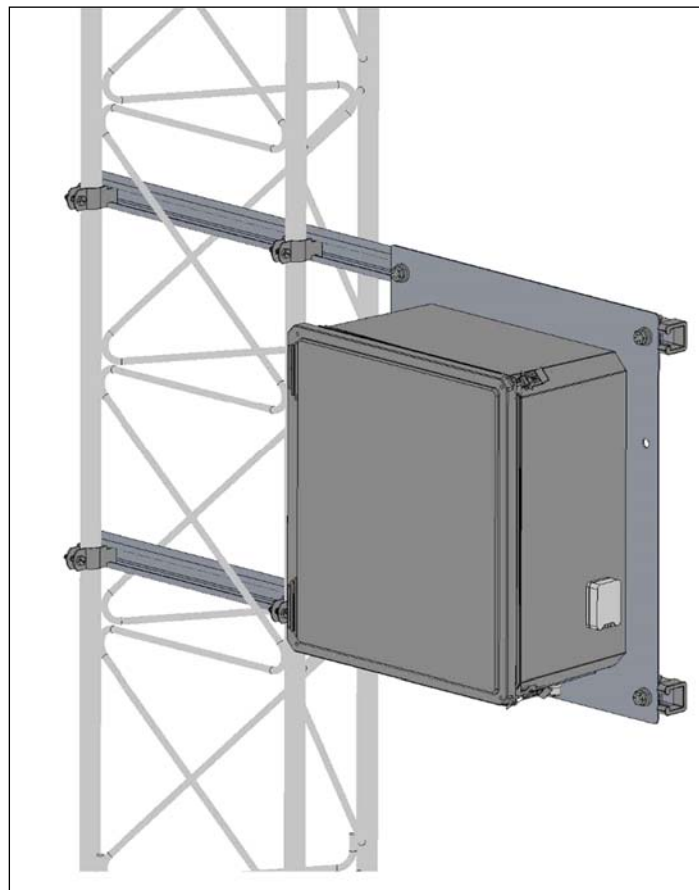


Figure 3-36 ESA electronics installed sensor 920-00400

3.20 Fault Isolation, LRU Inspection

The following paragraphs describe fault isolation procedures. For all LRUs, fault analysis uses a combination of BIT diagnostic data and hardware measurements presented in this section. For all failures, carefully examine the BIT data record and error log messages to identify the troubleshooting starting point, and usually, a probable failure point.

Start with:

- Reviewing logs.
- Pinging all Bases (from either the MI or DOS window) to determine if any of the LRUs are not responding.

3.20.1 User Interface Fault Isolation: RCU and MIU

RCU

- Does it power up? If not, make sure it is plugged into a working AC outlet.
- Can you communicate through Ethernet?
- Does a shorter Ethernet cable work correctly (if the RCU computer can be moved close to the TLS rack electronics)?
- Do the keyboard and mouse work correctly?
- Reboot the RCU media (fiber or wireless) converter or RF modem.
- Try pinging the other hosts.

MIU

- Does it power up?
- Can you communicate through Ethernet?
- Do the keyboard and mouse work correctly?
- Does BIT go into Track? If not, look at the logs of Base1 and Base2. What are the status colors for system monitor LRUs on the maintenance status display?

3.20.2 Network Switch Communications Fault Isolation

If the Network Switch box is suspected of being faulty (no communication between components), perform the following test:

- On the MIU, open the MI application.
- From the MI, select "System Status" and then "Network Test." Run the communication check, and verify that each of the base station computers and sensors responds. This performs a communication check of each host on the network and returns the status.
- If anyone host does not reply, switch the connections of that network connection at the Network switch box with a like connection (AOA sensor with AOA sensor or Base with another Base), and rerun the Network Test.
- If the unit not responding switches with the cable connection switch, then the Network Switch box is probably faulty.
- If the fault remains with the same system element, try exchanging cables.
- Finally, replace the LRU that has the communication fault.

3.20.3 Ethernet Communications Fault Isolation

1. Verify that card has the correct IP address and a valid MAC address. Type "ipconfig /all" using the command prompt (DOS window).

2. “Netstat –an”
3. Try pinging other hosts.
4. Type the following address in Internet Explorer: “192.168.28.1”. This is the HP Procurve Switch’s address. Use its diagnostics to troubleshoot further.
5. Verify host and services by opening the files. Check that all host addresses are some version of “192.168.28.xxx”. The host and services files can be found in c:\windows\system32\drivers\etc\ on Windows 7 and 10. On Windows 10 LTSC 2019 they are stored in c:\Windows\system32\drivers\etc. And on QNX, they are stored in /etc/.
6. Try reseating the card.
7. Check the operating system’s error logs to make sure it is loading.
8. Try using other ports in the Ethernet switch. Make sure to use an available port of the same bandwidth speed. Generally, ports 1 to 12 are configured for 1GbaseT communications, and ports 13 to 24 are configured for 100baseTx.
9. Run the following commands on base2: “ping baseX” or “ping 192.168.28.xxx”
10. If a sensor will successfully ping, but the base software will not complete the boot sequence and is stuck at the “initializing” mode, then reboot the sensors by turning the power off at the circuit breaker panel that controls power to each sensor. Turn on the sensor power and wait two minutes for the sensor boot sequence to complete. This should resolve the problem.
11. If the network connections are incorrectly positioned, the system may suffer perpetual T0 timeouts because the sensor replies are not passing through the switches. This can be caused by hooking the Common port of the Rackswitch to the ".4" (RCU link and Rackswitch traffic mitigation) ethernet switch instead of the 100BaseTX port of the ".3" switch.

3.20.4 Base Computer Hardware/Software Fault Isolation

If either base Base1 or base Base2 software fails to start, is not performing correctly, or if the hardware is suspected of being faulty (no communication with one or the other base, or failure of TLS software to complete initialization), three different types of faults can be responsible:

- Computer hardware (hard drive, etc.) Failures:
This can be determined by switching the (Keyboard Video Mouse) KVM switch to the suspect Base and determining if the computer is running or if the operating system has halted. If re-booting the Base fails to start the computer properly, the Base can be considered a failed LRU. Replace.
- Base Station LAN Failures:
This can be determined by performing the Network Test from the MI System Status Window. This isolates which LRU is not communicating on the network. Perform network failure isolation to determine if the fault is a connector, the network switch, or a faulty network adaptor in the computer.
- TLS Software Failures:
Typically, this is a failed initialization. Examine the Base Log or Startup Log files using the System Status window in the MI to determine this. Check for faults due to bad system data files or configurations.

If the software failed after initialization, refer to the assert.log and/or base.log files on both bases. These files would contain an error if the TLS software halted on either Base computer. Switch to Base1 and check to see if there is a message indicating that base software asserted. Type in 'sf' at the command prompt and press the Enter key, which will shut down & restart, then log on. Switch to Base2. If Base2 software is asserted, there will be a message dialog box. Use standard Windows menus to restart Base2 and log on.

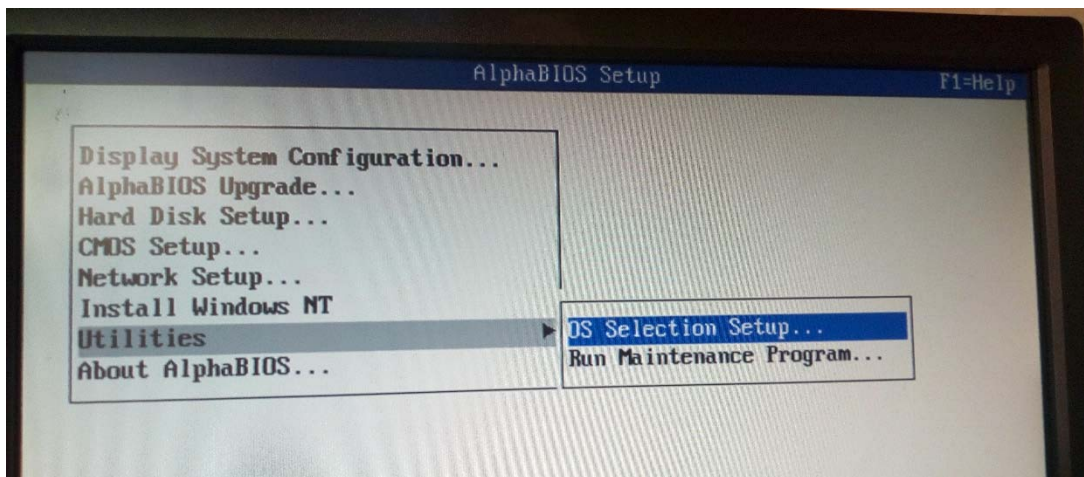
After both Bases have rebooted, the MIU should say "NO COMM WITH BASE SW." Next, the data archive should be created in accordance with procedure 6.3. If a software Assert occurred, there will be a base1_assert.zip file and/or a base2_assert.zip file in the archive that will have captured the event. Now the TLS system software can be restarted.

3.20.4.1 General Troubleshooting for Base1

- Does it power up? If not, verify that the power supply switch is on.
- Do keyboard and mouse (if applicable) work? Try swapping with a keyboard that is known to work.
- Log in as "root" with the supplied password.
- Run the "sin" command: is "su_server" running?
- Ping Base1, Base2, MIU, ESA, ASA, RCU.
- Reboot by executing the "sf" command.

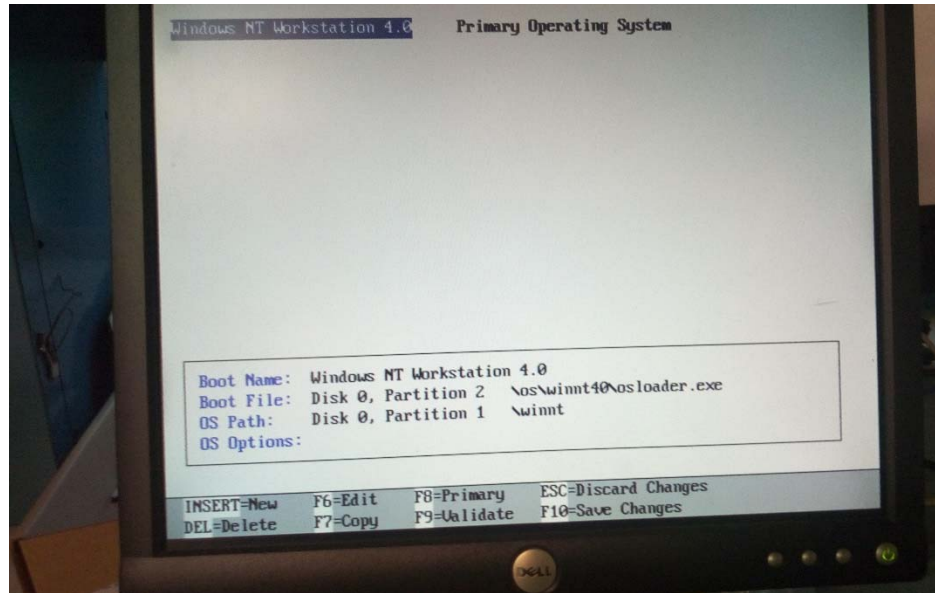
3.20.4.2 General Troubleshooting for Base2

- Does it power up? If not, verify that the power supply switch is on.
- Do keyboard and mouse (if applicable) work? Try swapping with a keyboard that is known to work.
- Is su_server running? If not, reboot. Also, verify that it is the correct version of the su_server.
- If the problem with NT booting on Base2 shows an error message that "The highlighted Operating System Selection is not valid," perform the following change to the Alpha BIOS.
 1. During the boot process, a message will display at the bottom of the screen that says press F2 to enter setup.
 2. Press F2, and then the screen below will appear. Use the arrow keys to select "utilities" and then select "OS selection setup."



3. On the OS selection setup page, press F8 to make windows NT the primary operating system
4. Then press f10 to save changes
5. Then follow the bios instructions to set NT as the primary OS. Go to utilities, select OS selection setup and then press F8 to set NT as the primary OS.

6. Save and exit the bios utility software and reboot the computer; the problem should be resolved.



3.20.4.3 Base1 Timing Signal Fault Isolation

1. If Base1 fault is suspected with respect to timing signal generation, perform the test presented in Signal Test Points Panel.
2. If a signal is not present at the Signal Test Points panel, connect the oscilloscope to the timing signal outputs at the back of Base1. Repeat the verifications of step 1.
3. If the signal faults are still present at the Base1 signal outputs, replace the Base1 assembly, and skip the remainder of this test.
4. If the signals at the Base are correct (step 2), repeat step 1, connecting the oscilloscope to the timing signal cables at the end that attach to the rear of the Signal Test Points panel.
5. If the signal faults are still present at the cable ends, replace (or correct routing error) the coax cable from the Base1 assembly to the Signal Test Points panel. If the signals at the cable ends are correct, replace the Network Panel.

CAUTION

If a Base(Base1 or Base2) is replaced, all *.dat files MUST be copied from the other Base's Run directory. A software tool (Sendtobases) is supplied on the MIU to automate this process.

3.20.5 Guidance Transmitter Unit (GTU) Fault Isolation

If a faulty Guidance Transmitter Unit (GTU) is suspected, BIT data is the best indication of the failure type. Use the MultiBIT tool to examine the GTU BIT data. This tool provides manual control of all GTU functions and read-backs.

1. Turn the GTU OFF. Check for visible damage to the transmitter, guidance antennas, and cables.
2. Turn the GTU ON. Verify that the ON indicator light is lit and that the fan is drawing air.
3. If the problem seems to be communication to the GTU from Base2, reboot Base2 and choose the keyboard selection during the boot process to enter the Bios. Navigate to peripherals, select the parallel port and verify the EPP mode is set for the parallel port. Depending on the keyboard configuration, it may be necessary to attach a detected keyboard directly to Base2 (and not use a KVM).
4. On the MI, stop the system software if it is running.
5. On the MI, select GTU Test in the Tools menu.
6. If the BIT failures indicate all GTU function is lost (Both GS and Loc faults, 24 V failures and/or TØ fault), perform the following steps:
 - a. In the GTU Test interface, click the "Loc On" button.
 - b. Verify that the localizer section Base1 and Base2 read-back indication "boxes" are checked. Compare all readings and boxes. All readings for Base1 and Base2 should match, and all boxes should be checked.

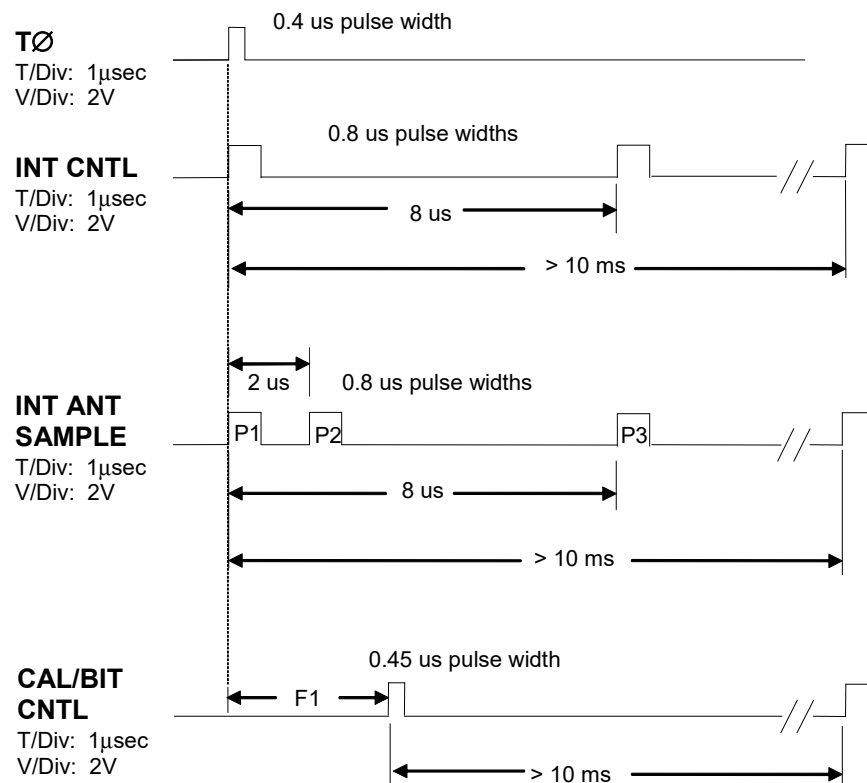


Figure 3-37 Signal Test Points Panel Timing Relationships

- c. The Health section of the window provides status and power supply monitor data. Both "T0 Period OK" boxes should be checked. Both "Freq Pgm Verify" boxes should be checked. Power supply data for both bases should match.
- d. If the "Freq Pgm Verify" box is not checked, then program the GTU frequency; this may resolve the problem.

- e. If any of the Base1 vs. Base2 boxes do not match, switch the Base inputs to the GTU at the Base1 Control Input and the Base2 Control Input of the GTU.
 - f. If the readings and checked boxes remain the same, the GTU has failed. If the readings and boxes from base1 switch to base2 and vice versa, the Base or cabling is bad.
 - g. If, in step (c), the base status boxes match, but neither "T0 Period OK" box is checked, check the T0 test point on the Signal Test Points panel. If the T0 test point is good, then either the GTU or the T0 cable between Base1 and the GTU is bad. Repeat the test with the oscilloscope at the GTU end of the cable - if the fault does not exist, replace the GTU.
7. If the BIT faults indicate an SWR fault, which will have an SWR alarm, bad 24 V reading, and bad carrier power and modulation data, perform the following steps:
- a. If both the localizer Transmitter ON and SWR OK indicators are OFF, isolation of the fault between the transmitter and uplink cable/antenna is required.
 - b. Remove the localizer RF Output cable from the GTU, and connect 10 dB of attenuators to the RF connector. On the GTU, re-enable the transmitter by depressing the LOC ON button. If the Transmitter ON indicator is still not checked, the GTU is the faulty LRU. If the Transmitter ON indicator is checked, the fault is in the Cable/Antenna – by changing the location of the attenuator in the antenna path, and the fault can be isolated to a single item.
 - c. Enable the glide slope by clicking on the "GS On" button. Repeat steps a and b above for the glide slope section.
8. If the transmitter is enabled (no SWR fault), but there are Carrier power faults that indicate NO output power (carrier power < 100), perform the following steps:
- a. Turn the GTU OFF. Remove both the faulty signal RF Output and faulty signal Monitor input cables, and connect the RF Out to the Monitor In through a string of attenuators (44 ± 3 dB for localizer, 54 ± 3 dB for glide slope) - keeping the smaller attenuators at the RF Out end.
 - b. Turn the GTU ON. On the MI, select GTU Test in the Tools menu. Re-enable the transmitter by depressing the LOC ON button.
 - c. If the GTU Test monitor window indicates that the carrier power is NOT present, the GTU should be replaced. Skip the remainder of this test.
 - d. If the GTU Test monitor window indicates that the carrier power is present, the fault is in the antenna/cables.
 - e. Install a PIR 10-20 meters in front of the guidance antenna array - be sure not to block the signal path from the transmit antenna to the monitor antenna. With the transmitter enabled, measure the signal power with the PIR. If the PIR power measurement is between -25 and -10 dBm, the fault is in the antenna or cables on the monitor side. If the PIR does not measure signal power, the fault is in the antenna or cables on the transmit side.
9. If the transmitter is enabled (no SWR fault), but there are Carrier power faults that indicate unstable or intermittent output power, perform the following steps:
- a. Turn the GTU OFF. Remove both the faulty signal RF Output and faulty signal Monitor input cables, and connect the RF Out to the Monitor In through a string of attenuators (44 ± 3 dB for localizer, 54 ± 3 dB for glide slope) - keeping the smaller attenuators at the RF Out end.
 - b. Turn the GTU ON. On the MI, Settings window, set the BIT interval to 1 minute. Start system software and verify that BIT data is being collected.

- c. Collect BIT data long enough to have captured the unstable or intermittent connection (generally twice as long as it takes the event to happen under normal setup conditions).
- d. Analyze the BIT data. If the fault condition is still present, the GTU should be replaced. Skip the remainder of this test.
- e. If the fault condition is NOT still present, the fault is in the antenna/cables.
- f. Reconnect the normal antenna connections to the GTU. Replace the cables and antenna on the transmit side and repeat steps c and d. If the fault condition is still present, replace the cables and antenna on the monitor side.

3.20.6 Interrogation Transmitter and Suppression Transmitter Fault Isolation

If a faulty Interrogation Transmitter is suspected, perform the verification steps below. There are two pulse amplifiers built into the Interrogation Transmitter: The Interrogation pulse amplifier (for P1 and P3 Interrogation output pulses) and the Suppression pulse amplifier (for P2 Suppression output pulses).

*Some systems are comprised of an Interrogation Transmitter without the P2 (Suppression) pulse amplifier.

1. Set the TLS for continuous interrogations by tracking a 0000 transponder code from the Test Mode window from the MI Tools menu.
2. Connect channel 1 of the oscilloscope to the TØ test point. Use channel 1 (TØ) as the trigger.
3. On the Signal Test Points Panel, disconnect the jumper between INT ANTENNA SAMPLE IN and INT ANTENNA SAMPLE OUT. Connect channel 2 of the oscilloscope to the INT ANTENNA SAMPLE IN test point on the Signal Test Points panel.
4. Referring to Figure 3-37, Signal Test Points Panel Timing Relationships, use the oscilloscope and the test points on the front panel of the Signal Test Points Panel to verify that the Interrogation transmitter sample signals for P1, P2, and P3 are correct.
5. If the proper pulses are not present on channel 2, verify that the control signals are OK, being careful to validate timing relationships accurately. If control pulses are OK, one of the transmitters, control card, or control cables is faulty.
6. If pulses are present but do not meet specifications (amplitude, timing, or shape), the faulty LRU is the pulse transmitter that produces the faulty pulses.
7. If one transmitter's pulses are present, but the other transmitter's pulses are not, power down the Interrogation Transmitter and swap RF cables at the Interrogation Transmitter output ports only. Power on the Interrogation Transmitter and repeat steps 1 through 4 above. Proceed per the following table:

Table 3-8 Interrogation Transmitter Fault Isolation 1

Fault Observed	Post Step 7 Swap Fault	Bad LRU
P1 and P3 are OK, but no P2.	P2 OK, but no P1 or P3	Interrogation Transmitter (Interrogation pulse amplifier)
P1 and P3 are OK, but no P2.	P1 and P3 are OK, but no P2.	Cable or Base1– go to step 8
P2 OK, but no P1 or P3	P1 and P3 are OK, but no P2.	Interrogation Transmitter (Suppression pulse amplifier)
P2 OK, but no P1 or P3	P2 OK, but no P1 or P3	Cable or Base1– go to step 8

8. Return control cables at the transmitters to their original locations.
9. In the MI Test Mode window, abort the 0000 track. Exit Test Mode.

3.20.7 Cal/BIT Transmitter Fault Isolation

Review the base log and BIT data. If an identical Cal/BIT signal measurement problem is occurring in both the ESA and the ASA, then the problem is a faulty Cal/BIT Transmitter.

3.20.8 Measurement Sensor Fault Isolation

If a faulty sensor is indicated, review the BIT log files of the suspect measurement sensor for out-of-tolerance data. If a particular RF input path is identified as the fault from BIT and base log data, then proceed with fault isolation of a measurement sensor antenna and the associated sensor.

Things to check:

- Does the sensor power up?
- Can it communicate through Ethernet fiber?
 - a. If not, can you communicate by using a short Ethernet cable (bring sensor electronics to the base station)? Check by connecting the sensor directly to the hub using a short straight Ethernet cable. If the sensor is now able to communicate, there's a problem with the fiber portion of the cable. Replace or repair.
 - b. Reconnect fiber. Verify that the connectors are attached and tightened correctly at sensor and base station terminations.
 - c. Use fiber meter to check attenuation. Loss should be less than 2-4 dB; if greater than 10dB, replace the cable or use spare.
- Do the tilt sensors work? Are tilts plugged in? Note: only check plugs with power off.
- Are the power supplies correct? Measure voltages to verify.
- Is the sensor seeing Cal/Bit pulse?
 - a. Does it measure 65 out of 65 pulses? If not, set TOA near from TOA far - 390000
- Is TOA jitter > 4ns?
 - a. Set the F1 timing pulse to minimize TOA jitter.
- Phase movement/noisy?
 - a. Check if truck/airplane/personnel are in the critical area. Remove if so.
 - b. Tighten tower guys and check if the performance improves.

Further troubleshooting can be performed by repeating the Measurement Matrix for the LRU Fault Isolation test.

3.20.8.1 Measurement for LRU Fault Isolation

1. From the MI, stop the system software.
2. At the ASA, swap antenna connections to the sensor RF inputs as follows:

<u>Move Cable from Input</u>	<u>T0 Input</u>
Reference	Low
Low	High
High	Reference

3. At the ESA, swap antenna connections to the sensor RF inputs as follows:

<u>Move Cable from Input</u>	<u>T0 Input</u>
Reference	Low
Low	Reference
High	Medium
Medium	High

4. From the MI, start the system software. Wait for the system to complete "Diagnostic" mode and enter "Maintenance" mode.
5. From Test Mode, command BIT at least 5 times and exit Test Mode to MI main screen. Stop the system software.
6. Using MultiBIT data tool, open \\Base2\C\Run\BIT\esa.bit. Review each data display element and verify that the swapped cable data is stable. Using averages of the measured swapped-cable data ranges, record the ESA amplitude, phase, and delta phase values.
7. Using MultiBIT data tool, open \\Base2 \ C \Run \BIT\asa.bit. Review each data display element and verify that the swapped-cable data is stable. Using averages of the measured swapped-cable data ranges, record the ASA amplitude, phase, and delta phase values.
8. At the ASA and ESA, swap antenna connections back to their original sensor RF inputs.
9. From the MI, start system software. Wait for the system to complete "Diagnostic" mode and enter "Track" mode.
10. From Test Mode, command BIT at least 5 times and exit Test Mode to MI main screen.
11. Using MultiBIT data tool, open \\Base2 \ C \Run \BIT\esa.bit. Review each data display element and verify that the data returned to the pre-swap nominal levels.
12. Using MultiBIT data tool, open \\Base2 \ C \Run \BIT\asa.bit. Review each data display element and verify that the data returned to the pre-swap nominal levels.
13. At the ASA and ESA, seal the RF connections per the coaxial weatherproofing procedure.

Compare the results to the baseline data.

- If the BIT data fault remains on the same sensor channel (example: Low phase contains the original fault, and after the swap, the Low phase data is different from the baseline data), the sensor is at fault and needs to be replaced.
- If the fault follows the swap (for example, the Low phase contains the original fault, and after the swap with the High antenna cable, the High phase is data that is different from the baseline data), the fault is with the cable and/or antenna and needs to be resolved.

3.20.9 Coaxial Cables Fault Isolation

3.20.9.1 Visual Inspection

If a coaxial cable is suspected as being the cause of the system fault, visually inspect the cable(s) for damage. Check for kinks, twists, and cuts. Re-run cable to remove kinks and twists. Minor cuts that do not cut into cable conductors may be repaired with the cable repair kit. Replace any cable with cuts into the cable conductor. Check connectors for corrosion and waterproof seal. If corrosion is present, remove with a wire brush and waterproof using the cable repair kit. If no corrosion is noted but moisture is present, remove old waterproofing and replace it.

3.20.9.2 Antenna Cable Isolation

Continuity checks can be made by removing the cable, shorting the center conductor of one end to the outer shell, and taking a resistance reading with a DMM between the center conductor and outer shell of the other end. This should read as a short. Next, remove the shorting wire from the opposite end and taking a reading with the DMM between the center conductor and outer shell. This reading should indicate an open. If either test fails, a bad cable is indicated.

3.20.10 ASA Antenna Re-alignment (Transportable)

Conduct this procedure if the ASA antenna position has moved and the CAL/BIT elevation has not changed.

3.20.10.1 Overall General Description

The procedure to fix the alignment requires manually adjusting the ASA antenna beam and then using test mode at the base station to run Built-In-Test (BIT). The Multibit software is used to examine BIT results for assessing the results of the change and verifying the desired resolution.

NOTE: If the phase has shifted by 20 counts, then the antenna must have moved $20/256 \times (\text{wavelength of } 27\text{cm}) \sim 2.0 \text{ cm}$.

To determine the direction the assembly needs to be moved, you can follow these general guidelines. If the high-resolution phase has increased, then the ASA antenna beam structure must be rotated slightly clockwise (when viewed from above) in order to resolve the phase measurement alert or alarm. If the high-resolution phase decreased, then the ASA antenna beam must be rotated slightly counterclockwise.

3.20.10.2 Step-by-step Instructions

1. At the TLS maintenance interface in the shelter, open a File Explorer window and navigate to `base2\c:\run\bit` and open the `asa.bit` file with Multibit.
2. Select the Monitor Limits button (circled red in Figure 3-37 below) and the high-resolution phase parameter button to display the High-resolution phase data. Verify the High-resolution phase is in alert or alarm by reviewing the screen and looking for the dark data line to exceed the limits of the alert (yellow) or alarm (red) lines.



Figure 3-38 Multibit software with monitor limits and high-resolution phase selected and properly centered in the monitor limits

3. To set up your test and verify the alert or alarm can be identified, display the Maintenance Interface software and select Test Mode from the File menu.
4. On the Test, Mode window, use the BIT button to command BIT, then wait 10 seconds for BIT to complete.
5. Using File Explorer, double click again on the asa.bit file to reopen the file and verify that the last bit cycle was recorded and the high-resolution phase is in alert or alarm.
6. To realign the beam in a counter-clockwise direction at the ASA, you will rotate the beam physically in tiny steps. First, loosen the ratchet strap on the approach side (see Figure 3-38) of the beam next to the antenna with the obstruction light enough to relieve the tension on the beam. Next, tighten the ratchet strap on the departure side next to the obstruction light. You should observe a slight adjustment of the beam position to produce the desired counter-clockwise beam rotation described above. This procedure is a step-wise procedure, and it is best to start small and verify the change before making larger adjustments.

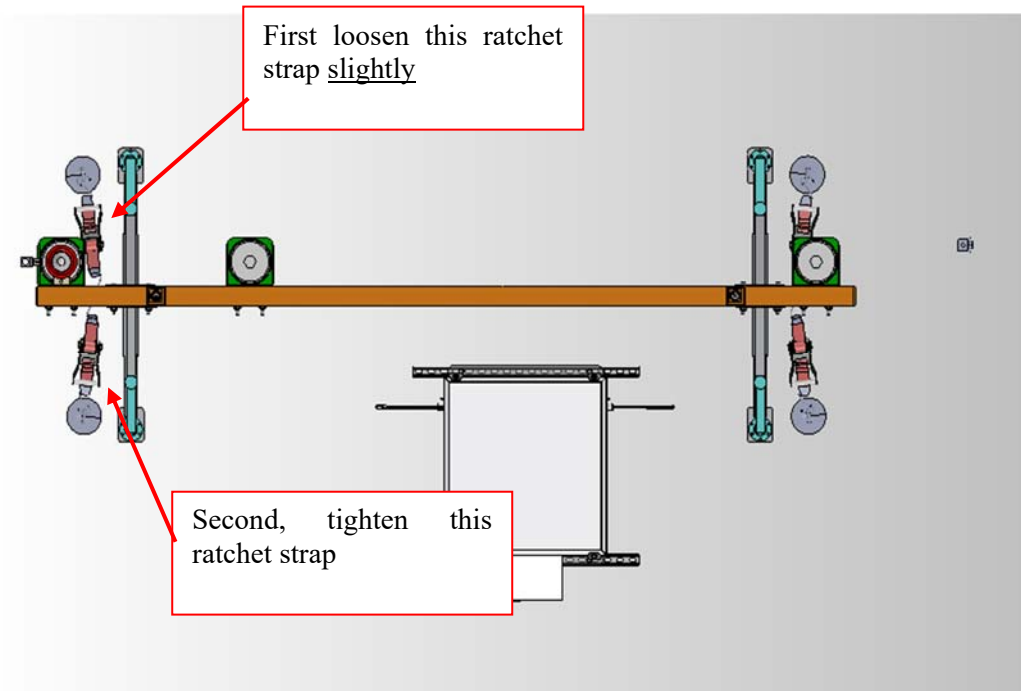


Figure 3-39 Top view of ASA sensor

7. Check to verify the BIT status for the ASA by performing the following steps:

- At the base station, command another BIT from test mode, wait 10 seconds for BIT to complete.
- Reopen the BIT file by double-clicking on the asa.bit file to reload the Multibit graphic of the bit data file.
- Compare the new values with the antenna moved slightly versus the previous value for the high phase. You should see the phase move towards the center of the Alert/Alarm limit lines.
- Repeat this process of moving the antenna and reloading the asa.bit file with the Multibit software tool until the high-resolution phase is centered within the monitor limits.

Note: If the phase values move the opposite direction and go further into alarm when the array is rotated counter-clockwise, then the antenna array should instead be rotated in the opposite direction, clockwise. This problem should take less than 30 minutes to resolve following the instructions.

3.20.11 ESA Antenna Re-alignment (Transportable)

Conduct this procedure if the ESA antenna position has moved, and the CAL/BIT elevation has not changed.

Perform the steps described in 3.19.10.2 to gather data. Adjust an ESA turnbuckle tension 2 turns and then repeat the data-gathering steps. Determine if the tower has moved closer or further to the center of the monitor limits. Make further adjustments with not more than 2 turns at one time to the first turnbuckle based on the data gathered during BIT. Make adjustments to the other 5 ESA turnbuckles as needed until the ESA phase monitor limits are well centered. Take notes on which turnbuckle was adjusted and the result in the BIT file.

This problem should take less than 60 minutes to resolve following the instructions.

3.20.12 Tilt Sensor Fault Isolation

If BIT data indicates a tilt sensor error, fault isolation must be done to determine if it is the tilt sensor, the ESA sensor, or the tilt sensor cable.

1. Swap the tilt sensor cables at the ESA sensor connector end. Run BIT a sufficient number of times to see the fault. If the fault indication is for the same tilt sensor (ex: the initial fault was parallel tilt data, and post swap fault is seen in parallel tilt data), the faulty component is the AOA sensor.
2. Return the cables at the ESA sensor end to their original locations. Swap the tilt sensor cables at the tilt sensors. Run BIT a sufficient number of times to see the fault. If the fault indication is for the same tilt sensor (ex: the initial fault was parallel tilt data, and post swap fault is seen in parallel tilt data), the faulty component is the tilt cable. If the fault moves to the other tilt sensors data, the faulty component is the tilt sensor.

3.20.13 RCU Troubleshooting Procedures

3.20.13.1 RCU Troubleshooting

If the RCU display indicates a NO-COMM status, the problem may be a communication link between the RCU computer and the Base station computers. Perform one of the procedures below based on the type of RCU installed; see Remote Control Unit Installation for set-up instructions.

1. Fiber or Twisted Pair Ethernet:

1. On the RCU, exit RCUx.
2. Verify that the communication fiber lines are connected and are not damaged.
3. Double-click the desktop shortcut to DOS Prompt.
4. At the DOS prompt, type in "ping base2" (NO QUOTES) and hit the enter key.
5. Verify that base2 responds and that ping requests do not timeout.
6. If pings are unsuccessful, close the DOS shell and contact the TLS maintenance technician. Otherwise, continue with the next step.
7. From the RCU, open the RCUx software. The RCU should display INITIALIZATION mode followed by Track.
8. If Track is displayed, the TLS system is online and ready for normal operation.
9. If these steps did not succeed in establishing an RCU link to the Base station, or the link was successful, but MAINTENANCE is displayed, the RCU Operator should contact the TLS maintenance technician.

2. Wireless Ethernet Connection:

Verify that the RCU modem is powered on and properly connected to the system by checking the following.

1. The "POWER" LED is illuminated. If not, check that the power supply is connected.
2. The 3 switches (accessible through an opening in the modem) are all pulled out toward the entrance to the opening. If not, remove the power from the modem, adjust the switch positions and re-apply power to the modem.
3. The "LINK" LEDs are illuminated. If they are not illuminated, remove the power from the modem and then re-apply power to the modem.
4. The Ethernet cable between the computer and modem is securely connected and in good condition. Tighten any loose connections and replace any damaged cables as required.

5. The coaxial cable connections between the modem and antenna are secure and in good condition. Tighten any loose connections and replace any damaged cables as required.
 6. The antenna is not damaged. If it has signs of damage, have it replaced.
 7. There is an unobstructed path between the RCU and base antennas. Any obstruction should be cleared. If clearing the obstruction is not possible, the antennas may need to be relocated.
 8. The small ends of the RCU antenna (end with the shortest elements) must be pointed toward the TLS base. Reposition the antenna if necessary.
3. **If the system is equipped with Ubiquiti radios, perform the following procedure.**
1. Verify that the POE injectors are powered on and that all ethernet cables are in good working condition.
 2. Check that the radio dishes are able to be pinged from a terminal that has a hardwired path to the dish. Use the correct IP address for the connected dish; refer to ANPC document 600-00017 (current Rev). If the dish is powered on and the dish is not responsive, contact ANPC
 3. Repeat lines 1 and 2 for each of the paired devices.
 4. If the above yields positive results, adjust the link such that the dishes are pointed in the correct direction. Use the following procedure.

To better adjust the remote links to ensure that they have the best possible signal strength given the installed locations, follow the process detailed below.

Tools required:

- Philips screwdriver
- ½ inch wrench
- Small crescent wrench
- Handheld radios
- Climbing harness for safety while on TLS shelter.
- Two technicians

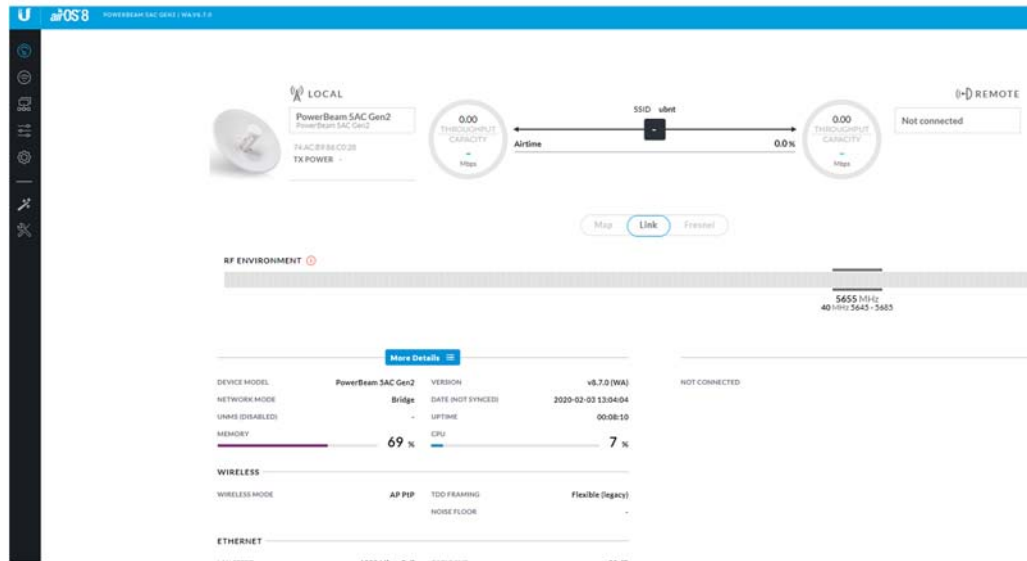
The following process will need to be completed at all installed radio dishes.

1. Open the web User Interface from one of the linked RCU's. Go to one of the following IP addresses (192.168.28.10, 192.168.28.11, 192.168.28.12, 192.168.28.13).

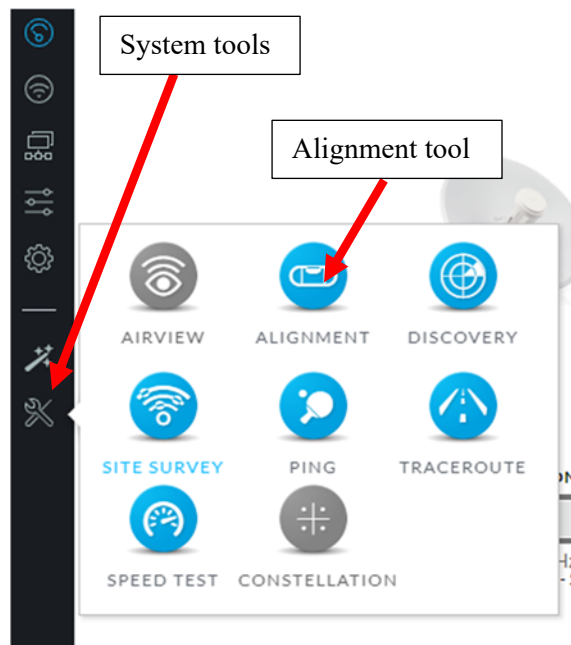
Username: administrator

Password: austinaustin

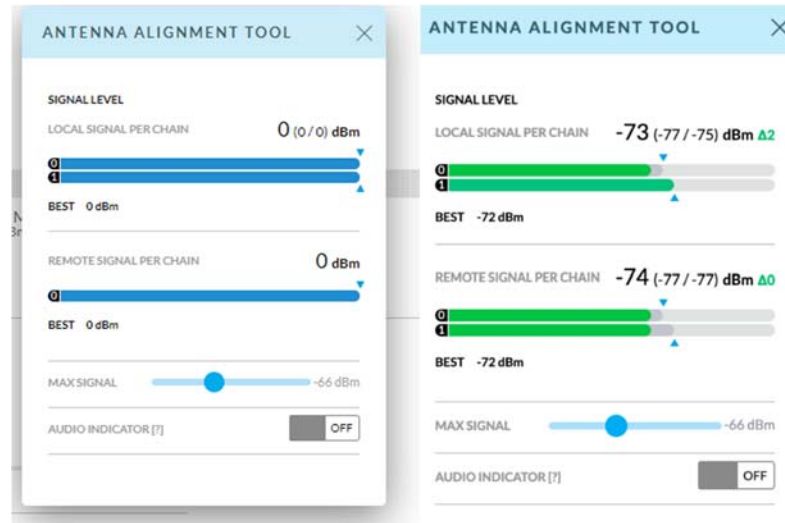
The following web page should be visible.



2. Find the tools tab in the figure above; it is on the left-hand side of the page. Another place that it may be is in the top right of the page.
3. Open the tools and open the alignment tool.



4. Now with one technician at the radio with the other monitoring the Alignment Tool. The technician will make slowly adjust the azimuth on the antenna until the signal gets improves, and then it begins to worsen.



The figure above on the left is an unpaired antenna. The figure on the right is what to expect. Since these are negative measurements, we are looking for a larger number, e.g., $81\text{dBm} < -72\text{dBm}$. In this case -72dBm is a better reading. Ideally, the deltas are as small as possible.

5. Once you notice you're losing signal, start backing up the azimuth until you reach the peak signal. Tighten all fasteners, and ensure that the signal has not dropped any during tightening.
 6. Repeat steps 4 and 5 for the elevation angle.
 7. Repeat this for all antennas installed.
4. If these steps did not succeed in establishing an RCU link to the Base station, or the link was successful, but MAINTENANCE is displayed, the RCU Operator should contact the TLS maintenance technician.

3.21 LRU Replacement

3.21.1 Field Return Material Authorization

The Field Return Material Authorization (RMA) procedure must be followed when replacing an LRU. The Field RMA procedure ensures that the LRU is tracked properly through shipping and repair, that system configuration and documentation are maintained, and that data records are completed.

1. Prior to returning a faulty LRU to ANPC, call or email ANPC Customer Service (Email FSG@anpc.com) and receive an RMA number that will allow tracking of the faulty equipment. Document the RMA number, equipment description, and serial number in the Facility Maintenance Log Book. Mark the shipping container clearly with the RMA number. Please do not return equipment without an assigned RMA number.

FSG@anpc.com / phone: 1 (541) 386-1747

2. Upon receipt of the replacement LRU, and after the equipment is installed and configured, call ANPC Customer Service and inform them that the equipment is installed. Document the information defining the new LRU in the Facility Maintenance Log Book. Documented information shall include the description, part number, revision level, and the serial number of the new LRU.

3.21.2 Base Station Rack LRU Replacement

1. Power down the TLS according to the Power Down procedure. At the main service panel breakers, ONLY turn off the rack power breakers.

WARNING

Tag the main service panel RACK circuit breaker with LOCK-OUT/TAG-OUT warning tag or device.

The UPS Battery Pack weighs 94 lbs, and the UPS weighs 69 lbs. Multiple personnel is required to install and service these components.

2. From the back of the equipment rack, ensure all cables going into the LRU being removed are properly marked as to where they are connected on the LRU, and remove each cable. Be especially careful with the fiber optic cables (if applicable).

*Maintenance technicians must remove the rack-mount brackets from an LRU before returning it to ANPC for servicing. ANPC does not provide extra brackets, so each site will need to remove and retain the brackets from the defective LRU so that they can be installed on the replacement unit.

At the front of the equipment rack, remove the four screws securing the LRU in the rack. Pull the LRU drawer to the full extension of the slides. Being careful to support the LRU drawer, release the slide catches on each slide, and pull the drawer fully out, disengaging the drawer from the slides and set aside. Return the slides to the closed position in the rack until a replacement LRU is received.

3. To install the replacement LRU in the equipment rack, first, replace the rack-mount brackets. Next, fully extend the slides out from the rack. Ensuring proper support for the drawer, insert the slide mounts on the drawer into the slides from the rack and push the drawer fully into the equipment rack. Secure the drawer in the rack with four screws through the front panel. At the rear of the rack, reconnect all cables to their proper connector, again taking extra care with the fiber optic cables.
4. After LRU replacement, power up the equipment according to the Power-Up Procedure.
5. If the replaced LRU is a Base1 or Base2, perform the following steps to update the software *.dat files.
 - a. If the replaced LRU is Base1:
 - i. On the MIU, exit the MI application if it is running.
 - ii. On the MIU, start the SendToBases tool from the c:\tools folder.
 - iii. Press the “Select All” button on the left side of the Files section.
 - iv. Press the “Select Directory” button, and navigate to the \\Base2\\c\\run folder. Press “OK.” Verify "Path of File to Send" for all files is the Base2 run folder and that all files have a valid checksum.
 - v. Next to the “Send to” label, check BASE1 and uncheck BASE2.
 - vi. Press the “Send” button at the bottom of the dialog.

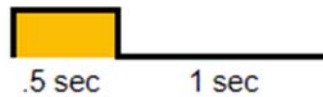
3.22 Troubleshooting Rack Switch

If the Rack Switch is suspected to be at fault, the following perform the following to determine if the Rack Switch is functional. There are two (2) possible types of failure modes, coax, and fiber. Check the LEDS located on the front of the Rack Switch, if they are blinking, determine the pattern to notice the type of fault. If there are faults replace the Rack Switch.

- 1 blink is fiber switch error
- 2 blink is rf switch error
- 3 blinks for both

The patterns are as follows:

- **One Blink**



Pattern:

On for a half-second

Interval:

Off for one (1) second

- **Two (2) Blinks**



Pattern

On for a half-second

Off for a half-second

On for a half-second

Interval:

Off for one (1) second

- **Three (3) Blinks**



Pattern:

On for a half-second

Off for a half-second

On for a half-second

Off for a half-second

On for a half-second

Interval:

Off for one (1) second

3.23 Troubleshooting Combiner

If there are Carrier Power related errors and the Combiner is suspected to be at fault, perform the following to determine if the combiner is functional.

1. Stop the system software, then run Archive.
2. Switch off the GTUs not in use to protect them from SWR
3. Select one GTU to use
4. Test Forward power with PIR, note power level.
5. Select one GTU to use, take the combiner out of the RF path
6. Turn on the GTU
7. In GTU test adjust the DSA setting as the carrier power should have increased the target range is 2200mv to 2300mv.
8. Test Forward power with the PIR
9. If the forward power changes more than 3 dB of gain vs. the previous test swap out the combiner.
10. If the forward power changes 3 dB but is still out of spec, replace the combiner and refer back to the trouble-shooting tables for further diagnostics.

3.24 Procedures Required for Return to Service

Replacement LRUs require performance validation before the TLS can be returned to service. After the configuration of the system with the new LRU(s) is verified, procedures from Chapter 6 (Installation) and Chapter 5 (Maintenance) shall be completed as defined in Table 3-9 below.

When a procedure is completed to validate system performance after an LRU replacement, copy the entire system validation data sheet (Appendix C), complete only the sections required per Table 3-9 below, sign off results, and file per requirements.

3.24.1 Calibration Status Stickers

Test equipment that requires calibration is used in many of the procedures performed to return the TLS to service. Verify that the equipment that is required to have current calibration status stickers are so marked.

After a measurement sensor, Cal/BIT, or any of the antennas or antenna cables for a sensor, the change to system phase and TOA measurement nominal values determines the tests that must be done to complete validation and return the TLS to service. Table 3-9 below details the tests required. The following steps shall be performed to determine the size of any change to phase or TOA nominal values:

3.24.2 BIT Monitor Limits Analysis

1. On the MI, start the system software. Wait for the system to transition to Maintenance or Track mode.
2. On the MI, select Test Mode from the Tools menu. From “Test Mode,” set the BIT interval for 1 minute, and allow BIT data to run for 10 minutes. Wait until BIT finishes, exit “Test Mode,” returning back to the MI main menu, and stop system software.
3. Use the MultiBIT tool to open the esa.bit file located on Base2 in the following directory “c:\run\bit.”
4. Examine the phase nominals (all) and compare them to the phase data from before the LRU fault condition was present.

NOTE: Temperature affects the phase nominals, and comparisons should be made to similar conditions within the pre-replacement data.

5. Repeat steps 3 and 4 above for the ASA using the asa.bit file located in the same directory. Close the MultiBIT tool.

6. Repeat steps 3 and 4 above for the ATA using the ata.bit file located in the same directory. Close the MultiBIT tool.
7. If the phase data indicates the need for a nominal value change (>2 counts), additional tests are required per Table 3-9.
8. If the phase data does NOT indicate the need for a nominal value change (change ≤ 2 counts from current value), do NOT change the nominal phase values.

As noted in the Introduction. The ANPC TLS is accompanied by a group of four primary manuals:

- (CM) Calibration Manual, # 020-00071
- (OM) Operations Manual, # 020-00072
- (IM) Installation Manual, # 020-00073
- (MM) Maintenance Manual, # 020-00074

Sections of each manual are referenced in Table 3-9. Effective use of this table requires that the user have the three other manuals in the four manual series at hand.

Table 3-9 LRU Performance & Return to Service Validation Procedures

LRU	Procedure Step Required for Return to Service	Section Number
ESA Sensor Electronics	Install spare sensor electronics and configure	3.21.3 (MM) 3.7.4/3.8.1.3/ 4.9 / 4.9.1 if necessary (IM)
	Set Cal/BIT amplitude limits (do not change ESA TOA or Phase limits), then go to step (a)	4.18 (IM)
ESA Antenna	Install spare ESA antenna, then go to step (a)	3.8.1 (IM)
ESA Perpendicular Tilt Sensor ESA Parallel Tilt Sensor	Install spare tilt sensor	3.8.1.2 (IM)
	Level tilt sensor to achieve +/- 0.01 degrees of last known good angle recorded by the failed tilt sensor, then go to step (a)	4.17 steps 1 – 16 (IM)
ESA Antenna Cable	Install replacement antenna cable then go to step (a)	3.8.1.3 (IM)
ASA Sensor Electronics	Install spare sensor electronics and configure	3.20.3 (MM) 3.7.4 / 4.9.1 / 4.19..2 (if necessary (IM)
	Set Cal/BIT amplitude limits (do not change ASA TOA or Phase limits) then go to step (a)	4.18 / 4.19.2 / 4.19.3 (IM)
ASA Antennas	Install spare ASA antenna then go to step (a)	3.7.2, 3 (all that apply) (IM)
ASA Antenna Cable	Install replacement antenna cable, then go to step (a)	3.7.2 steps 2 – 11 for a single antenna cable (IM)
ATA Antenna	Install spare ATA antenna, then go to step (a)	3.6.2
Steps for above LRUs:		
(a)	BIT monitor limits analysis	3.21.2 (MM)
(b)	If the phase measurements for each channel changed by less than or equal to 2 counts and the TOA near changed by less than or equal to 2 ns:	
(c)	Update facility maintenance book	
(d)	Return to service	

LRU	Procedure Step Required for Return to Service	Section Number
(e)	If phase measurement for any channel changed by more than 2 counts or if the TOA changed by more than 2ns, then:	
(f)	Sensor monitor limits	4.18 (IM)
(g)	Calibrate	2.3.3 (CM)
(h)	Setting Validation	2.4 (CM)
(i)	Conduct Periodic flight inspection	Section 3 (CM)
(j)	Update facility maintenance book	
(k)	Return to service	
Cal/BIT Transmitter		
	Install spare Cal/BIT Transmitter, then go to step (a)	3.20.3 (MM) 3.6.1 (IM)
Cal/BIT Antenna or Antenna Cable		
	Install spare antenna or antenna cable, then go to step (a)	3.6.2/4.5 step 5/ (IM)
(a)	Cal/BIT Amplitude Setting	4.19.2 (IM)
(b)	TOA F1 Tuning	4.19.3 (IM)
(c)	BIT Nominal Change Analysis	3.21.2 (MM)
(d)	If the phase measurements for each channel changed by less than or equal to 2 counts and the Differential TOA (ESA –ASA) near changed by less than or equal to 2 ns, then:	Note: the new TOA nominal value can change up to 50 ns as long as the DTOA is less than or equal to 2 ns; for example, +10ns at the ESA and +11 ns at the ASA is a DTOA difference of only 1 ns.
(e)	Update facility maintenance book	
(f)	Return to service	

LRU	Procedure Step Required for Return to Service	Section Number
(g)	If phase measurement for any channel changed by more than 2 counts or if the TOA changed by more than 2ns, then:	
(h)	Sensor monitor limits	4.18 (MM)
(i)	Calibrate	2.3.3 (CM)
(j)	Setting Validation	2.4 (CM)
(k)	Conduct Periodic flight inspection procedure	Section 3 (CM)
(l)	Update facility maintenance book	
(m)	Return to service	
GTU		
(a)	Base station rack LRU replacement	3.20.2 Steps 1 to 4 (MM)
(b)	GTU Cal, Validation, and Monitor Limits.	4.15, 4.15.2.1, 4.18 (IM) C-11 (MM)
(c)	Settings Validation	2.4 (CM)
(d)	Update facility maintenance book	
(e)	Return to service	
GTU Glide Slope or Localizer Cables		
(a)	GTU Cal, Validation, and BIT Monitor Limits.	4.15, 4.15.2.1, 4.18 (IM) C-11 (MM)
(b)	Settings Validation	2.4 (CM)
(c)	Update facility maintenance book	
(d)	Return to service	

LRU	Procedure Step Required for Return to Service	Section Number
GTU Glide Slope or Localizer Antenna		
(a)	Install antenna	3.5.2 for GS or 3.5.3 for LOC (IM)
(b)	GTU Cal, Validation, and BIT Monitor Limits.	4.15, 4.15.2.1, 4.18 (IM) C-11 (MM)
(c)	Settings Validation	2.4 (CM)
(d)	Conduct Periodic flight inspection procedure	Section 3: check service volume only/field strength (CM)
(e)	Update facility maintenance book	
(f)	Return to service	
Interrogation Transmitter		
(a)	Base station rack LRU replacement	3.20.2 Steps 1 to 4 (MM)
(b)	Interrogation Transmitter Validation	C-10 (MM)
(c)	Interrogator BIT nominal setting (set Interrogator nominal values only)	4.18 (IM)
(d)	Update facility maintenance book	
(e)	Return to service	
Interrogation Suppression Transmitter		
(a)	Base station rack LRU replacement	3.20.2 Steps 1 to 4 (MM)
(b)	Interrogation Transmitter Validation	C-10 (MM)
(c)	Interrogator BIT nominal setting (set Interrogator nominal values only)	4.18 (IM)
(d)	Update facility maintenance book	

LRU	Procedure Step Required for Return to Service	Section Number
(e)	Return to service	
Interrogation Cables or Antennas		
(a)	Replace the faulty interrogation antennas or cables	3.9.1 all that apply (IM)
(b)	Interrogator BIT nominal setting (set Interrogation nominal values only)	4.18 (IM)
(c)	Update facility maintenance book	
(d)	Return to service	
Base1		
(a)	Base station rack LRU replacement	3.20.2 step 1 to 5 (MM)
(b)	System Clock Synchronization	4.13 (IM)
(c)	System Software Startup and Setup	4.14 (IM)
(d)	Signal Test Points Panel	C-10 (MM)
(e)	Settings Validation	2.4 (CM)
(f)	Update facility maintenance book	
(g)	Return to service	
Base2		

LRU	Procedure Step Required for Return to Service	Section Number
(a)	Base station rack LRU replacement	3.20.2 step 1 to 5 (MM)
(b)	System Clock Synchronization	4.13 (IM)
(c)	System Software Startup and Setup	4.14 (IM)
(d)	Settings Validation	2.4 (CM)
(e)	Update facility maintenance book	
(f)	Return to service	
MIU		
(a)	Base station rack LRU replacement	3.20.2 step 1 to 4 (MM)
(b)	System Clock Synchronization	4.13 (IM)
(c)	Copy configuration files	3.3 (IM)
(d)	System Software Startup and Setup	4.14 (IM)
(e)	Settings Validation	2.4 (CM)
(f)	Update facility maintenance book	
(g)	Return to service	
System Software		
(a)	System Clock Synchronization	4.13 (IM)
(b)	System Software Startup and Setup	4.14 (IM)
(c)	Signal Test Points Panel	C-10 (MM)

LRU	Procedure Step Required for Return to Service	Section Number
(d)	Settings Validation	2.4 (CM)
(h)	Update facility maintenance book	
(i)	Return to service	
Network Panel, Monitor, Keyboard, Mouse		
(a)	Power on	4.7 (IM)
(b)	RCU Startup	4.12.1 (IM)
(c)	Update facility maintenance book	
(d)	Return to service	
UPS, Battery Pack		
(a)	Install replacement	
(b)	Power on through UPS Battery Operational Check	4.11 / 4.11.1 (IM)
(c)	Update facility maintenance book	
(d)	Return to service	
RCU (optional)		
(a)	Installation	3.10 (IM)
(b)	Power On	4.12.1 (IM)
(c)	Update facility maintenance book	
(d)	Return to service	

LRU	Procedure Step Required for Return to Service	Section Number
Rack Switch		
(a)	Install replacement	
(b)	Power on	
(c)	Update facility maintenance book	
(d)	Return to service	

3.24.3 Flight Inspection Prior To Return To Service

Civil and Military Aviation Authorities flight inspections should use a specially-equipped aircraft that can measure the signals radiated by the TLS and then compare them to a truth reference. Refer to Table 3-9 prior to returning TLS to service to understand the requirements for flight inspection after repair.

Appendix A Abbreviations and Definitions

Abbreviation	Definition
μs	Micro Seconds
ANPC	Advanced Navigation & Positioning Corporation
ASA	Azimuth Sensor Assembly (previously called LAOA)
ATA	Azimuth Time-Of-Arrival (TOA) Antenna
ATC	Air Traffic Control
BIT	Built-In-Test
Cal/BIT	Calibration/Built-in-Test
CNTL	Control
CPU	Central Processing Unit
dB	Decibel
DDM	Differential Depth of Modulation
ESA	Elevation Sensor Assembly (previously called GAOA)
FTM	Flight Test Monitor
GA	General Aviation
GAOA	Glide slope Angle-of-Arrival (previous term for ESA)
GS	Glide slope
GTU	Guidance Transmitter Unit
Hz	Hertz
ID	Identification
IF	Intermediate Frequency

Abbreviation	Definition
ILS	Instrument Landing System
IM	Integrity Monitor
KVM	Keyboard Video Mouse
LAOA	Localizer Angle-of-Arrival (previous term for ASA)
LOC	Localizer
LRU	Line Replaceable Unit
nm	nanometer
MHz	Mega Hertz
MI	Maintenance Interface
MIU	Maintenance Interface Unit
nSec	Nanosecond
PM	Preventive Maintenance
PRC	Phase Receiver Card
PRM	Phase Receiver Module
RCU	Remote Control Unit
RF	Radio Frequency
RMA	Return Material Authorization
RSA	Runway Safety Area
RSC	Rack Switch Control
SCI	Software Configuration Item
SDM	Sum of Depth of Modulation
TLS	Transponder Landing System
TOA	Time-of-Arrival
UHF	Ultra High Frequency
UPS	Uninterruptible Power Supply
UPS-R	Rack UPS
UPS-S	Sensor UPS
VAC	Volts Alternating Current
VHF	Very High Frequency
VMC	Visual Meteorological Conditions

Appendix B Technical Performance/Periodic Maintenance Records

The following record sheets, TLS Technical Performance Record and TLS Periodic Maintenance Record are intended to be used as master copies from which to make working copies of the records for use during required maintenance and checks.

The following Periodic Maintenance records are included in this appendix:

Table C-1 - Data Review and Recording	Datasheet provided for recording data nominals as part of the Data Archive PM
Table B-2 - RCU functions check and monitor shutdown validation	Datasheet provided for recording data nominals as part of the RCU functions check
Table B-3 - Critical Area and Obstruction Light	Datasheet provided for inspection and repair of cables and obstruction lights
Table B-4 - UPS Output voltage and battery check	Datasheet provided for recording UPS validation data as part of the UPS output voltage and battery check
Table B-5 - Air Filters	Datasheet provided for recording UPS validation data as part of the UPS output voltage and battery check
C-6 - Guy Wire Tension	Datasheet provided for recording guy wire tension data
Table B-7 - RSC Periodic Maintenance	Datasheet provided for 90-day rack switch control system inspection
Table B-8 - Antenna System Maintenance	Datasheet provided for recording condition of the antenna system
Table B-9 - Clean and Inspect Equipment	Datasheet provided for recording cleaning and inspection of equipment
C-10 - Interrogation Transmitters Check	Datasheet provided for recording Interrogation Transmitters validation data as part of the Interrogation Transmitters Check preventive maintenance
Table B-11 - GTU Check	Datasheet provided for recording GTU validation data as part of the GTU Check preventive maintenance
Table B-12 - Cable Repair and Weatherproofing	Datasheet provided for recording cable repair and weatherproofing data

Customer Initials _____

ANPC Initials _____

Table B-1 Data Files Archive

Archive of Data Files		TLS Facility Identifier:
Location:	Date/Time:	Technician:

Para	Test	LRU	Indication of Problem
3.3.2- 4	Review of log files for indications of system problems.		

Para	Test	Current Nominal	Data Range	Notes / Remarks
3.3.2-7 Interrogation 3.3.2-11	P1 Amp (mV)			
	P2 Amp (mV)			
	High Phase (cnts)			
ESA	Med Phase (cnts)			
	Low Phase (cnts)			
	High Amp (mV)			
	Med Amp (mV)			
	Low Amp (mV)			
	Ref Amp (mV)			
	TOA Near (ns)			
	Frequency (MHz)	1090		
	Tilt Para (deg)			
	Tilt Perp (deg)			

Customer Initials _____

ANPC Initials _____

Para	Test	Current Nominal	Data Range	Notes / Remarks
3.3.2-12 ASA	High Phase (cnts)			
	Low Phase (cnts)			
	High Amp (mV)			
	Low Amp (mV)			
	Ref Amp (mV)			
	TOA Near (ns)			
	Frequency (MHz)	1090		
3.3.2- 13 ATA	Medium Amp (mV)			
	TOA Near (ns)			
3.3.2-15 GTU Uplink	Localizer DDM (all four tests)	0		
	Localizer SDM (all four tests)	40%		
	Localizer carrier (all four tests) (mV)			
	Localizer reflected power (mV)			
	glide slope DDM (all four tests)	0		
	glide slope SDM (all four tests)	80%		
	glide slope carrier (all four tests) (mV)			
	glide slope reflected power(mV)			

Para	Test	Verified	Units	Notes / Remarks
3.3.2-19	Chkfsys verification of Base1		✓	
Remarks				
Review		Date		

Customer Initials _____

ANPC Initials _____

Table B-2 RCU Functions Check/Monitor Shutdown Validation

RCU Functions Check/Monitor Shutdown Validation		TLS Facility Identifier:			
Location:		Date/Time:	Technician:		
Para	Test	Min	Measured	Max	Units
3.4.3 -1	Verify Track indication and Green Screen at RCU.				✓
3.4.3 -2	Verify transition to Acquire state and White screen at RCU.				✓
3.4.3 -3	Verify successful reset from Acquire				✓
3.4.3 -6	Verify MI Indication of Maintenance Mode				✓
3.4.3 -7	Verify Maintenance Mode indication and Red screen at RCU.				✓
3.4.3 -10	Verify MI Indication of Track Mode				✓
3.4.3 -11	Verify Track indication and Green screen at RCU.				✓
Remarks					
Review		Date			

Customer Initials _____

ANPC Initials _____

Table B-3 Critical Area and Obstruction Light

	TLS Facility Identifier:		
	Location:	Time:	Technician:
	Walk/Inspect and address designated critical area perimeter for objects, vegetation, snow higher than 1 foot for a 15' foot radius around the Cal/BIT and ATA. Do this for a 40' radius around ESA, Uplink, and Base Shelter assemblies	Yes	No
	Glide path angle is less than 2.5°, and critical area is maintained to max. height of 2 inches?	Yes	No
	Clear line of sight between Cal/BIT and sensor antennas.	Yes	No
	Objects cut, shoveled, or removed	Yes	No
	Ob-light Clean/Clear and working	Yes	No
	Ob-light replacement	Yes	No
	Signage/markings in Place	Yes	No

Customer Initials _____

ANPC Initials _____

Table B-4 UPS Output Voltage/Battery Check

Para	Test	Min	Measured	Max	Units
3.6.5 - 1	UPS-R Output voltage	108		132	VAC
3.6.5 - 5	Battery auto-test pass				✓
3.6.5 - 6	System time on UPS-R battery	20			Minutes
3.6.5 - 7	UPS-R Output voltage	108		132	VAC
3.6.5 - 1	UPS-S Output voltage	108		132	VAC
3.6.5 - 5	Battery auto-test pass				✓
3.6.5 - 7	UPS-S Output voltage	108		132	VAC
Remarks					

Table B-5 Air Filters

Air Filter Check						TLS Facility Identifier:			
Location:				Date/Time:		Technician:			
Interrogation Transmitter		ECU		GTU		Base1		Base2	
Replace Front Panel Foam Filter		Replace Foam Filter		Replace Front Panel Foam Filter		Replace Front Panel Foam Filter		Replace Front Panel Foam Filter	
Yes	No	Yes	No	Yes	No	Yes	No	Yes	No

Customer Initials _____

ANPC Initials _____

Table B-6 Guy Wire Tension

Guy Wire Tension		TLS Facility Identifier:	
Location:	Date/Time:	Technician:	
Top Guy Wire Tension: 660 pounds	Yes	No	
Bottom Guy Wire Tension 400 pounds	Yes	No	
ESA Tower Tilt within Limits; ± 0.05 degrees of the nominal setting but no more than ± 0.1 degrees of nominal	Yes	No	
If no, has ESA Tower Tilt been adjusted to appropriate tension and tilt values	Yes	No	
Turnbuckles have at least 1" of threads exposed inside of the coupler	Yes	No	

Customer Initials _____

ANPC Initials _____

Table B-7 RSC Periodic Maintenance

Rack Switch Controller (RSC)		Airfield Designator:		
180 day Periodic Inspection and Performance Review This procedure assumes the current Rack is A		Technician:		
		Date:		
Step	Procedure	Complete	Pass	Fail
1	Notify the appropriate airfield authority that the TLS is going out of service for a 10 minute maintenance procedure			
2	Ensure both racks are powered up			
5	In the Control menu on the RCU, select Switch Racks			
6	Verify the system software is automatically started on the alternate Rack			
7	Check the Rack Switch and UPS-S status. Verify the RS status is green and shows Rack B is the current Rack. Verify the UPS-S status is green, and has 20 minutes remaining.			
11	Verify Rack A is displayed as the current rack in the RCU's LRU status bar, and that the RS LRU is green			
12	Notify the appropriate airfield authority that the TLS maintenance procedures are complete and system is back in service			

Customer Initials _____

ANPC Initials _____

Table B-8 Antenna System Maintenance

Antenna System Maintenance		TLS Facility Identifier:	
Location:		Date/Time:	Technician:
Para.	Physical Inspection		Pass Fail
Physical Inspection of System			
Damage or deterioration			
Corrosion			
Loose bolts			
Inspection of Antenna Cables for kinks, cuts, twists			
Connections Corrosion and waterproof seal			
Deicing equipment warm to the touch?			
Inspect for between heater power pins A and B is 45 ohms for all heated antennas			

Table B-9 Clean and Inspect Equipment

Clean and Inspect Equipment		TLS Facility Identifier:	
Location:		Date/Time:	Technician:

Customer Initials _____

ANPC Initials _____

Table B-10 Interrogation Transmitters Check

Interrogation Transmitters Check		TLS Facility Identifier:			
Location:		Date/Time:		Technician:	
3.12	Signal Test Points				
	T0 Pulse Width	375		425	nSec
	T0 Amplitude	2		5	V
3.12.3 - 5	Cal/BIT CNTL Pulse Width	425		475	nSec
	Cal/BIT CNTL Amplitude	2		5	V
3.12.3 - 7	INT CNTL Pulse Width	775		825	nSec
	INT CNTL Amplitude	2		5	V
3.12.3 -9	Cal/BIT F1 Delay from 6.3.21 - 18	-0.050		+0.050	μSec
3.12.3 -13	T0 to P1 Delay			25	nSec
3.12.3 -15	T0 Period				✓
3.12.4	Interrogator Transmitter Output Verification				
3.12.4 2-3	Int. Trans. P1 Pulse Width	700		900	nSec
	Int. Trans. P1 Amplitude				mV
	Int. Trans. P3 Pulse Width	700		900	nSec
	Int. Trans. P3 Amplitude				mV
3.12.4 -3	Int. Trans. P1 to P3 Spacing	7800		8200	nSec
3.12.4 -12	Int. Trans. T0 to P1 Spacing			500	nSec

Customer Initials _____

ANPC Initials _____

Para	Test	Min	Measured	Max	Units
3.12.4 .1	Suppression Pulse Characteristics				
3.12.4 .1-3	P2 Trans. Pulse Width	700		900	nSec
	P2 Trans. Amplitude				mV
3.12.4 1-2	T0 to P2 Delay	2000		2500	nSec
3.12.4 1-4	P1 to P2 Delay = T0 to P2 Delay – T0 to P1 Delay = _____ - _____ = _____				
	P1 to P2 Delay	1850		2150	nSec

Customer Initials _____

ANPC Initials _____

Table B-11 GTU Check

GTU Check				TLS Facility Identifier:		
Location:		Date/Time:		Technician:		
Para	Test	Min	Measured	Max	Units	
GTU Tests						
3.13 1 -6	Localizer Carrier Monitor level	2.0		2.5	V	
3.13 1 -6	Glide slope Carrier Monitor level	2.0		2.5	V	
3.13	GTU calibration complete				✓	

Para	DDM Commanded	Mon. SDM	Mon. DDM	PIR SDM	PIR DDM
3.13 1 -13	Localizer +0.20				
Localizer SDM Spec 40 +/- 1 % DDM Spec In +/- .002	Localizer +0.18				
	Localizer +0.15				
	Localizer +0.12				
	Localizer +0.09				
	Localizer +0.06				
	Localizer +0.03				
	Localizer +0.00				
	Localizer -0.03				
	Localizer -0.06				
	Localizer -0.09				
	Localizer -0.12				
	Localizer -0.15				
	Localizer -0.18				
	Localizer -0.20				

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Para	DDM Commanded	Mon. SDM	Mon. DDM	PIR SDM	PIR DDM
3.13 1-14	glide slope +0.24				
glide slope SDM Spec 80 +/- 2 % DDM Spec In +/- .004	glide slope +0.22				
	glide slope +0.20				
	glide slope +0.18				
	glide slope +0.15				
	glide slope +0.12				
	glide slope +0.09				
	glide slope +0.06				
	glide slope +0.03				
	glide slope +0.00				
	glide slope -0.03				
	glide slope -0.06				
	glide slope -0.09				
	glide slope -0.12				
	glide slope -0.15				
	glide slope -0.18				
	glide slope -0.20				
	glide slope -0.22				
	glide slope -0.24				

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Table B-12 Cable Repair and Weatherproofing

Antenna System Maintenance				
Section	Location:	Date/Time:	Technician	
	Physical Inspection of System		Pass	Fail
	Cut/Sliced Cable		Replace Cable	
			Yes	No
	Waterproofing		Replace Waterproofing	
			Yes	No

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Table B-13 Generator periodic service record

Date	Hours meter	Service Description	Location	Technician

Appendix C Heated Antenna Control

D .1.1 Description

The antenna heater controller supplies AC power to the heated antennas in the TLS. When the potential for ice build up on the antenna exists, the heater controller will turn power on to the antenna heaters. The heater controller keeps the power on until the weather changes and ice formation is no longer expected.

As shown in the figure below, the antenna heater controller consists of two main parts. The large enclosure on the right is the power switch, which houses the AC power switches, and is typically mounted inside the TLS shelter. The small grey box on the left is the rain and snow sensor controller, which monitors the weather and turns the AC power switches on and off. It must be installed outside of the TLS shelter.

D .1.2 No Protection against Overheating

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Note that the antenna heater controller does not provide protection against overheating of the antennas. This protection is by adjusting the power using the variable power controller described below.



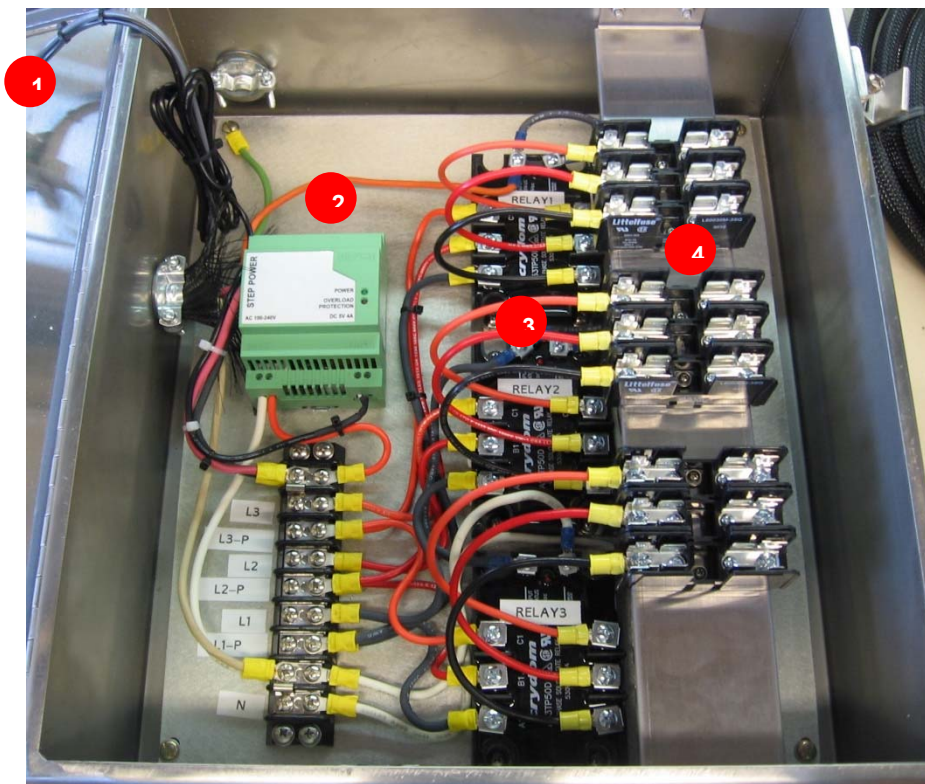
Figure C-1 Antenna Heater Controller

D .1.3 Power Switch

The power switch consists of three 3 Φ solid state relays, a 5V, 4A AC to DC power supply and a display unit. These components are mounted into a 14"x16" NEMA 4 rated (rain proof) stainless steel enclosure. A parts list for the key components is provided in the table below.

Table C-1 Power Switch Components

Item	Description	Quantity	Manufacturer	Manufacturer Part Number
1	Enclosure	1	Hoffman	A1614CHNFSS
2	Solid state relay, 3 phase, 530VAC, 50A	3	Crydom	A53TP50D
3	Power supply, 100 to 240 VAC, 5VDC at 4A	1	Phoenix Contact	STEP PS 100-240AC/5DC/4 part # 2938918
4	De-icing controller display panel	1	Automated Systems Engineering	DP-7B
5	Three phase fuse holders	3	Littelfuse	M6-10R/SX6



1. Display Panel
2. Power Supply
3. Solid State Relays
4. Fuse Holders

Figure C-2 Power Switch Interior

D.1.4 Power Switch Wiring

Figure C-3 shows how these components are electrically connected. Each of the main three phase power inputs (L1, L2, and L3) is connected to the inputs (A1, B1, C1) of the relay 2 and to the inputs of the variable power controller. The pulsed outputs from the variable power controller (L1-P, L2-P, L3-P) are connected to relays 1 and 3. The outputs from each of the relays (A2, B2, C2) are connected to the antenna heaters through 15 A fuses. The enclosure housing is connected to the AC power ground. The neutral connection is common to all circuitry.

Phase L3 is also connected to the AC/DC power supply input. The 5VDC output from the AC/DC power supply provides power to the display panel. Phase L3 also provides power to the sensor controller. When

triggered, the sensor controller powers up the control inputs on each of the relays simultaneously. Hence, they are either all-on or all-off.

The main AC inputs (L1, L2, and L3) connect to the variable power controller inputs where they pass through three 30 A fuses and solid state circuits. The fuses are very fast acting and intended to protect the solid state circuits from permanent damage when there is too much load on the controller outputs (L1-P, L2-P, and L3-P).

The 3 phase AC Power enters through an opening in the bottom of the power switch enclosure. A large opening in the back of the enclosure is provided for routing the wiring that goes from the antenna side of the 15 A fuses to the antenna heaters. The opening in the top of the enclosure is for connecting a 3 phase surge protector to the power lines (L1, L2, and L3).

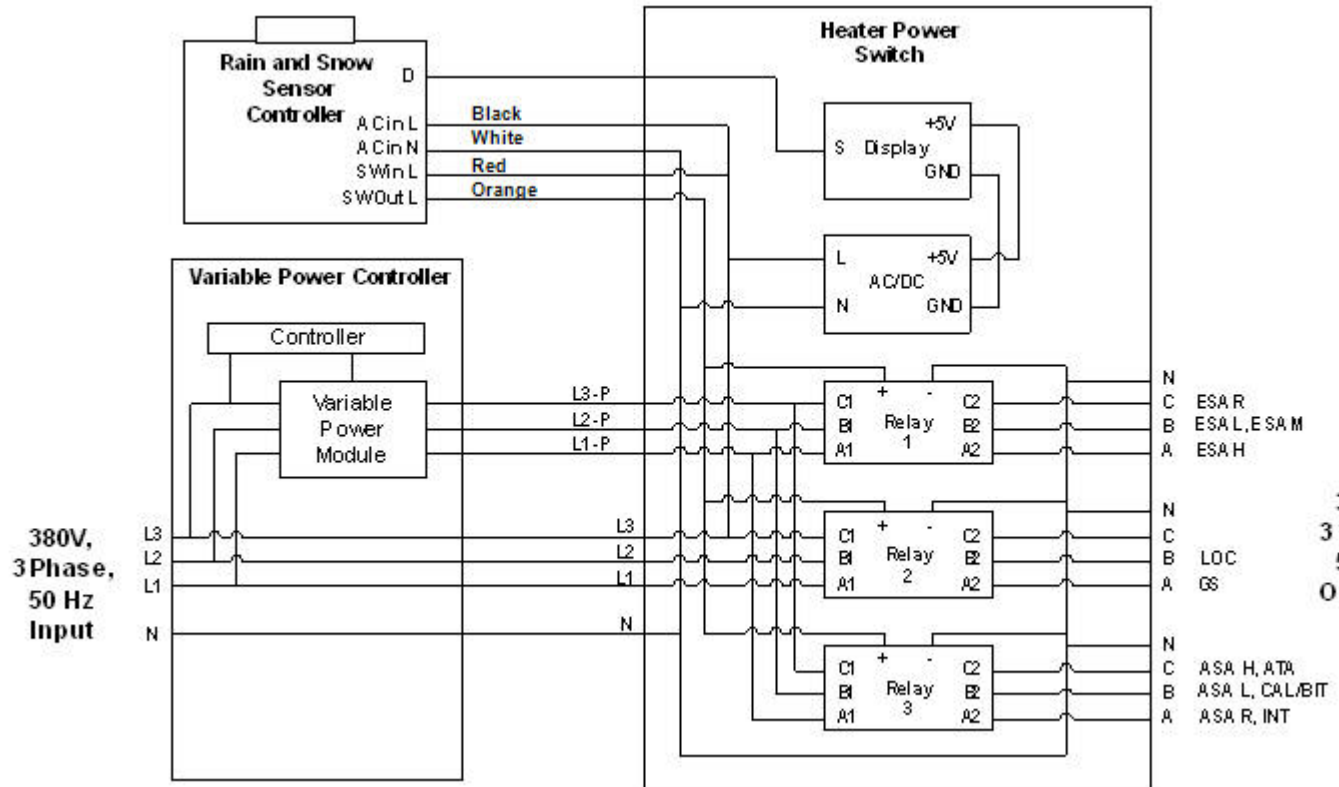


Figure C-3 Antenna Heater Controller Wiring

D.1.5 Display Panel

The display panel shown in Figure C-4 has been integrated into the front of the power switch enclosure. Normally it should be set to automatic and LEDs are illuminated when the display panel is powered on (POWER), and when AC power is switched to the antennas (DEICE ON).

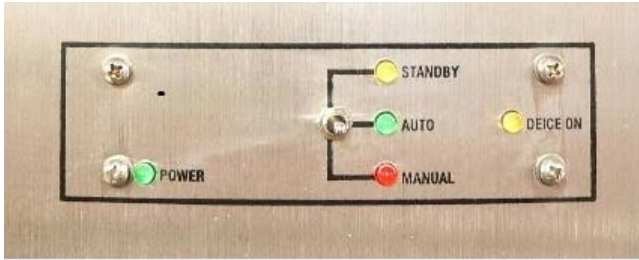


Figure C-4 Display Panel

D.1.6 Rain and Snow Sensor Controller

The sensor controller is a model DS-2B manufactured by Automated Systems Engineering. It consists of a precipitation sensor and a temperature sensor. It should be mounted outside with the round precipitation sensor having a clear view of the sky, in a location where snow and ice will not bury it. The temperature sensor is under the rubber boot at the bottom of the sensor controller. The switch on the right side is redundant with the switch on the display panel, so it is not used and has been disconnected.



Figure C-5 Sensor Controller

D.1.7 Sensor Controller

In general the sensor controller is designed so that it will power up the antenna heater relays when the temperature falls below a specified set point and snow is detected. There are several adjustments that can be made so it is imperative that the requirements for the installation site be understood and then the unit be tweaked to best meet these requirements.

An outdoor-rated cable comes permanently attached to the sensor controller. This cable is connected between the power switch and sensor controller. If the length of the cable needs to be reduced it is possible to remove the wire connections inside the rack switch, trim them to the desired length and then reconnect them. Figure C-3 shows the AC power connections between the power switch and the sensor controller.

Appendix D OPERATIONAL THEORY

2) Surveillance Mode

The system incorporates SSR interrogation schemes to identify and track aircraft within the service volume. The system can resolve and identify the transponder codes from synchronous replies. It maintains a position track for the transponder codes of interest.

a) Surveillance Measurements

In surveillance mode, the system uses Time-of-Arrival (TOA) measurement information and mode C transponder replies to determine the position of aircraft. Range and a rough azimuth calculation are based on the TOA of the transponder reply pulses. Mode C transponder replies are used for gathering altitude information.

i) Elapsed Time

The amount of time that elapses between the transmission of the interrogation signal and the reception of the transponder reply can be used to determine the aircraft's range. Accounting for the time duration of the aircraft's transponder to respond to the interrogation and delays in the receiver, the amount of time it takes a signal to travel to the aircraft and back, can be calculated. Halving this amount and knowing the propagation speed of the signal (light speed!) gives the aircraft's range.

ii) TOA Elliptical Measurement

Since the interrogator-aircraft-receiver pairing forms the elements of an ellipse, this is referred to as the TOA Elliptical measurement. Using two receiving antenna arrays on either side of the interrogator gives the system the ability to refine its calculations. Differing spacing between the interrogator and each antenna array corresponds to overlapping ellipses of different sizes. Using an additional TOA measurement taken perpendicular to the line formed by two arrays and the interrogator allows the system to determine from which side of the system the aircraft is approaching, and identifies the range and azimuth of the approaching aircraft.

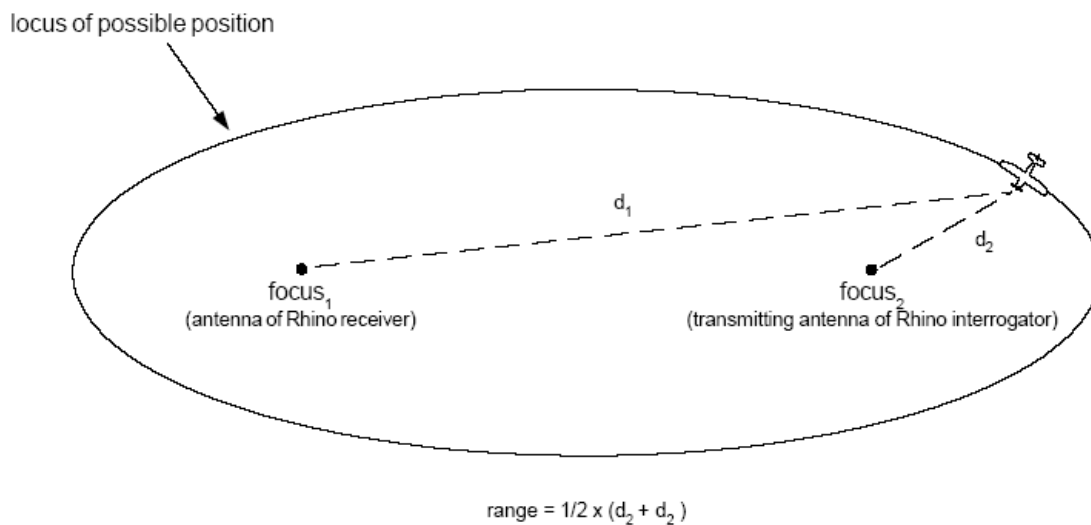
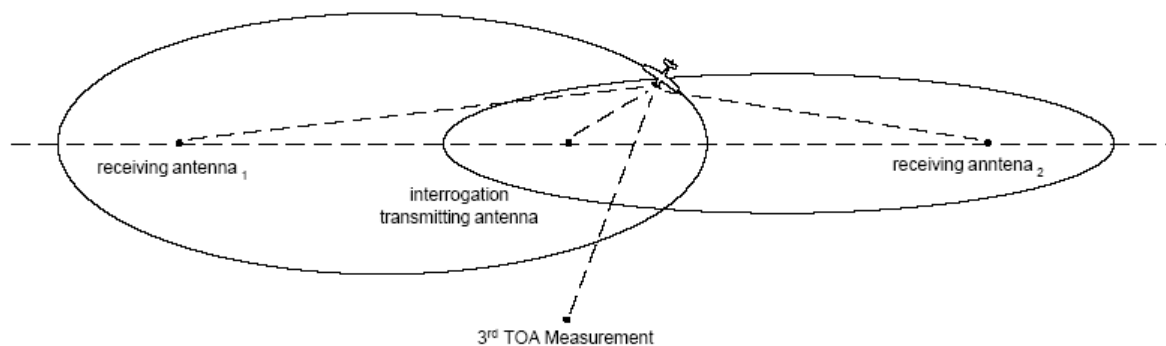


Figure D-1 Raw TOA Measurement Focus of Possible Position

- iii) The points at which the two ellipses intersect give two potential range solutions. By using an additional TOA measurement from the ATA that is perpendicular to the line formed by the ASA and ESA, the software determines which side of the system the aircraft is approaching from and identifies the range and azimuth of the approaching aircraft.



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Figure D-2 TOA Elliptical Measurement

iv) TOA Hyperbolic Measurement

The difference between TOA (DTOA) measurements gathered at the receiving antenna arrays is called to TOA Hyperbolic measurement. Most importantly, common measurement variables between the two elliptical measurements cancel out. Common to both is the transponder encoding delay and any jitter on the interrogation trigger.

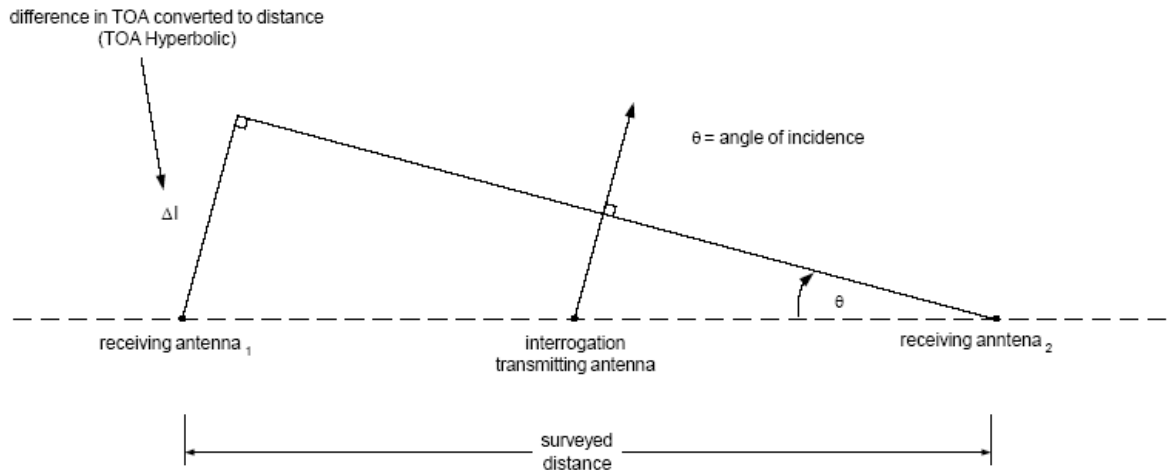


Figure D-3 TOA Hyperbolic Measurement

3) Guidance Subsystem

The guidance subsystem serves as the system's interface with the aircraft. The system offers two methods of providing guidance to the aircraft: guidance via Ground Controlled Approach (GCA) and guidance by the transmission of ILS-like guidance signals.

a) GCA- ILS

In the GCA display, the system presents a Precision Approach Radar (PAR) display format to a GCA controller, enabling the controller to execute a ground controlled approach by communicating directly with the pilot. For ILS, the system transmits emulated ILS guidance signals to the aircraft (i.e., localizer and glide slope valid for the tracked aircraft position). Instead of transmitting an interference pattern from static beams like conventional ILS, the TLS guidance subsystem synthesizes a guidance signal based on the aircraft's position. The aircraft's receiver decodes the guidance information and displays it on the course deviation indicator (CDI). From the perspective of the aircraft's receiver, the resulting signal has identical characteristic of a conventional ILS signal.

b) Sensor Array Design

The differing spacing between the measurement antennas and the reference antenna for each array allow the system to determine the location of the aircraft. The distances between the pair are determined from the relationship between the wavelength of the transponder reply, λ , and the desired cycle size Θ , for the high, medium (if applicable) and low accuracy measurements.

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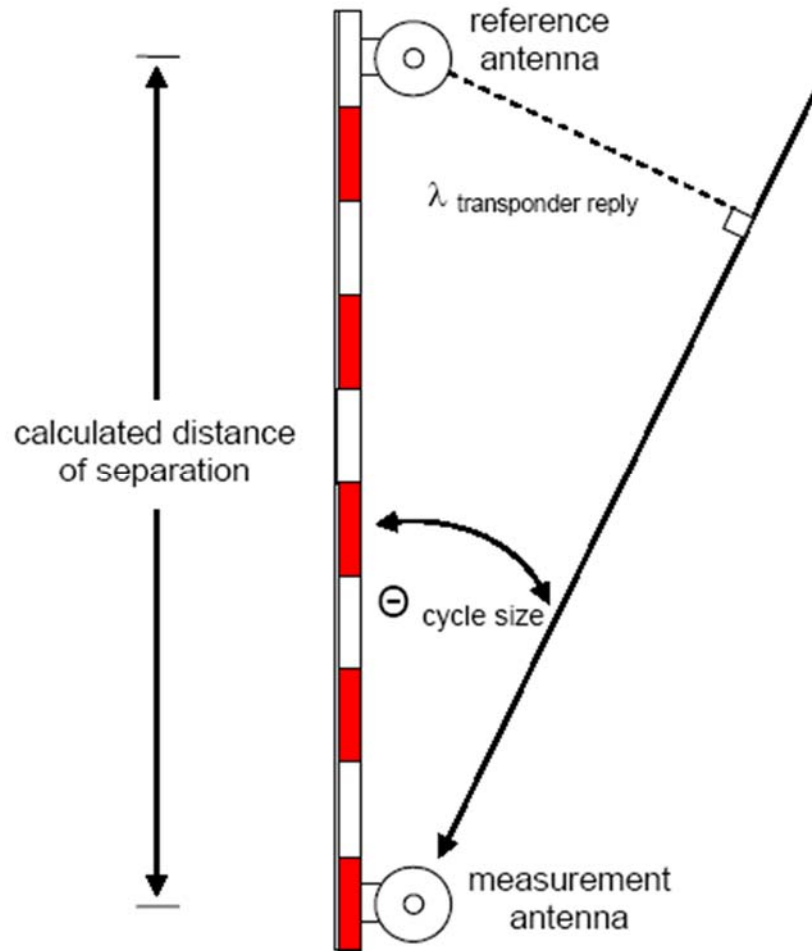


Figure D-4 Designing Separation Distance for Antenna Pairs

Table E-3-10 Cycle Size: ASA/ESA

Cycle Size	ASA	ESA
Low accuracy pair	21°	16°
High accuracy pair	5.17°	2.88°

- c) Desired cycle size is a product of three major factors: selecting a cycle size that will meet the specified accuracy requirements, selecting a cycle size that is broad enough to span the system's service volume, and physical constraints (i.e., "footprint") of the system's components. For example, if the desired low accuracy cycle size for the ESA is 16° and the wavelength of the transponder reply signal is 0.275 meters, the distance between the LOW and the REF antennas will be approximately 39 1/4".

d) Sensor Measurements

Each receiving antenna array has two or more antenna pairs. For each pair, one antenna behaves as the reference antenna, taking a measurement of the signal's time of arrival and carrier frequency. The other antenna, the measurement antenna, provides the system with a phase measurement.

i) Transponder Reply Signal

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When a transponder reply signal is received, each measurement will be a fraction of the desired cycle size. Comparing two or more pairs, a location (azimuth and elevation) can be calculated without ambiguity.

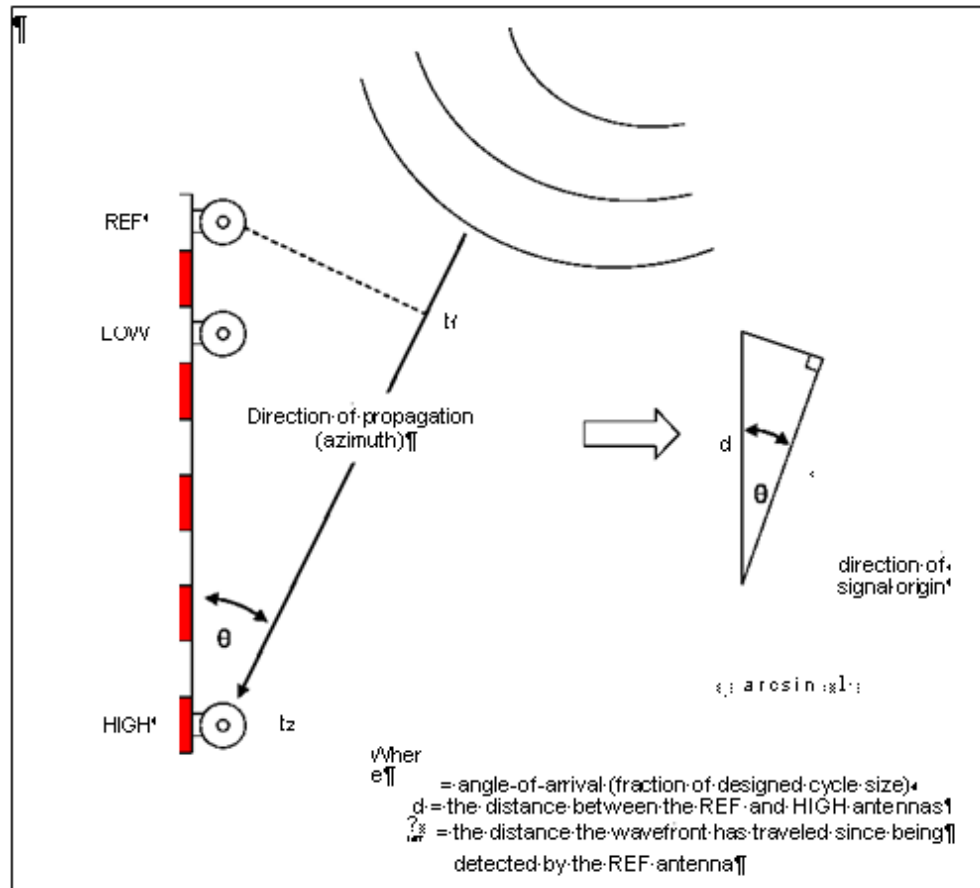


Figure D-5 AOA Measurement for the ASA (seen from above)

ii) Sensor Measurement

The sensor measures $\Delta \ell$ indirectly by measuring the difference in phase, $\Delta \phi$, between the signal at the reference antenna and the measurement antenna. Conversion of this measurement to a length is accomplished by using the following equation:

$$\Delta \ell = c (t_2 - t_1) = \lambda (K + \Delta \phi / 2\pi)$$

Where -

- λ is the wavelength of received signal (transponder reply)
- $\Delta \phi$ is the differential phase (range: $0 - 2\pi$)
- k is the integer number of wavelengths (cycle number)
- c is the speed of light

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The cycle number increments whenever the angle of arrival increases to the point where the difference in path length is the same length as the wave length of the transponder reply frequency. As a result, Δl will be expressed as some multiple of whole wavelengths (starting from zero) plus any fraction.

iii) Angle of Incidence

The Δl is converted into a distance measurement. Combined with the known separation of the antennae, the angle of incidence relative to the array can be calculated. Knowing the angle of incidence allows the direction of propagation of the reply signal to be determined. The spacing of the antennas will determine the resolution of the measurement and the measurement cycle span in degrees.

iv) Planar Wave Front

because of the range of the aircraft when compared to the distance separating the reference and measurement antennas, the wave front is assumed to be planar. One antenna array, the ASA, has three antennas arranged horizontally. The other, the ESA, has four antennas arranged vertically.

e) ESA Antenna Spacing and Measurement Uncertainty

The ESA's glide slope estimate is broken into two primary portions: an initialization stage and a track stage. The initial glide slope estimate is computed based on the low accuracy elevation estimate and time of arrival information. This measurement is refined using the high accuracy elevation estimate before guidance can be provided. A typical installation also includes a medium resolution antenna (MED). This antenna is located just below the HIGH measurement antenna and is used to improve the accuracy of the high accuracy measurement

- REF to LOW spacing: 39 ¼ inches (99.7 cm) → 16°
- REF to MED spacing: 119 inches (302.3 cm) → 5.2°
- REF to HIGH spacing: 216 inches (548.6 cm) → 2.9°

i) Unambiguous Angle-Of-Arrival vs. ESA Antenna Spacing

Figure 11-6 shows the full range of unambiguous angle-of-arrival resolution for various antenna spacing. From this we see that for the ESA spacing the low resolution system is unambiguous up until 16° while the high resolution system is unambiguous to 2.88°. Each of these unambiguous regions is referred to as a cycle.

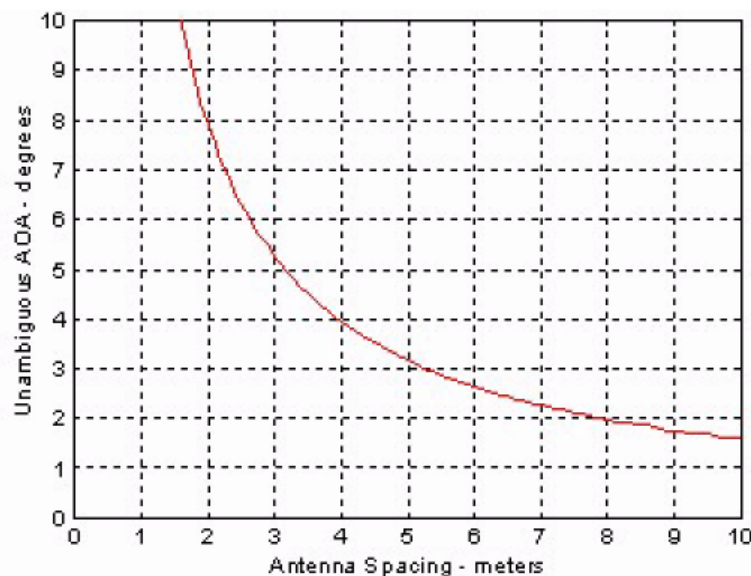


Figure D-6 Unambiguous AOA vs. ESA Antenna Spacing (Cycle Size)

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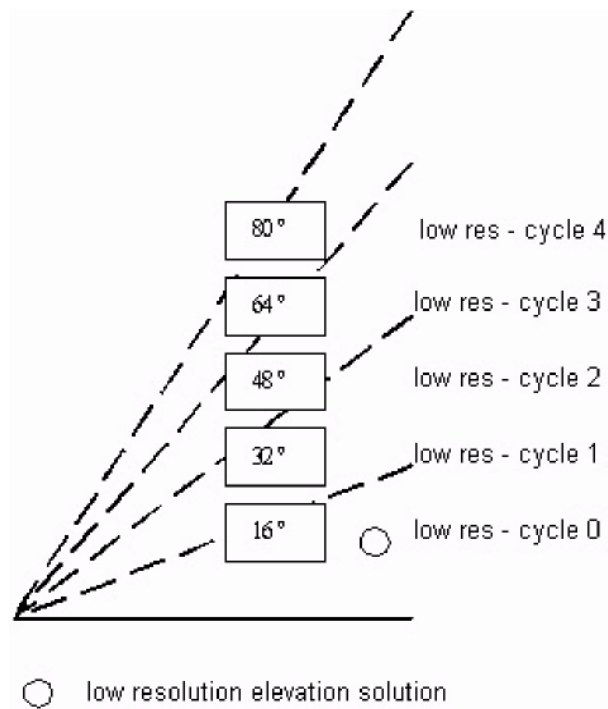


Figure D-7 Initial ESA Elevation Solution

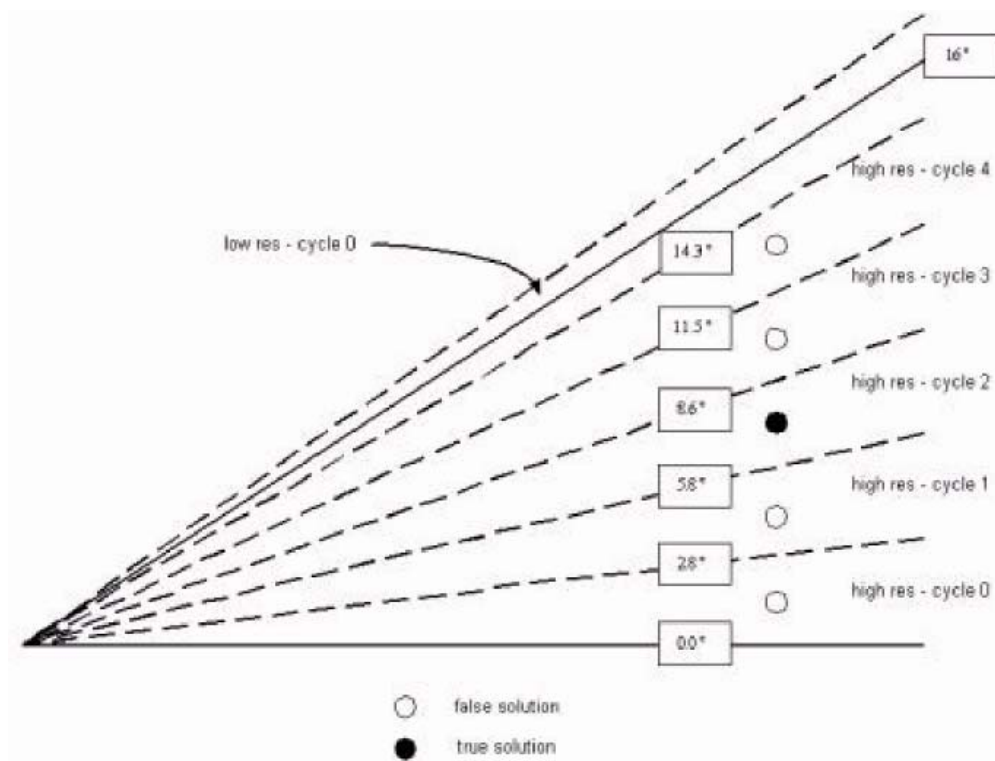


Figure D-8 ESA High Resolution vs. Low Resolution Spacing

f) ASA Antenna Spacing and Measurement Uncertainty

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The system estimates aircraft azimuth position using a sensor similar to the one shown above in Figure 11-5. An example of an array installation, where the sensor measures the difference in phase of the aircraft's transponder response, the computed angle-of-arrival is shown above in Section Figure 11-5 above.

i) ASA Measurements: Three Primary Portions

The ASA measurements are broken into three primary portions: a time-of-arrival (TOA) initialization; an AOA initialization stage; and a track stage. The initial azimuth estimate is computed based on a low resolution pair of antennas with phase centers spaced 30 inches apart while the high accuracy azimuth estimate is computed based on a wider spaced pair of antennas 120 inches apart.

ii) Unambiguous Angle-of-Arrival Resolution

Figure 11-9 shows the full range of unambiguous angle-of-arrival resolution for various antenna spacing. For the LOW-REF spacing, the system is unambiguous up until 21° while for the HIGH-REF spacing, the system is unambiguous to 5.17° . Each of these unambiguous regions is referred to as a cycle.

- 30 inches = 0.762 meters $\rightarrow$ cycle size for low is 21 degrees
- 120 inches = 3.048 meters $\rightarrow$ cycle size of high is 5.17 degrees

iii) Low Resolution Cycle for the ESA

Figure 11-10 shows the ASA low resolution cycles centered on runway centerline. Figure 11-11 illustrates the overlap of the cycles for the low and high resolution antennas. This diagram shows the several high resolution cycles within one low resolution cycle (cycle 0). Four solutions for the high estimate are found. The one that is $\pm \frac{1}{2}$ high cycle (2.58 degrees) of the low solution is selected.

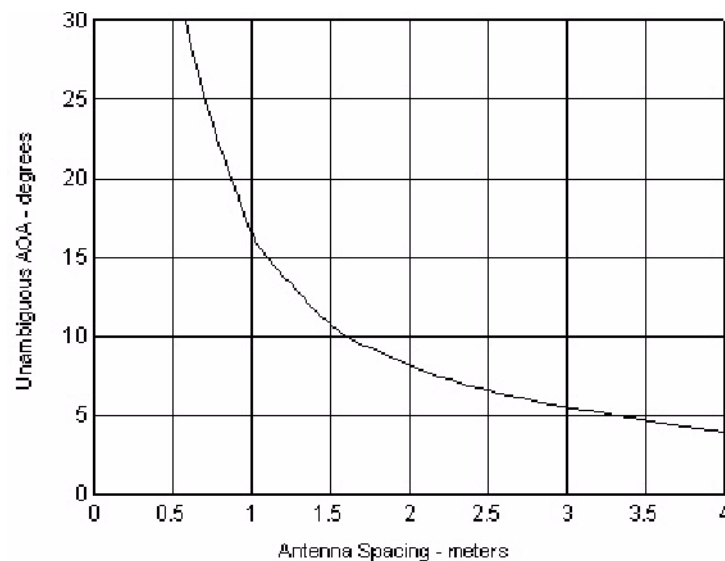


Figure D-9 Unambiguous AOA vs. ASA Antenna Spacing (Cycle Size)

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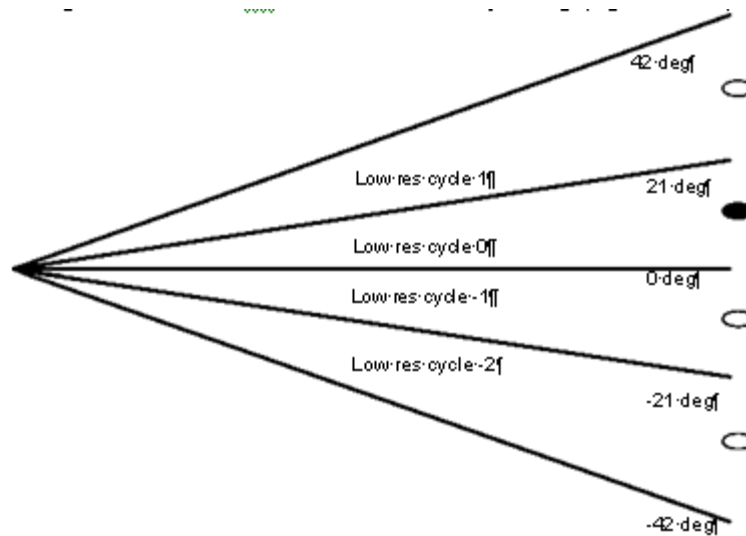


Figure D-10 Low Resolution Cycle for the ASA

iv) Accurate Initialization

Accurate initialization of the tracking system implies the low resolution system will be accurate enough to pick the right cycle for the high resolution system. The algorithm for finding the correct cycle is based on finding the minimum residual (minimum difference) between the low resolution angle-of-arrival estimate and the high resolution estimate, factoring in the integer cycle k . Again referring to Figure 11-10, the case where the low resolution measurement can be off by up to $\frac{1}{2}$ cycle is illustrated. There are multiple potential solutions for the high resolution measurement but the minimum residual between it and any low resolution AOA estimate within $\frac{1}{2}$ of a high resolution cycle will result in the correct cycle for the high resolution.

v) Graphic Verification of Error

From this illustration, we can graphically verify that the error of the low resolution system must be less than $\frac{1}{2}$ the cycle size of the high resolution system. In order to deal with noise and provide a robust localizer estimate, the low resolution and high resolution system are independently tracked using a linear Kalman filter.

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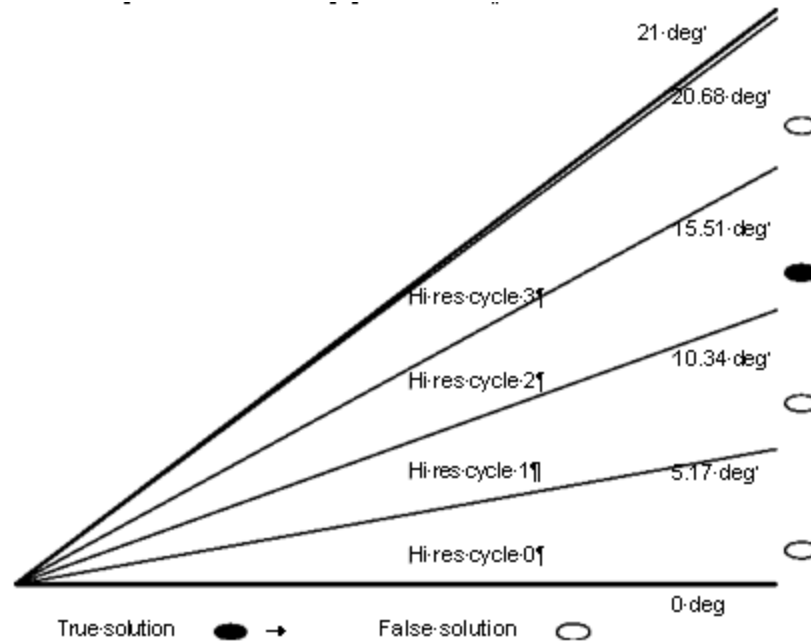


Figure D-11 High Resolution vs. Low Resolution Spacing

vi) TOA Data Measurements

Time-of-Arrival (TOA) data measurements generated by the ASA are used to generate a low resolution azimuth solution which is used to resolve the ambiguities of the low resolution cycle, created because the low cycle is not large enough to cover the tracking volume.

4) Overview: System's Signal Processing

a) Main

Main sends messages to or receives messages from other processes. These messages include:

- Input of desired transponder code to track,
- Output of sensor setup parameters,
- Output of track data for recordings, and
- Output of aircraft correction information for Uplink processing.

*Main also controls synchronization and data flow and internal process state (e.g., Acquiring, Tracking, etc.).

b) Pulse Processing

Pulse Processing receives the sensor data as input and determines the location of the transponder reply in the data using a combination of pulse phase and amplitude discrimination techniques.

c) Data Quality Monitor

Data Quality Monitor (DQM) provides integrity monitoring of the sensor acquisition by validating the raw phase and power measurements across channels. It also checks the frequencies of the Cal/BIT antenna and the aircraft transponder.

d) Calibration Processing

Calibration Processing updates and validates the calibration tracks, to monitor the calibration and data quality.

e) Track Processing

Track Processing does all of the processing required to maintain the aircraft track. These tasks are:

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- Conversion of raw measurements into physical measurements,
- Initializing system and secondary tracks,
- Filtering physical measurements into tracks, and
- Track maintenance bookkeeping functions.

f) Track Monitor

Track Monitor monitors the system and updates the track status. It performs integrity monitoring tests on track jitter and comparisons of estimates from different measurement sets.

g) Steer

Steer computes corrections to provide to the aircraft through the Uplink process. It uses track information computed by Track Processing and approach information for the selected approach to calculate the corrections.

h) Creating the Guidance Solution

The system treats the course line as a series of one or more segments. This method allows the system to provide guidance to an aircraft following a curved approach path. For each segment, the guidance subsystem computes course corrections and provides this information to the aircraft. The guidance subsystem uses track information computed by tracking subsystem and compares it to the nearest point on the known course line.

i) Segments and Waypoints

The course line is defined as a series of “segments” connected by “waypoints” with the first waypoint being the touchdown point. The guidance subsystem finds the segment on the approach nearest the location of the aircraft by calculating the distance from each waypoint to the aircraft and choosing the smallest. The system then computes the perpendicular distance between the aircraft position estimate and the closest approach leg (segment) in directions parallel to both the y and z axis.

ii) Guidance Volume

The guidance volume is an imaginary conical area with vertex at the glide slope apparent emanation point. The apparent emanation point for glide slope is determined by dividing the threshold crossing height by the tangent of the glide slope angle e.g. $55\text{ft}/\tan(3.0)$. Guidance is only calculated if the aircraft is in the guidance volume [Figure 11-12]. If the location of the aircraft is not within the guidance volume, the system does not provide guidance and possible failure status occurs.

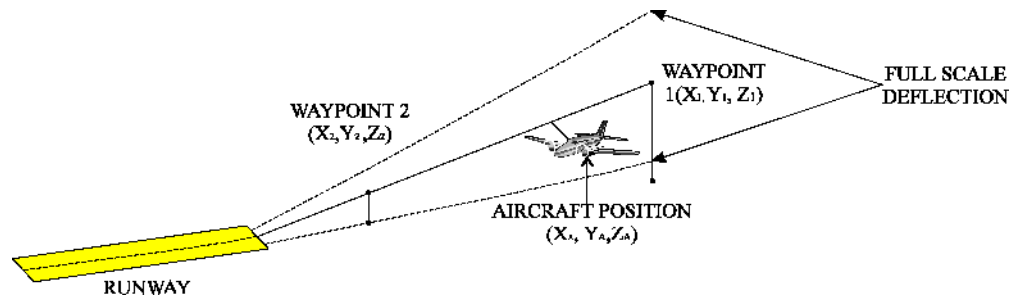


Figure D-12 Calculation of Course Offsets

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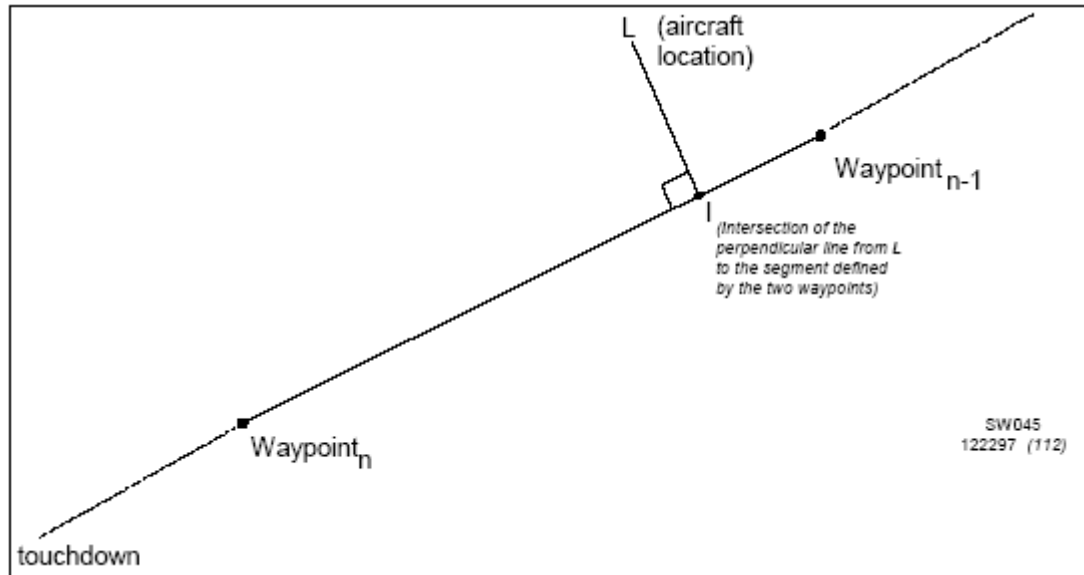


Figure D-13 Determining Closest Segment on Approach Course Line

iii) Localizer Signal

The localizer signal is based on the position of the aircraft relative to the desired course line.

i) Elevation Guidance

The system transmits a signal that emulates an ILS glide slope signal in order to provide guidance in elevation. The glide slope signal is identical to that which would be produced by a standard UHF glide path system with the exception that these signal properties apply at the position of the tracked aircraft only.

i) Glide Slope Transmission Pattern

The glide slope transmission creates a composite field pattern similar to an ILS glide slope signal at the aircraft position. The signal is amplitude modulated by a 90 Hz and a 150 Hz tone. As the aircraft nears the runway and the distance between the aircraft and the system's sensors lessens, minor changes in vertical position translate into significantly greater glide slope indicator needle deflections. In order to compensate for this effect, the system reduces glide slope displacement sensitivity when the aircraft is within its last 600 ft/183 m of descent.

j) System Monitoring

The monitoring subsystem is designed to monitor the essential elements of the system to preserve the safety and integrity of the performance of the system. Requirements for the monitoring subsystem were derived from the System Safety Assessment (SSA).

i) Fault Tree Analysis

The Fault Tree Analysis (FTA) and exposure time considerations contribute to determining whether a monitor element is scheduled to occur during BIT or whether it must be tested more often. Tests that require aircraft data can only occur during track – mode. The BIT cycle interval was determined from exposure time analysis in the FTA.

k) Health Monitoring

System health is monitored during Built-In-Test (BIT). BIT is an ongoing series of testing that functions during surveillance and guidance. During BIT, the system creates a reference signal, the Cal/BIT signal, which is then broadcast towards the receiving antenna arrays. The signal emulates a transponder reply signal allowing the system to ascertain its ability to accurately measure transponder replies. The BIT also utilizes additional hardware health monitors to verify other system component readiness.

l) Surveillance Monitoring

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During surveillance, the Cal/BIT signal is used to continuously monitor the tracking measurement quality. Along with testing the characteristics of the incoming transponder reply signal and the system's estimate of the aircraft's position, it ensures that the measurement and system tracks are accurate.

m) Guidance Monitoring

During guidance, the monitoring subsystem ensures that guidance signals are accurate as defined by ICAO Annex 10. The monitoring subsystem does the same monitoring for guidance tracks as that done for surveillance with the addition of track quality monitors. In the event of degradation to the guidance track integrity or if the monitoring sub-system itself should fail, guidance to an aircraft will cease.

Appendix A ANPC TLS Manual Recommended Change Form

The following form is intended to document recommended changes pertaining to the Technical Instruction Manual. The TLS TI Manual Recommended Change Form may be legibly handwritten or typed and must include the revision level of the TI Manual being discussed. The recommendation sheet is intended to be used as master copies from which to make working recommendation sheet.

Scan and email the completed form to Email CustomerSupport@anpc.com, or use standard mail and post to this address:

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489 N. 8th Street, Suite 203
HOOD RIVER, OR 97031
ATTN: Customer Support

Phone: (541) 386-1747

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